

31. (a) (i) A parallel plate capacitor A of capacitance C is charged by a battery to a potential ' V '. The battery is disconnected and an uncharged identical capacitor B is connected across it. Calculate for the capacitor A the new value of the :

(I) charge

(II) potential difference

(III) energy stored

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Justify your answers.

(ii) Draw the pattern of electric field lines due to :

(I) positively charged conducting sphere, and

(II) an electric dipole.

(i) (I) Charge will become half

Charge will flow from capacitors A (high potential) to capacitor B (low potential) till they achieve equilibrium.

Alternatively

$$q_A = cV \quad q_B = 0$$

$$V_{common} = \frac{q_A + q_B}{c_A + c_B} = \frac{cV}{2c}$$

$$= \frac{V}{2}$$

$$q'_A = cV_{common} = \frac{cV}{2} = \frac{q_A}{2}$$

(II) Potential difference will become half.

$$\begin{aligned} V_{common} &= \frac{q_A + q_B}{c_A + c_B} \\ &= \frac{cV + 0}{2c} = \frac{V}{2} \end{aligned}$$

(III) Energy stored will becomes one fourth

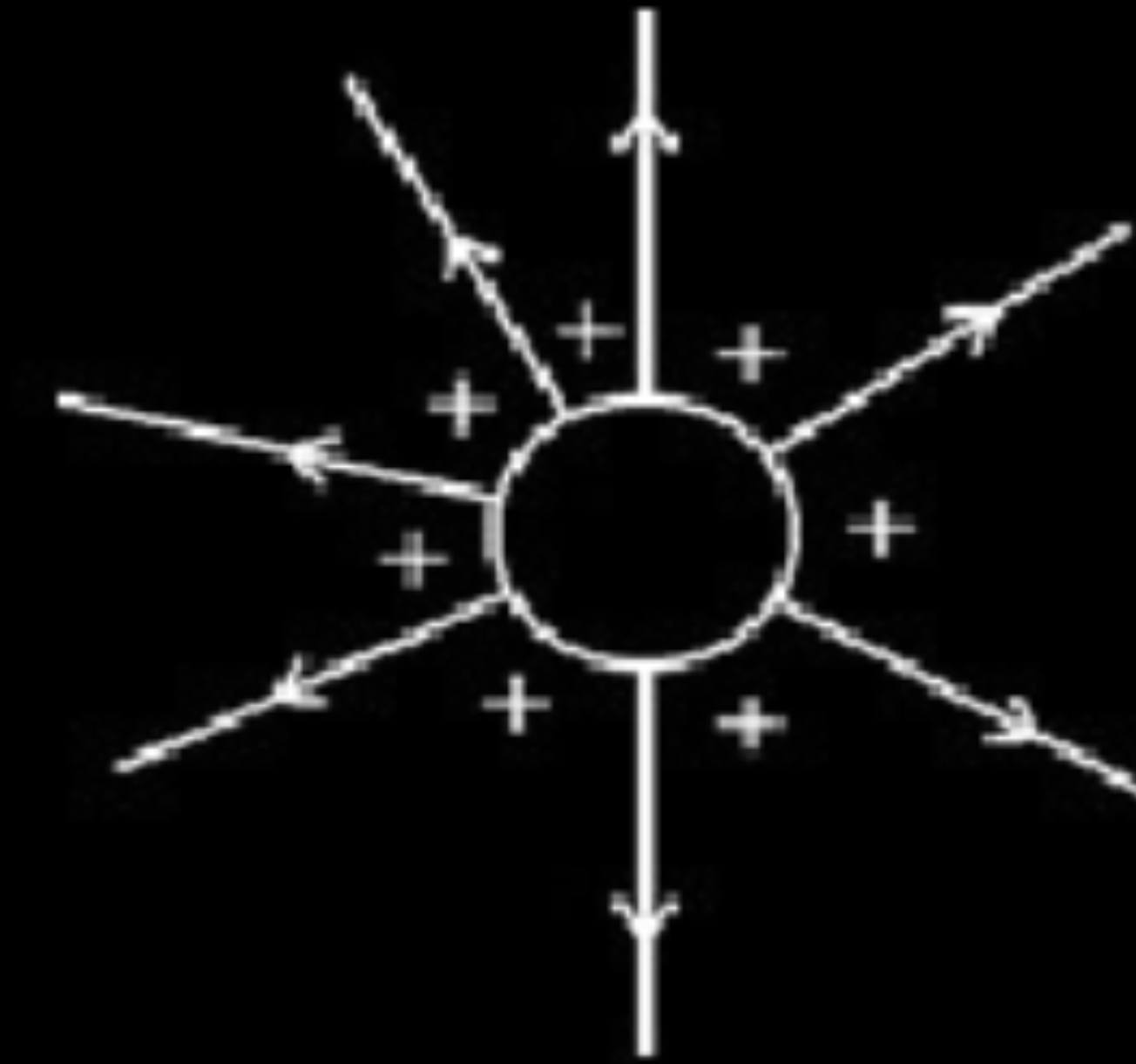
$$U_A = \frac{1}{2} CV^2$$

$$U'_A = \frac{1}{2} CV_{common}^2$$

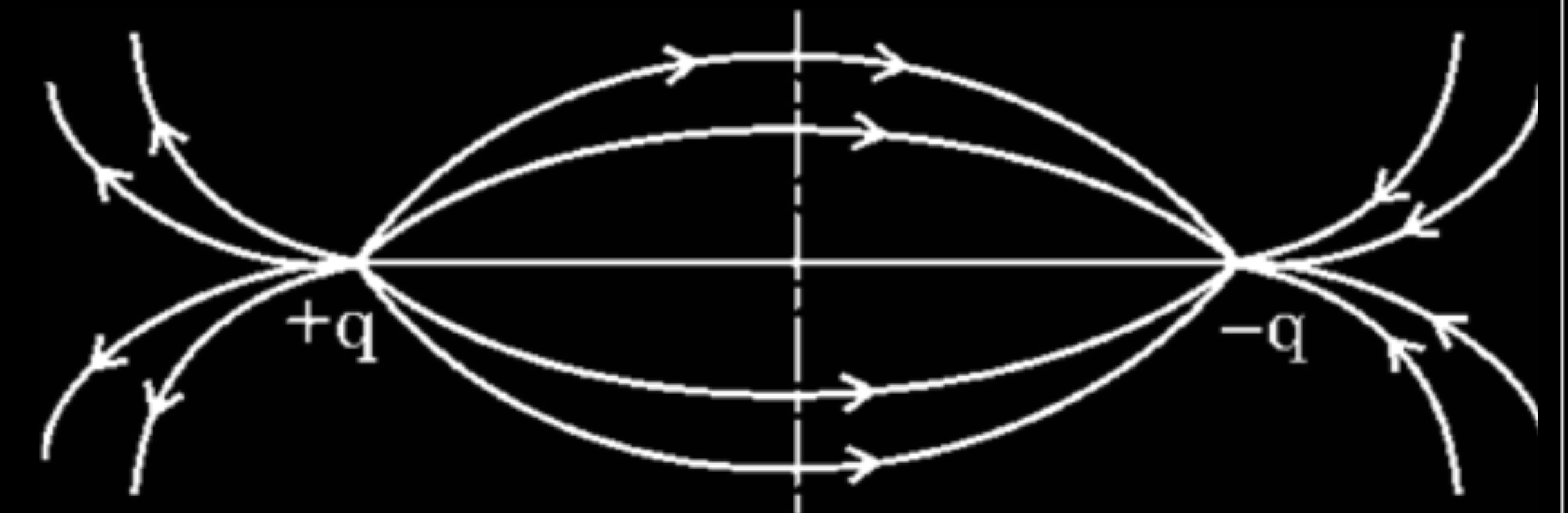
$$= \frac{1}{2} C \left(\frac{V}{2} \right)^2$$

$$= \frac{U_A}{4}$$

(ii)
(I)



(II)



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

1

- 18.** A 50 pF capacitor is charged by a 100 V battery. The battery is disconnected and the capacitor is connected to another uncharged 50 pF capacitor. Find the energy stored in the combination.

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2

$$C_1 = C_2 = 50 \times 10^{-12} \text{ F}$$

$$V_1 = 100 \text{ V} \quad V_2 = 0$$

$$\text{Common potential } V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$V = \frac{50 \times 10^{-12} \times 100}{100 \times 10^{-12}} = 50 \text{ V}$$

$$\text{Energy stored } U = \frac{1}{2} (C_1 + C_2) V^2$$

$$U = \frac{1}{2} \times 100 \times 10^{-12} \times 50 \times 50 \text{ J}$$

$$U = 1.25 \times 10^{-7} \text{ J}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

Alternatively:

$$q_1 = C_1 V_1$$

$$= 50 \times 10^{-12} \times 100$$

$$= 5 \times 10^{-9} \text{ C}$$

$$q_2 = 0$$

$$q = q_1 + q_2$$

$$q = 5 \times 10^{-9} \text{ C}$$

$$U = \frac{1}{2} \frac{q^2}{(C_1 + C_2)} = \frac{1}{2} \frac{(5 \times 10^{-9})^2}{100 \times 10^{-12}}$$

$$U = 1.25 \times 10^{-7} \text{ J}$$

- (b) (i) A parallel plate capacitor of capacitance C is connected to a battery. Explain how the plates of the capacitor acquire equal and opposite charges. Hence obtain expression for the energy stored in the capacitor.
- (ii) A parallel plate capacitor has $+4.0\ \mu\text{C}$ and $-4.0\ \mu\text{C}$ charges on its plates. Find the area of each plate so that an electric field of $1600\ \text{V/mm}$ is produced between the plates.

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(b) (i) As soon as we switch on the circuit the plate of the capacitor connected with positive terminal of battery loses electrons and gains positive charge. At the same time plate connected with negative terminal gains electrons and acquires negative charge till the potential difference between the capacitor plates is equal to potential difference between the terminals of battery.

1

Work done in charging the capacitor through charge dQ

$$dW = VdQ$$

As, $V = \frac{Q}{C}$

$\frac{1}{2}$

$$dW = \frac{Q}{C} dQ$$

$\frac{1}{2}$

Total work done

$$\int dW = \int \frac{Q}{C} dQ$$

$\frac{1}{2}$

$$W = \frac{Q^2}{2C} = \frac{1}{2} CV^2 \quad (\because Q = CV)$$

$\frac{1}{2}$

(ii) Electric field, $E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$

$\frac{1}{2}$

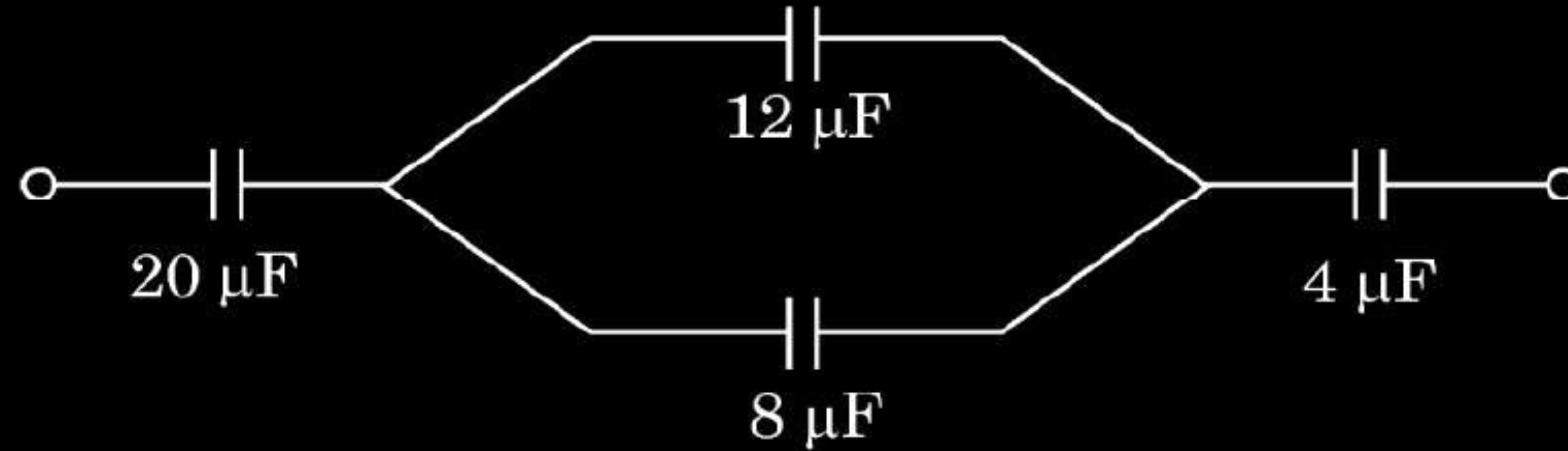
$$A = \frac{Q}{E\epsilon_0} = \frac{4.0 \times 10^{-6}}{1600 \times 10^3 \times 8.85 \times 10^{-12}} m^2$$

1

$$A = 0.28 m^2$$

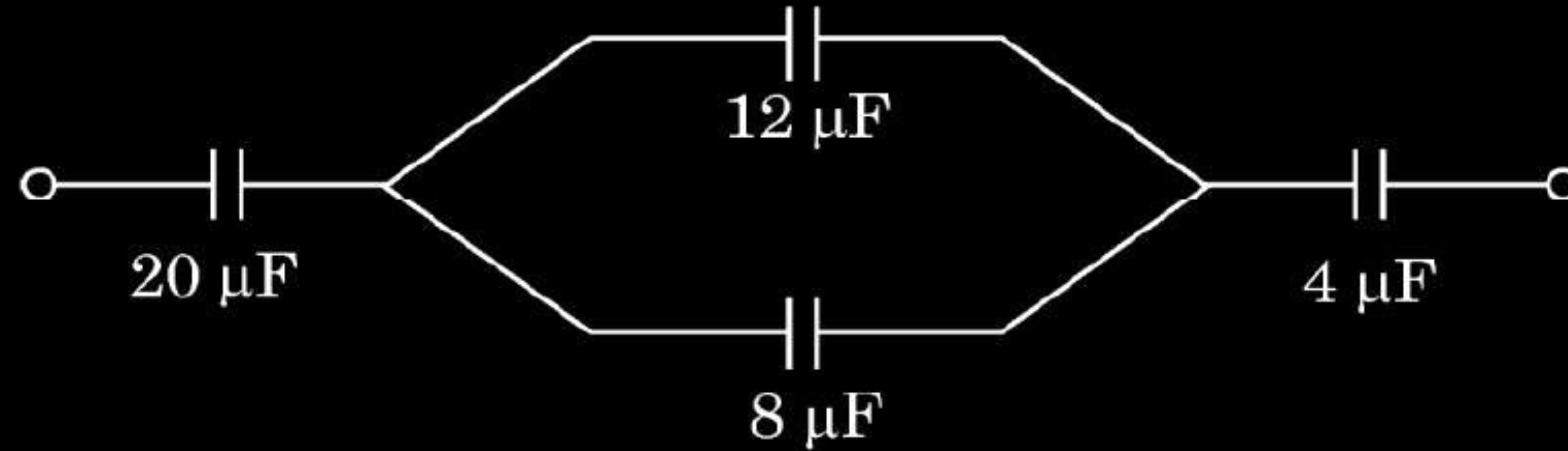
$\frac{1}{2}$

1. In the circuit shown, the charge on the left plate of $20\ \mu\text{F}$ capacitor is $-50\ \mu\text{C}$. The charge on the right plate of the $12\ \mu\text{F}$ capacitor is :



- (A) $20\ \mu\text{C}$ (B) $-20\ \mu\text{C}$
(C) $30\ \mu\text{C}$ (D) $-30\ \mu\text{C}$

1. In the circuit shown, the charge on the left plate of $20\ \mu\text{F}$ capacitor is $-50\ \mu\text{C}$. The charge on the right plate of the $12\ \mu\text{F}$ capacitor is :



- (A) $20\ \mu\text{C}$ (B) $-20\ \mu\text{C}$
(C) $30\ \mu\text{C}$ (D) $-30\ \mu\text{C}$

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(C) $30\ \mu\text{C}$

- 33.** (a) (i) A parallel plate capacitor with plate area A and plate separation d has a capacitance C_0 . A slab of dielectric constant K having area A and thickness $\left(\frac{d}{4}\right)$ is inserted in the capacitor, parallel to the plates. Find the new value of its capacitance.
- (ii) You are provided with a large number of $1\ \mu\text{F}$ identical capacitors and a power supply of $1200\ \text{V}$. The dielectric medium used in each capacitor can withstand up to $200\ \text{V}$ only. Find the minimum number of capacitors and their arrangement, required to build a capacitor system of equivalent capacitance of $2\ \mu\text{F}$ for use with this supply.

$$(i) \quad C_0 = \frac{\epsilon_0 A}{d}$$

$$C = \frac{\epsilon_0 A}{(d - t) + \frac{t}{K}}$$

$$t = d/4$$

$$C = \frac{\epsilon_0 A}{\left(d - \frac{d}{4}\right) + \frac{d}{4K}} = \frac{\epsilon_0 A}{d \left(\frac{3}{4} + \frac{1}{4K}\right)}$$

$$= C_0 \frac{4K}{(3K+1)}$$

(ii) Each capacitance can withstand 200V

$$\text{No. of capacitors in each row} = \frac{1200}{200} = 6$$

Net capacitance of each row= $1/6 \mu\text{F}$

Number of rows = n

$$C_{eq} = C_1 + C_2 + \text{-----} + C_n$$

$$C_{eq} = \frac{1}{6} + \frac{1}{6} + \dots + n$$

$$2 = \frac{n}{6}$$

$$\therefore n = 12$$

$$\begin{aligned}\text{Total no. of capacitors in the arrangement} &= 6 \times 12 \\ &= 72\end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

Alternatively: When dielectric is inserted, the electric field between the plates is $E = E_0/K$

The potential difference will be

$$V = E_0 \left(\frac{3d}{4} \right) + E \left(\frac{d}{4} \right)$$

$$= E_0 \left(\frac{3d}{4} \right) + \frac{E_0}{K} \left(\frac{d}{4} \right)$$

$$= V_0 \left(\frac{3}{4} + \frac{1}{4K} \right)$$

$$V = V_0 \left(\frac{3K+1}{4K} \right)$$

$$C = \frac{Q_0}{V} = \left(\frac{4K}{3K+1} \right) \frac{Q_0}{V_0}$$

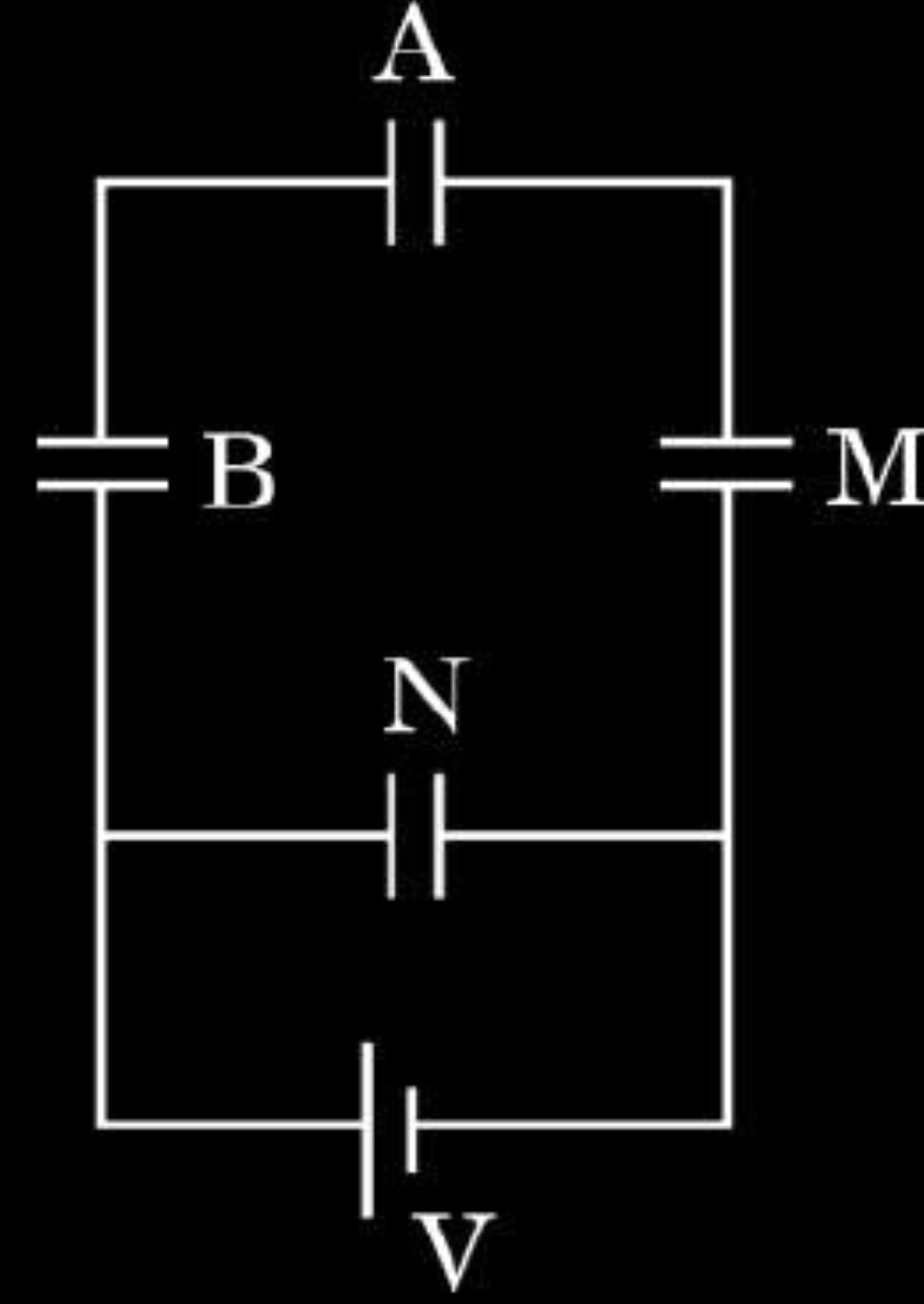
$$C = C_0 \left(\frac{4K}{3K+1} \right)$$

29. A capacitor is a system of two conductors separated by an insulator. In practice, the two conductors have charges Q and $-Q$ with potential difference $V = V_1 - V_2$ between them. The ratio $\frac{Q}{V}$ is a constant, denoted by C and is called the capacitance of the capacitor. It is independent of Q or V . It depends only on the geometrical configuration (shape, size, separation) of the two conductors and the medium separating the conductors. When a parallel plate capacitor is charged, the electric field E_0 is localised between the plates and is uniform throughout. When a slab of a dielectric is inserted between the charged plates (charge density σ), the dielectric is polarised by the field. Consequently opposite charges appear on the faces of the slab, near the plates, with surface charge density of magnitude σ_p . For a linear dielectric σ_p is proportional to E_0 . Introduction of a dielectric changes the electric field, and hence, the capacitance of a capacitor, and hence, the energy stored in the capacitor.

Like resistors, capacitors can also be arranged in series or in parallel or in a combination of series and parallel.

- (i) (a) Three capacitors A, B and M, each of capacitance C are connected to a capacitor N of capacitance $2C$ and a battery as shown in the figure. If the charges on A and N are Q and Q' respectively, then $\frac{Q'}{Q}$ is :

1



(A) $\frac{1}{6}$

(B) $\frac{1}{3}$

(C) 3

(D) 6

(b) A slab (area A and thickness $\frac{d}{2}$) of dielectric constant K is inserted in a parallel plate capacitor of plate area A and plate separation d . If C and C_0 are the capacitances of the capacitors with and without the dielectric, then $\frac{C}{C_0}$ is : *1*

(A) $\frac{K+1}{2K}$

(B) $\frac{2K}{K+1}$

(C) $\frac{K}{K-1}$

(D) $\frac{K-1}{K}$

- (ii) An electric field E is established between the plates of an air filled parallel plate capacitor, with charges Q and $-Q$. V is the volume of the space enclosed between the plates. The energy stored in the capacitor is :

1

(A) $\frac{1}{2} \epsilon_0 E^2$

(B) $\epsilon_0 Q^2 E$

(C) $\frac{1}{2} \epsilon_0 E^2 V$

(D) $\epsilon_0 E Q V$

- (iii) A slab (area A and thickness d_1) of a linear dielectric of dielectric constant K is inserted between charged plates (charge density σ) of a parallel plate capacitor [plate area A and plate separation d ($> d_1$)] and opposite charges with charge density of magnitude σ_p appear on the faces of the slab. The dielectric constant K is given by : 1

(A) $\frac{\sigma + \sigma_p}{\sigma}$

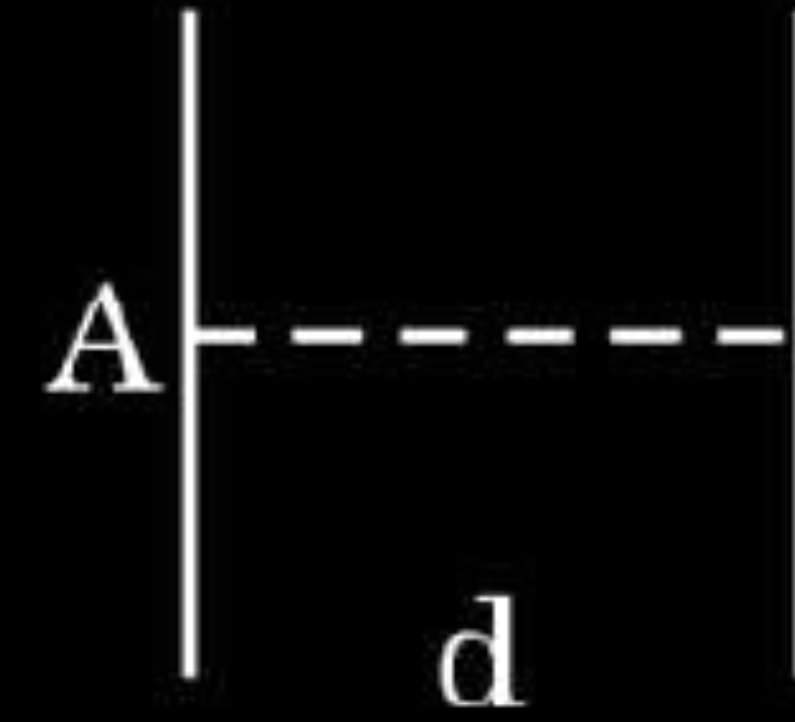
(B) $\frac{\sigma}{\sigma - \sigma_p}$

(C) $\frac{\sigma + \sigma_p}{\sigma_p}$

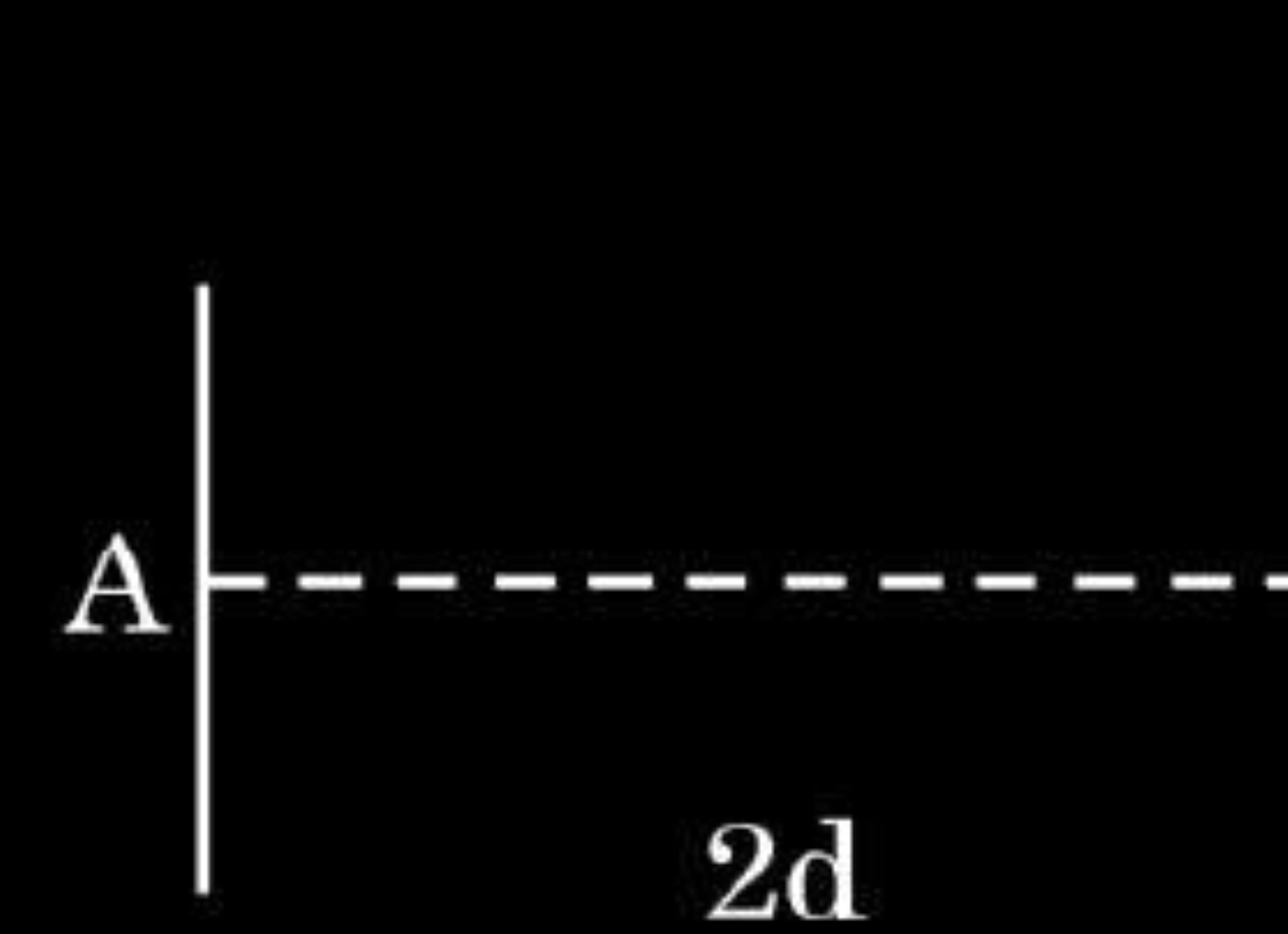
(D) $\frac{\sigma}{\sigma_p}$

- (iv) Consider a capacitor of capacitance C , with plate area A and plate separation d , filled with air [Fig. (a)]. The distance between the plates is increased to $2d$ and one of the plates is shifted as shown in Fig. (b). The capacitance of the new system now is :

1



(a)



(b)

(A) $\frac{C}{4}$

(B) $\frac{C}{2}$

(C) $2C$

(D) $4C$

29. A power supply is connected across the two plates of a capacitor. Consequently, one plate becomes positively charged and the other plate becomes negatively charged. Thus a uniform electric field is established between the two plates. If a charged particle is released between these two plates, it experiences a force. The potential difference between the plates affects its charge and the energy stored in the capacitor. $4 \times 1 = 4$

(i) A proton and an electron are released from rest in the space between the two plates of a charged capacitor. Assume that the proton and the electron do not interact with each other. The acceleration of the proton compared with that of the electron is

1

- (A) same in both magnitude and direction.
- (B) same in magnitude but in opposite direction.
- (C) greater in magnitude and opposite in direction.
- (D) lesser in magnitude but in opposite direction.

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- (A) same in both magnitude and direction.
- (B) same in magnitude but in opposite direction.
- (C) greater in magnitude and opposite in direction.
- (D) lesser in magnitude but in opposite direction.

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(D) Lesser in magnitude but in opposite direction.

(ii) An electron is released between the two parallel plates of a charged capacitor. Which of the following statements is incorrect ? 1

- (A) The electric field between the plates is directed from the positively charged plate to the negatively charged plate.
- (B) The magnitude of electric field is the same at all points between the plates.
- (C) The acceleration of the electrons will be constant between the plates.
- (D) The potential increases in the direction of the electric field between the plates.

(ii) An electron is released between the two parallel plates of a charged capacitor. Which of the following statements is incorrect ? 1

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- (B) The magnitude of electric field is the same at all points between the plates.
- (C) The acceleration of the electrons will be constant between the plates.
- (D) The potential increases in the direction of the electric field between the plates.

(ii) (D) The potential increases in the direction of the electric field between the plates.

(iii) The potential difference between the two plates of a capacitor is V . An electron of mass m and charge $-e$ is released from rest near the negative plate. The maximum speed gained by the electron is

1

- (A) $2 e V m$ (B) $\sqrt{\frac{e m}{2 V}}$ (C) $\sqrt{\frac{2 e V}{m}}$ (D) $\sqrt{\frac{2 m e}{V}}$

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(C) $\sqrt{\frac{2 e v}{m}}$

(iv) (a) The electric field between two parallel plates of a charged capacitor is 200 V/m . The plates are 5 mm apart. The potential difference between the two plates is

(A) 20 V

(B) 10 V

(C) 5.0 V

(D) 1.0 V

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1

(A) 20 V

(B) 10 V

(C) 5.0 V

(D) 1.0 V

(iv) (a) (D) 1.0 V

(b) A proton is located at a point between the above plates. The force acting on the proton is **1**

(A) $1.6 \times 10^{-19} \text{ N}$

(B) $2 \times 10^9 \text{ N}$

(C) $3.2 \times 10^{-17} \text{ N}$

(D) $4 \times 10^9 \text{ N}$

(b) A proton is located at a point between the above plates. The force acting on the proton is **1**

(A) $1.6 \times 10^{-19} \text{ N}$ (B) $2 \times 10^9 \text{ N}$

(C) $3.2 \times 10^{-17} \text{ N}$ (D) $4 \times 10^9 \text{ N}$

(C) $3.2 \times 10^{-17} \text{ N}$

4. A metal sheet is inserted between the plates of a parallel plate capacitor of capacitance C . If the sheet partly occupies the space between the plates, the capacitance :
- (A) remains C (B) becomes greater than C
(C) becomes less than C (D) becomes zero

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- (A) remains C (B) becomes greater than C
(C) becomes less than C (D) becomes zero

(B) becomes greater than C

- 22.** (a) “There is a limit to the amount of charge that can be stored on a given capacitor.” Explain.
- (b) A capacitor is charged by a battery to a potential difference V . It is disconnected from the battery and connected across another identical uncharged capacitor. Calculate the ratio of total energy stored in the combination to the initial energy stored in the capacitor.

- 22.** (a) “There is a limit to the amount of charge that can be stored on a given capacitor.” Explain.
- (b) A capacitor is charged by a battery to a potential difference V . It is disconnected from the battery and connected across another identical uncharged capacitor. Calculate the ratio of total energy stored in the combination to the initial energy stored in the capacitor.

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3

(a) When a conductor holds a large amount of charge its potential is also high. If electric field becomes high enough, the atoms or molecules of surrounding medium gets ionized. A breakdown occurs in medium and charge on conductor get neutralized or leaks away.

(b) The common potential after the connection of two capacitors

$$\text{Given } V_1 = V, V_2 = 0$$

$$C_1 = C_2 = C$$

$$= \frac{CV}{2C}$$

$$= \frac{V}{2}$$

$$U_i = \frac{1}{2}CV^2$$

1

$\frac{1}{2}$

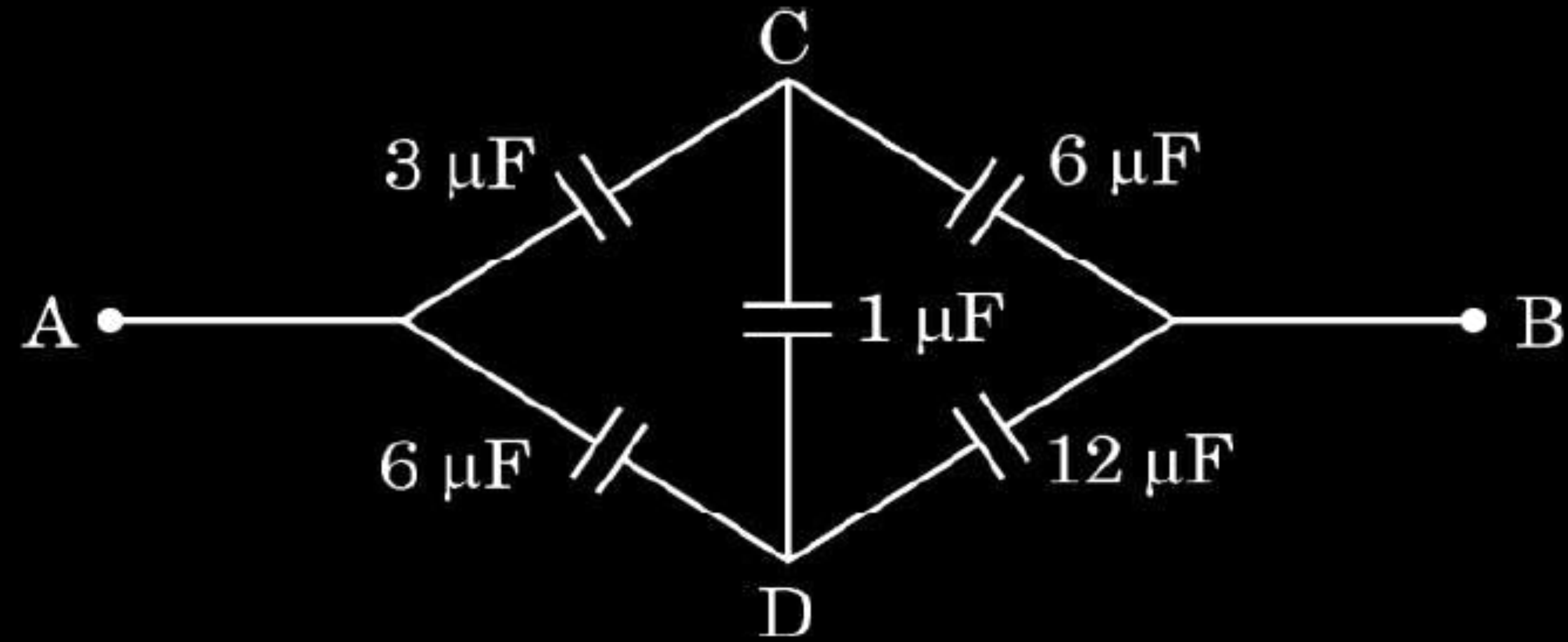
$\frac{1}{2}$

$$\begin{aligned} U_f &= \frac{1}{2}C'V'^2 \\ &= \frac{1}{2} \times 2C \times \left(\frac{V}{2}\right)^2 \\ &= \frac{1}{4}CV^2 \\ \frac{U_f}{U_i} &= \frac{1}{2} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

2. The effective capacitance of the network between points A and B shown in the figure is :



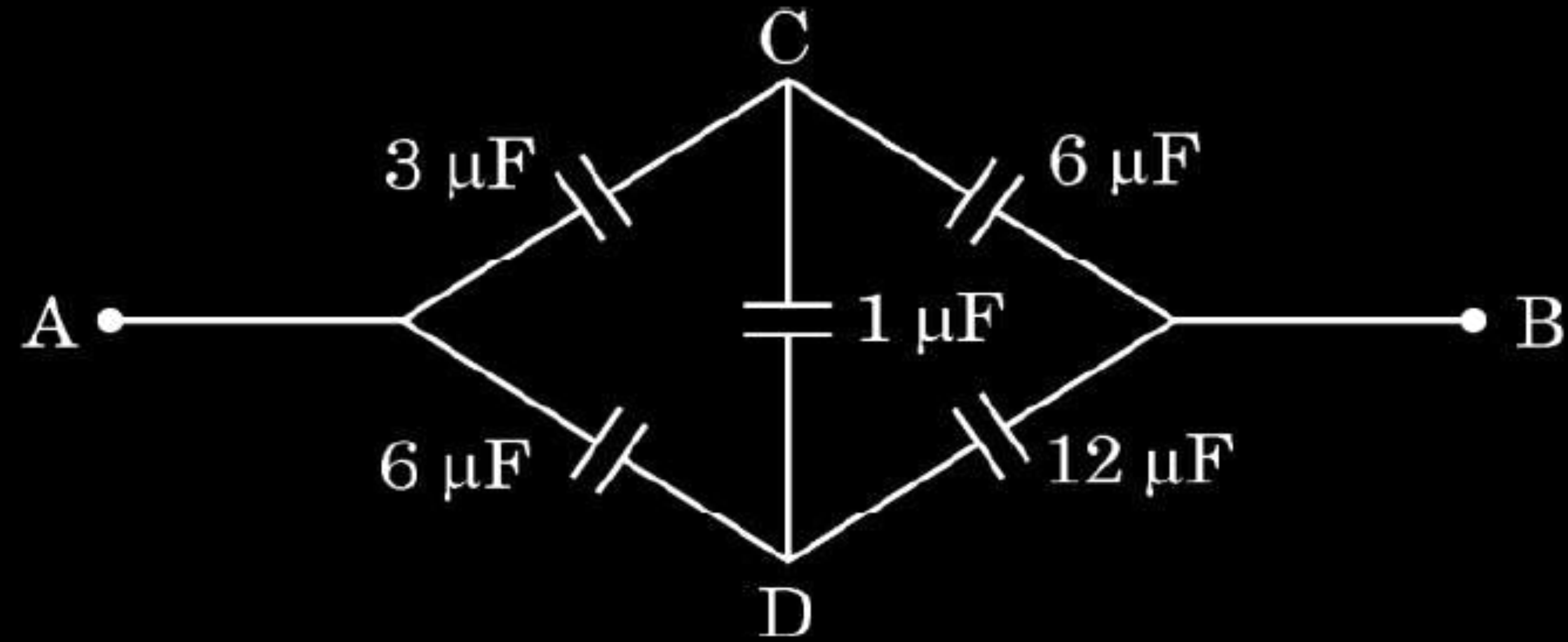
(A) $\frac{4}{3}\ \mu\text{F}$

(B) $2\ \mu\text{F}$

(C) $\frac{3}{2}\ \mu\text{F}$

(D) $6\ \mu\text{F}$

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(A) $\frac{4}{3}\ \mu\text{F}$

(B) $2\ \mu\text{F}$

(C) $\frac{3}{2}\ \mu\text{F}$

(D) $6\ \mu\text{F}$

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(D) $6\ \mu\text{F}$

24. A capacitor of plate area A and plate separation d is charged by a battery to voltage V . The battery is disconnected and plates are slowly pulled apart till the separation becomes $2d$. Find the value of :

3

- (a) potential difference between the plates,
- (b) electric field between the plates,
- (c) work done in pulling the plates apart.

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24. A capacitor of plate area A and plate separation d is charged by a battery to voltage V. The battery is disconnected and plates are slowly pulled apart till the separation becomes 2d. Find the value of:

3

- (a) potential difference between the plates,
- (b) electric field between the plates,
- (c) work done in pulling the plates apart.

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As battery is dis connected, charge remain same

$$C' = \frac{C}{2}$$

(i) $q = q'$

$$CV = C'V'$$

$$CV = \frac{C}{2} V'$$

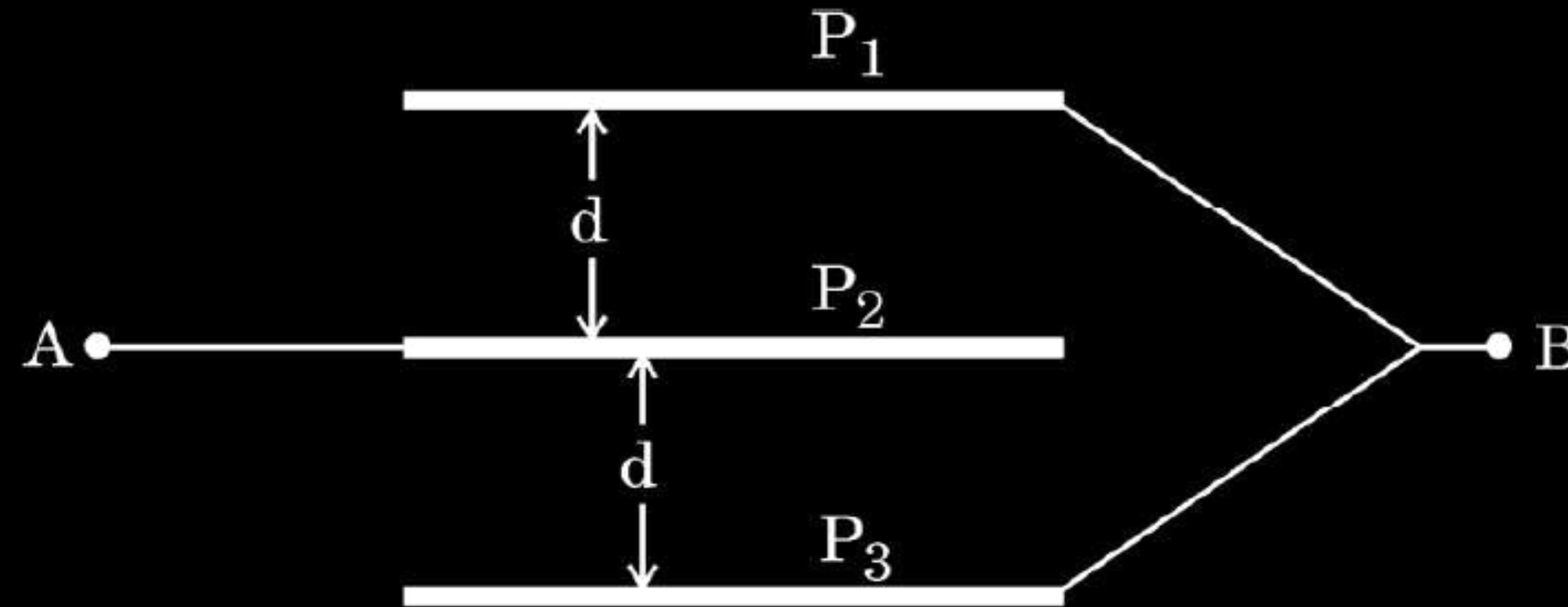
$$V' = 2V$$

$\frac{1}{2}$

$\frac{1}{2}$

(ii)	$E' = \frac{V'}{d'}$	$\frac{1}{2}$
	$= \frac{2V}{2d}$	
	$= \frac{V}{d} = E$	$\frac{1}{2}$
(iii)	$W = U' - U$	$\frac{1}{2}$
	$= \frac{1}{2} \times \frac{C}{2} \times (2V)^2 - \frac{1}{2} CV^2$	
	$= \frac{1}{2} CV^2$	$\frac{1}{2}$

30. A parallel plate capacitor consists of two conducting plates kept generally parallel to each other at a distance. When the capacitor is charged, the charge resides on the inner surfaces of the plates and an electric field is set up between them. Thus, electrostatic energy is stored in the capacitor. The figure shows three large square metallic plates, each of side 'L' held parallel and equidistant from each other. The space between P_1 and P_2 and P_2 and P_3 is completely filled with mica sheets of dielectric constant 'K'. The plate P_2 is connected to point A and other plates P_1 and P_3 are connected to point B. Point A is maintained at a positive potential with respect to point B and the potential difference between A and B is V.



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- (i) The capacitance of the system between A and B will be :

1

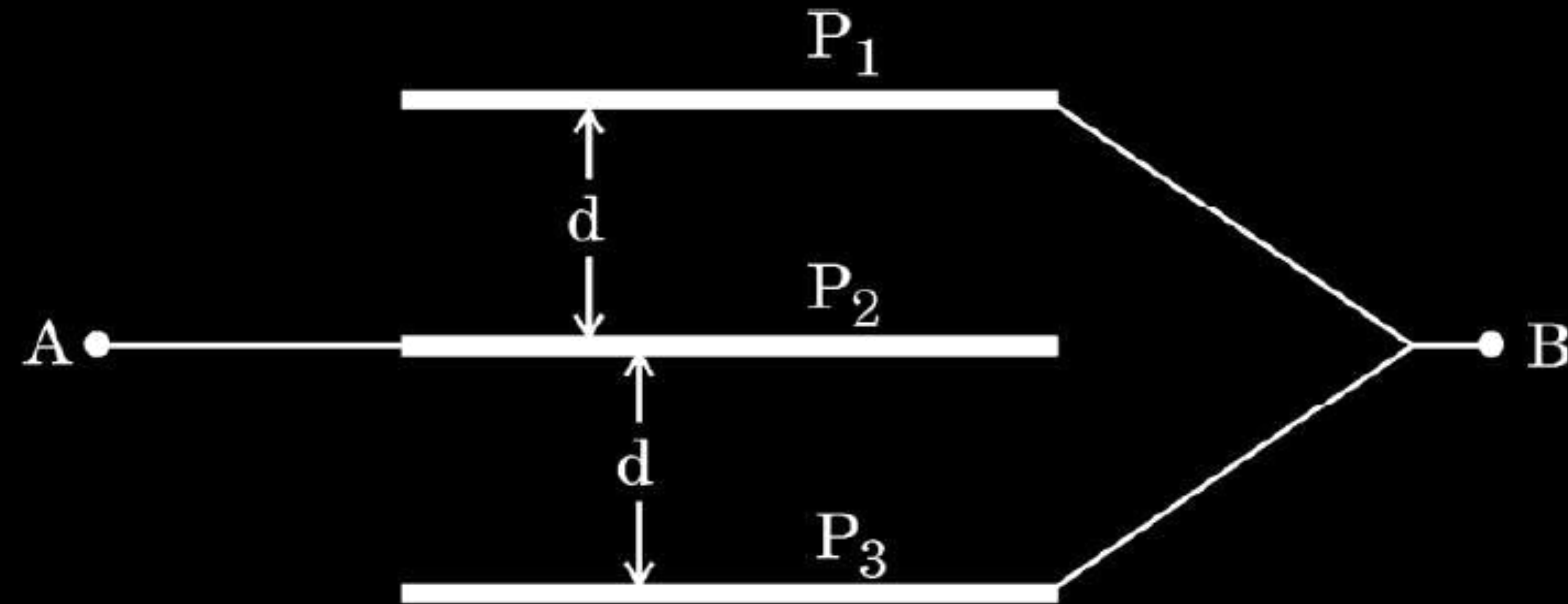
(A) $\frac{\epsilon_0 K L^2}{d}$

(B) $\frac{\epsilon_0 K L^2}{2d}$

(C) $\frac{2\epsilon_0 K L^2}{d}$

(D) $\frac{2\epsilon_0 K d}{L^2}$

30. A parallel plate capacitor consists of two conducting plates kept generally parallel to each other at a distance. When the capacitor is charged, the charge resides on the inner surfaces of the plates and an electric field is set up between them. Thus, electrostatic energy is stored in the capacitor. The figure shows three large square metallic plates, each of side 'L' held parallel and equidistant from each other. The space between P_1 and P_2 and P_2 and P_3 is completely filled with mica sheets of dielectric constant 'K'. The plate P_2 is connected to point A and other plates P_1 and P_3 are connected to point B. Point A is maintained at a positive potential with respect to point B and the potential difference between A and B is V.



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- (i) The capacitance of the system between A and B will be :

1

(A) $\frac{\epsilon_0 K L^2}{d}$

(B) $\frac{\epsilon_0 K L^2}{2d}$

(C) $\frac{2\epsilon_0 K L^2}{d}$

(D) $\frac{2\epsilon_0 K d}{L^2}$

(C) $\frac{2\epsilon_0 K L^2}{d}$

(ii) The charge on plate P_1 is :

1

(A) $\frac{\epsilon_0 V K L^2}{2d}$

(B) $\frac{\epsilon_0 V K L^2}{d}$

(C) $\frac{2\epsilon_0 V K L^2}{d}$

(D) $\frac{\epsilon_0 V K L^2}{4d}$

(ii) The charge on plate P₁ is :

1

(A) $\frac{\epsilon_0 V K L^2}{2d}$

(B) $\frac{\epsilon_0 V K L^2}{d}$

(C) $\frac{2\epsilon_0 V K L^2}{d}$

(D) $\frac{\epsilon_0 V K L^2}{4d}$

(B) $\frac{\epsilon_0 V K L^2}{d}$

(iii) The electric field in the region between P_1 and P_2 is : *1*

(A) $\frac{V}{d}$

(B) $\frac{2V}{d}$

(C) $\frac{V}{2d}$

(D) $\frac{d}{V}$

(iii) The electric field in the region between P_1 and P_2 is : 1

(A) $\frac{V}{d}$

(B) $\frac{2V}{d}$

(C) $\frac{V}{2d}$

(D) $\frac{d}{V}$

(A) $\frac{V}{d}$

- (iv) (a) The separation between the plates of same area (L^2) of a parallel plate air capacitor having capacitance equal to that of this system, will be :

1

(A) $\frac{d}{K}$

(B) $\frac{2d}{K}$

(C) $\frac{d}{2K}$

(D) $\frac{d}{4K}$

- (iv) (a) The separation between the plates of same area (L^2) of a parallel plate air capacitor having capacitance equal to that of this system, will be :

1

(A) $\frac{d}{K}$

(B) $\frac{2d}{K}$

(C) $\frac{d}{2K}$

(D) $\frac{d}{4K}$

(C) $\frac{d}{2K}$

(b) If the source of potential difference applied between A and B is removed, and then A and B are connected by a conducting wire, the net charge on the system will be : 1

(A) $\frac{\epsilon_0 V K L^2}{4d}$

(B) $\frac{\epsilon_0 V K L^2}{2d}$

(C) $\frac{\epsilon_0 V K L^2}{d}$

(D) Zero

(b) If the source of potential difference applied between A and B is removed, and then A and B are connected by a conducting wire, the net charge on the system will be : 1

(A) $\frac{\epsilon_0 V K L^2}{4d}$

(B) $\frac{\epsilon_0 V K L^2}{2d}$

(C) $\frac{\epsilon_0 V K L^2}{d}$

(D) Zero

(D) Zero

30. A parallel plate capacitor has two parallel plates which are separated by an insulating medium like air, mica, etc. When the plates are connected to the terminals of a battery, they get equal and opposite charges and an electric field is set up in between them. This electric field between the two plates depends upon the potential difference applied, the separation of the plates and nature of the medium between the plates. $4 \times 1 = 4$

(i) The electric field between the plates of a parallel plate capacitor is E . Now the separation between the plates is doubled and simultaneously the applied potential difference between the plates is reduced to half of its initial value. The new value of the electric field between the plates will be :

(A) E

(B) $2E$

(C) $\frac{E}{4}$

(D) $\frac{E}{2}$

30. A parallel plate capacitor has two parallel plates which are separated by an insulating medium like air, mica, etc. When the plates are connected to the terminals of a battery, they get equal and opposite charges and an electric field is set up in between them. This electric field between the two plates depends upon the potential difference applied, the separation of the plates and nature of the medium between the plates. $4 \times 1 = 4$

(i) The electric field between the plates of a parallel plate capacitor is E . Now the separation between the plates is doubled and simultaneously the applied potential difference between the plates is reduced to half of its initial value. The new value of the electric field between the plates will be :

(A) E

(B) $2E$

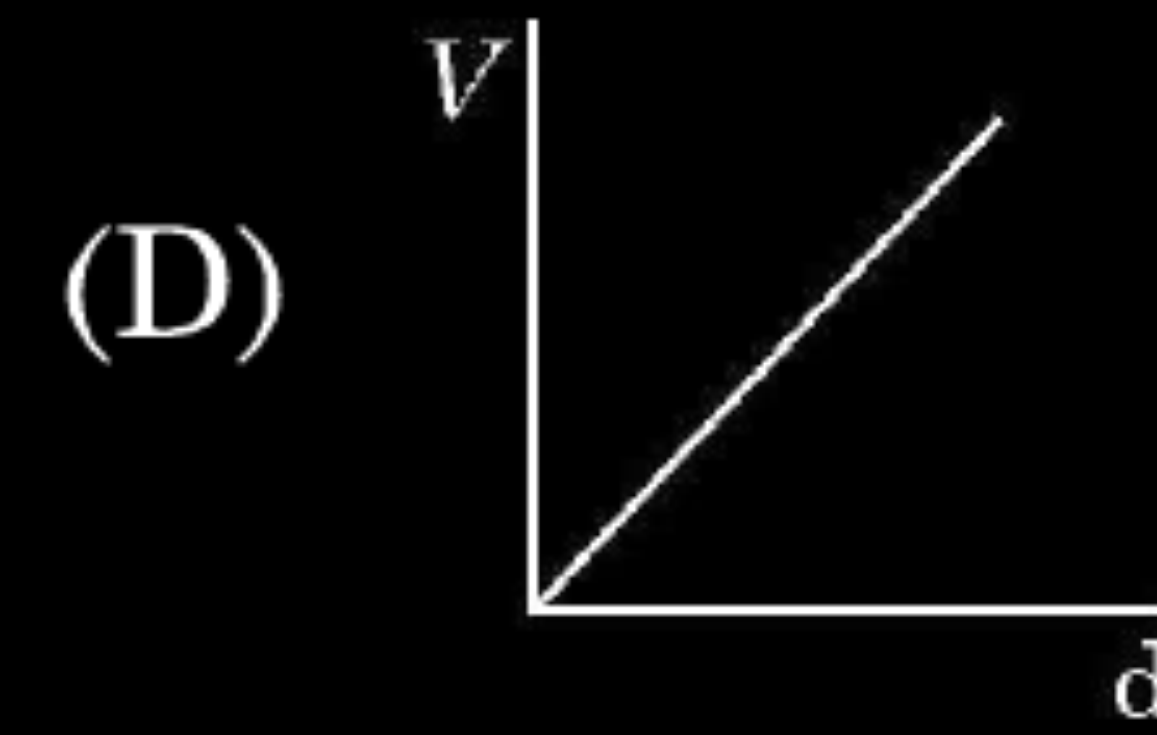
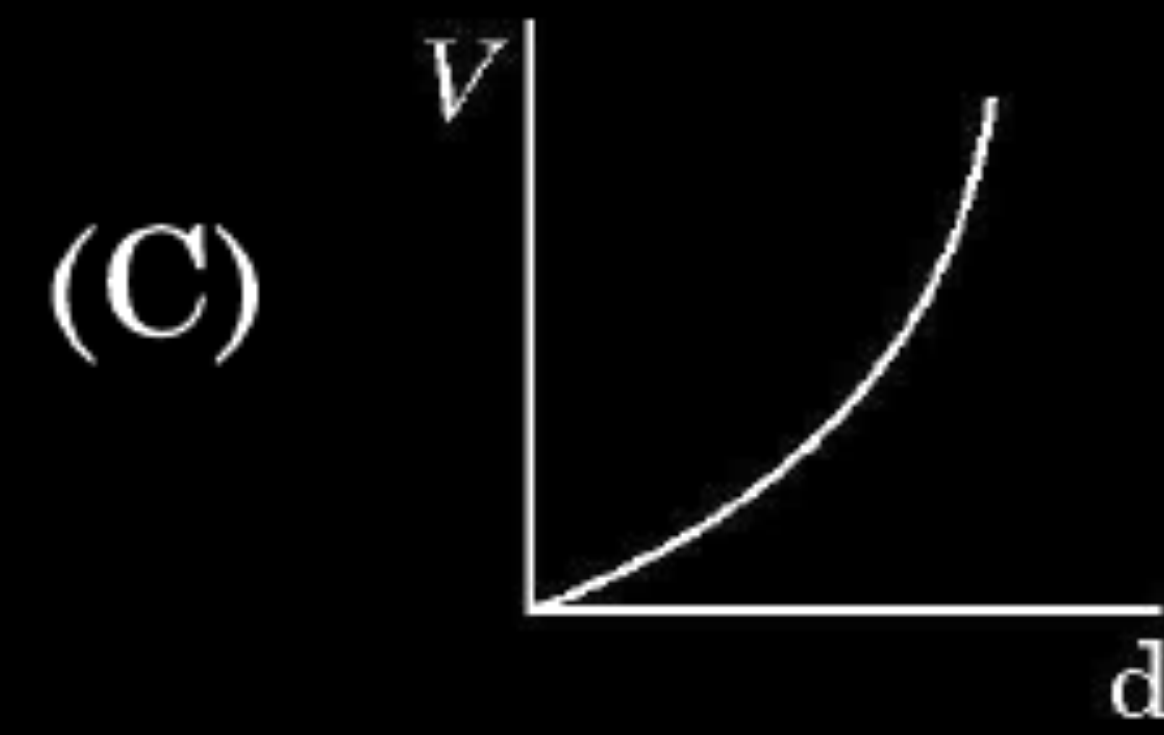
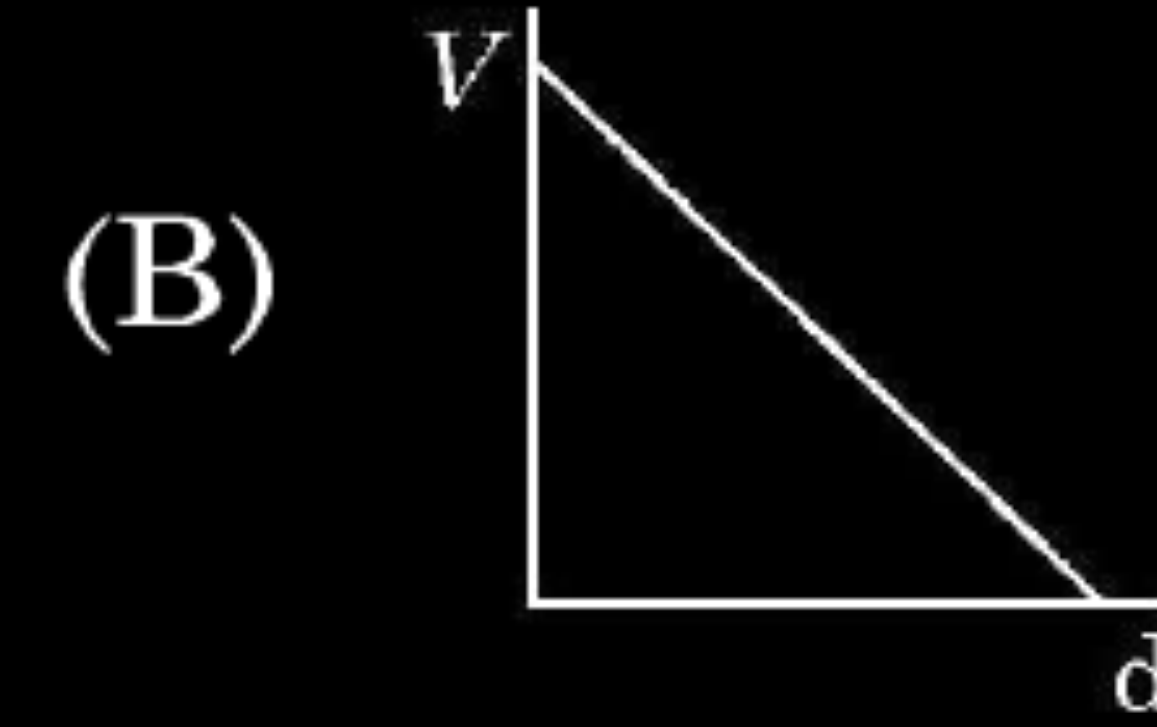
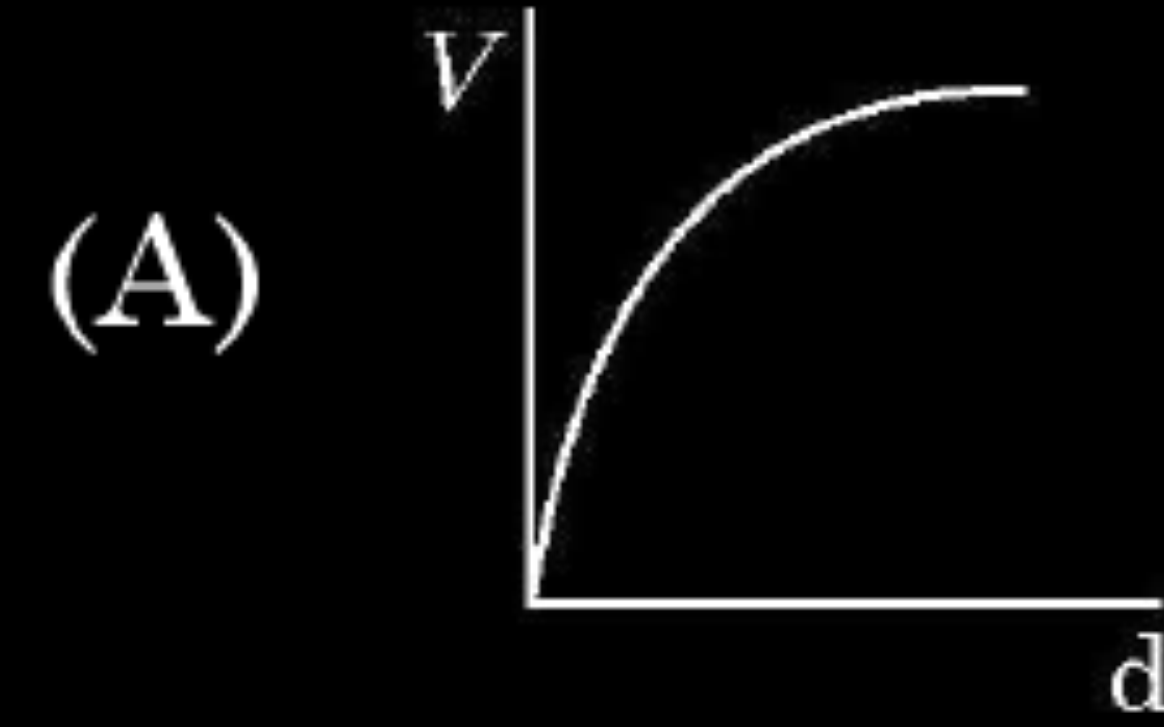
(C) $\frac{E}{4}$

(D) $\frac{E}{2}$

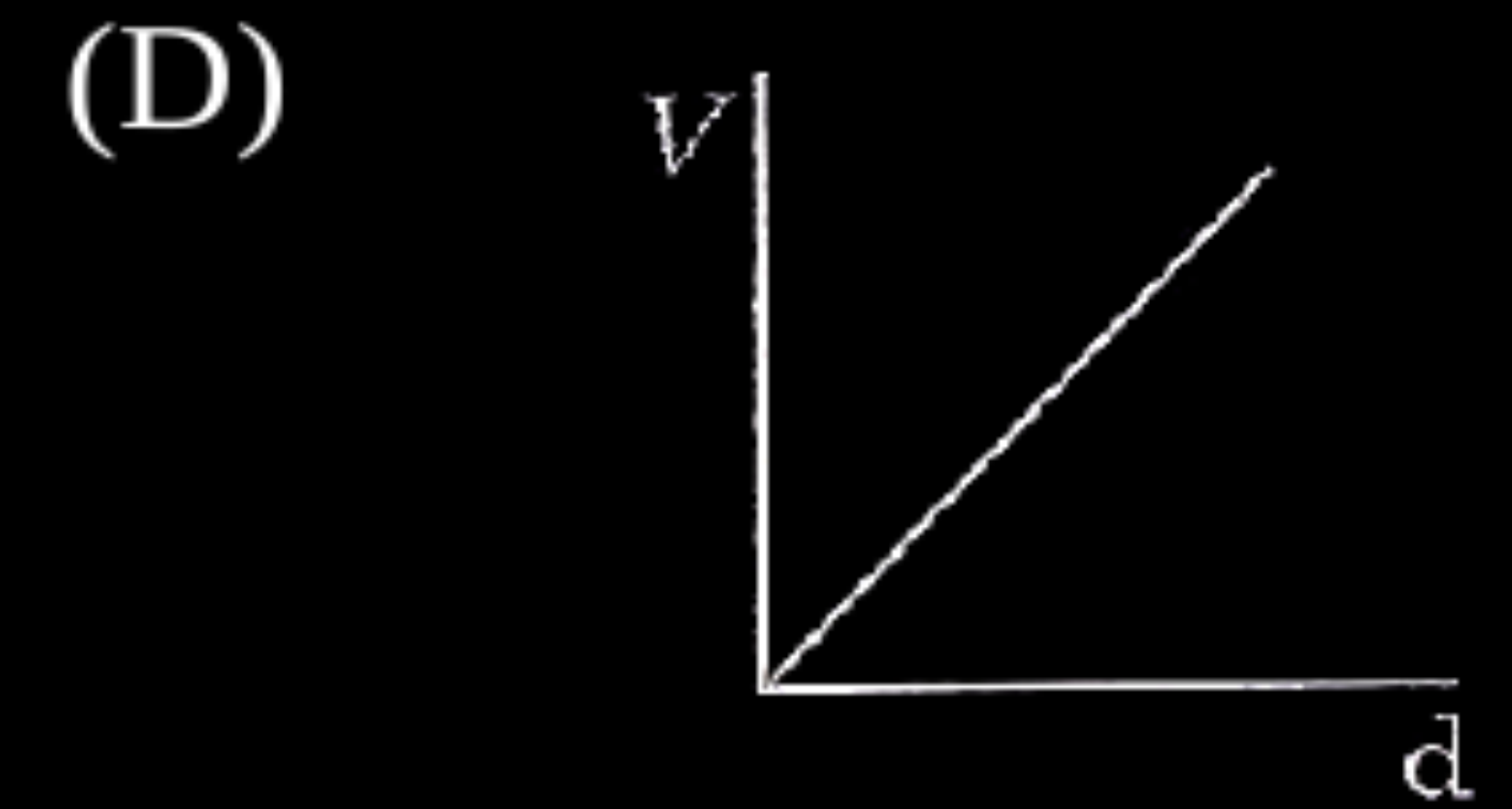
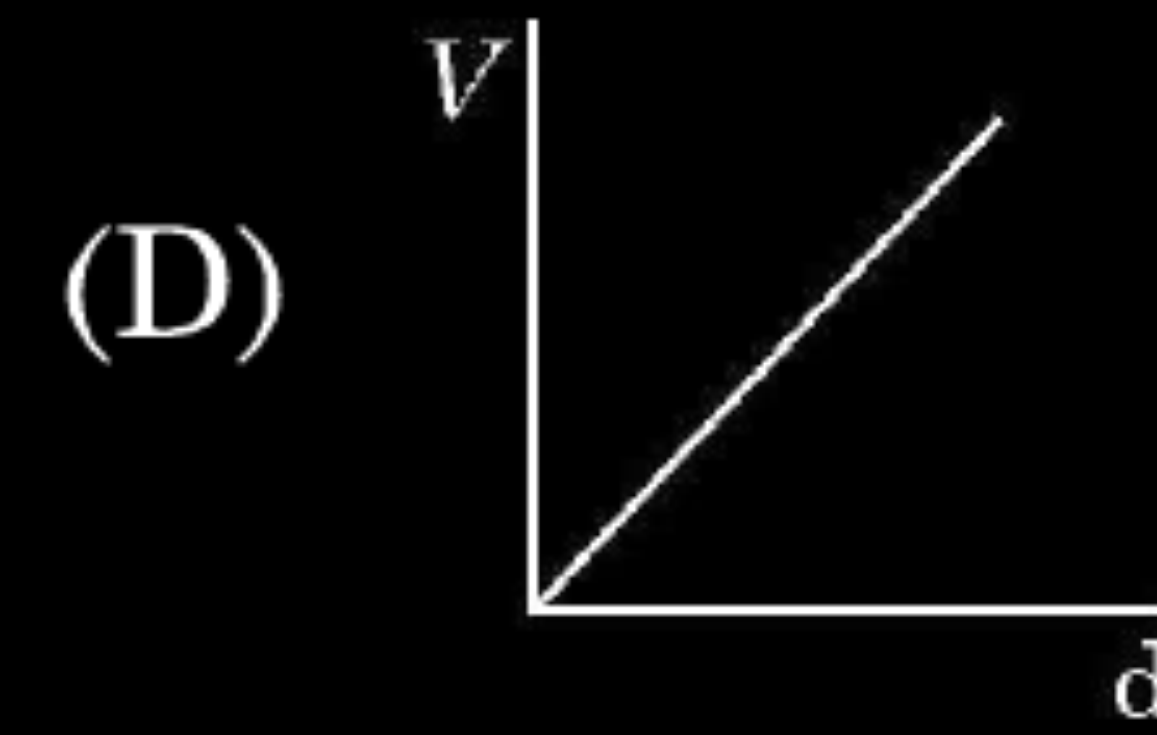
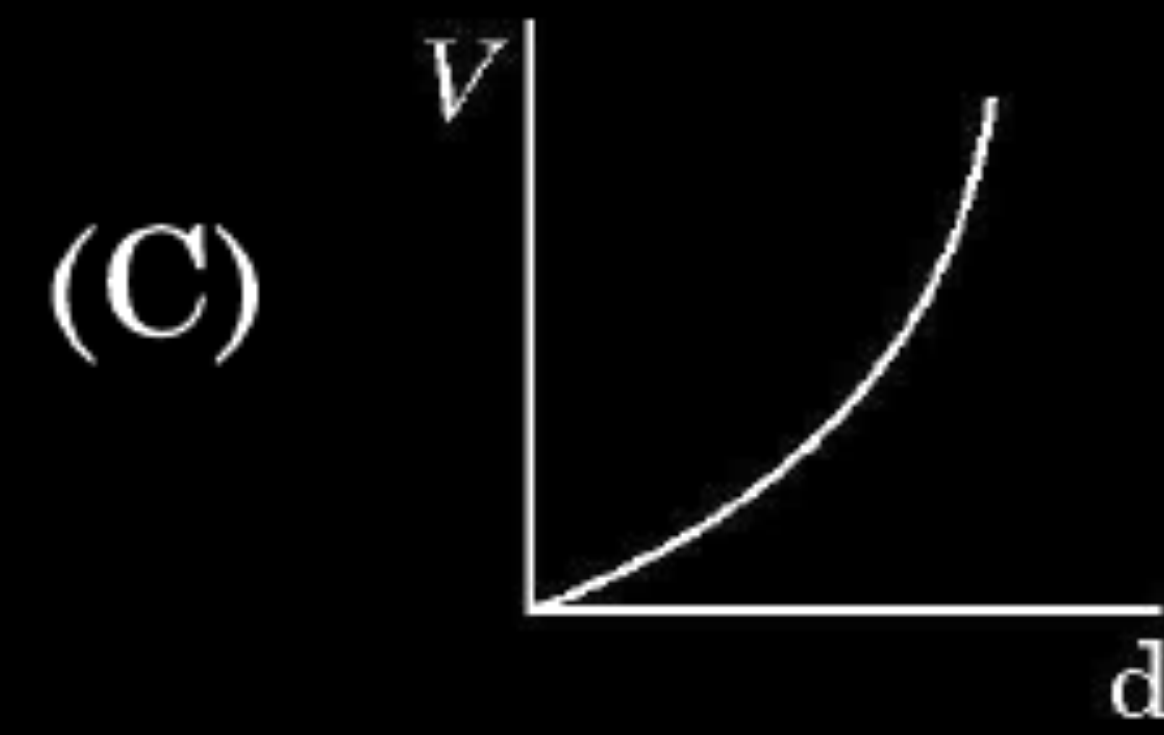
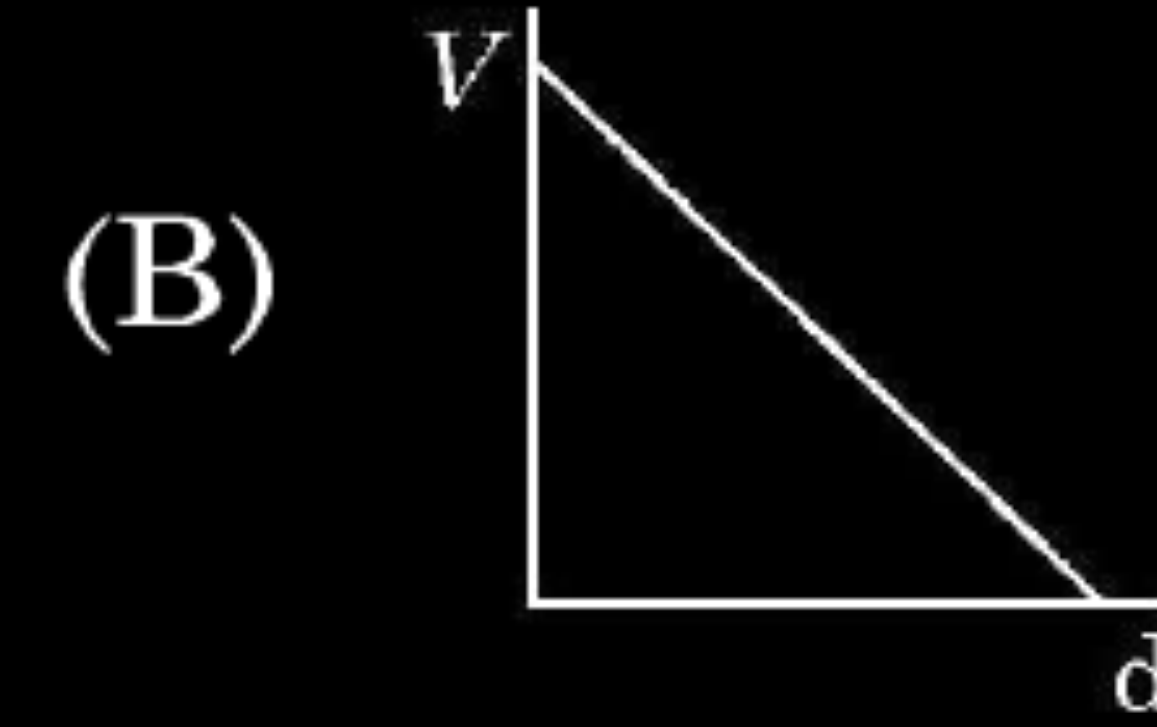
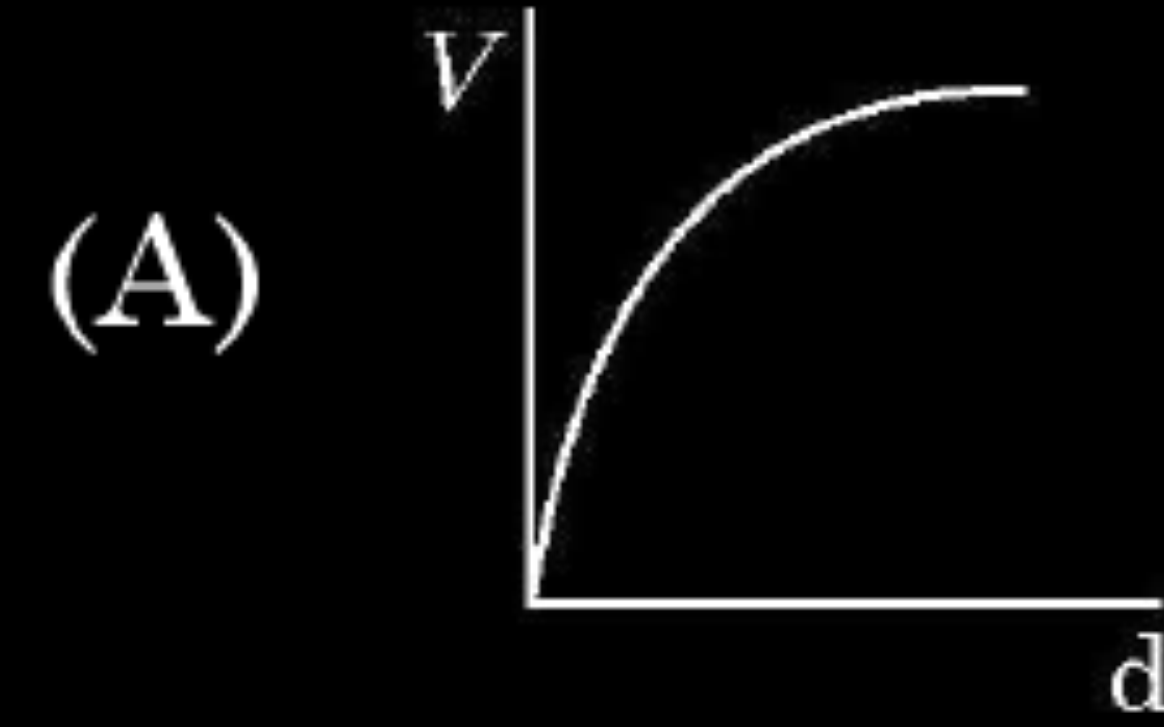
CBSE 25

(C) $\frac{E}{4}$

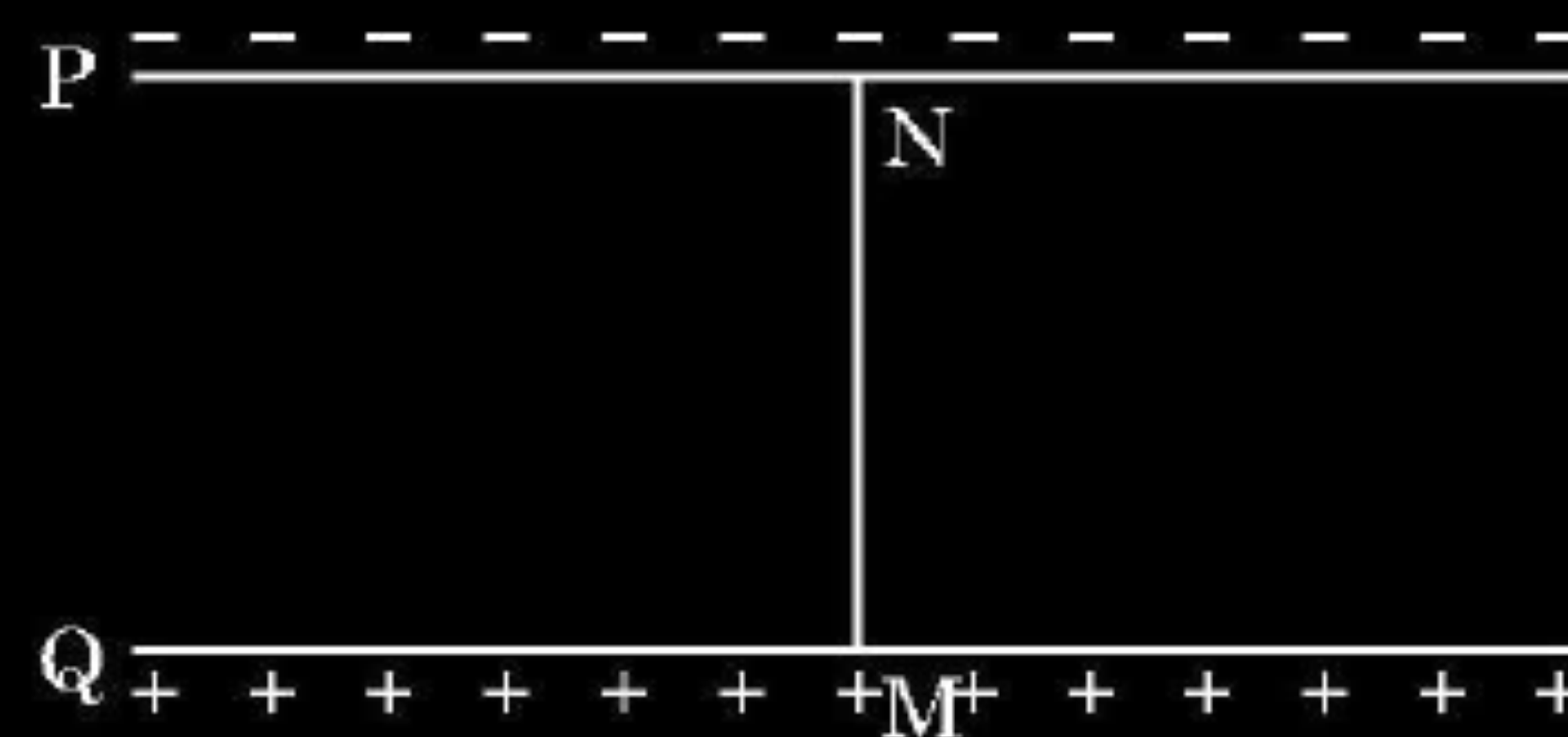
- (ii) A constant electric field is to be maintained between the two plates of a capacitor whose separation d changes with time. Which of the graphs correctly depict the potential difference (V) to be applied between the plates as a function of separation between the plates (d) to maintain the constant electric field ?



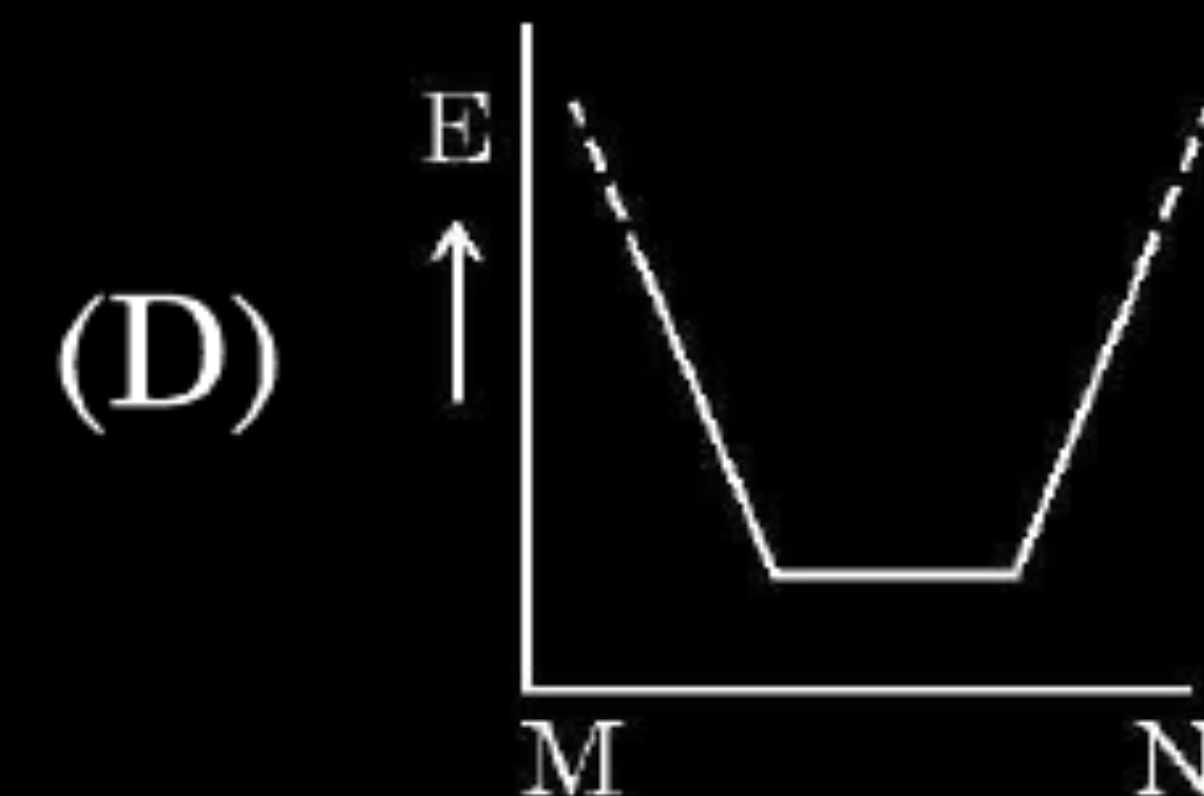
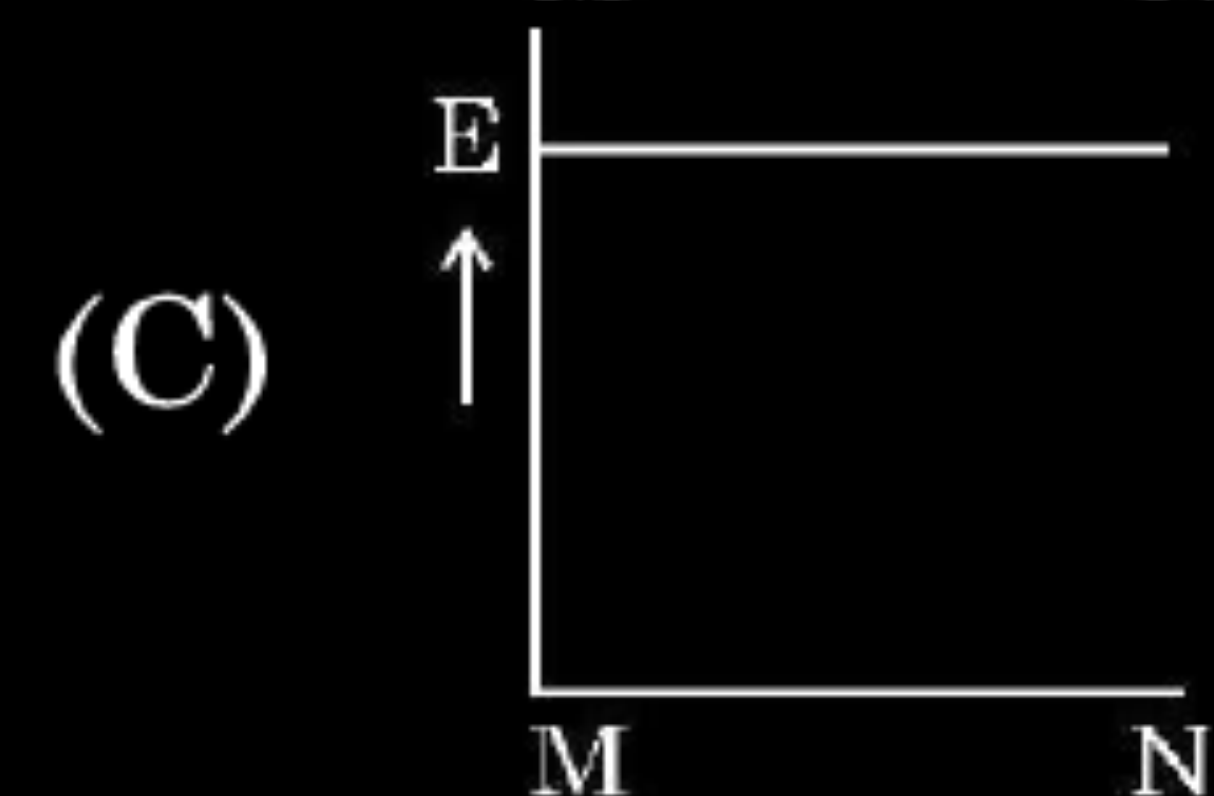
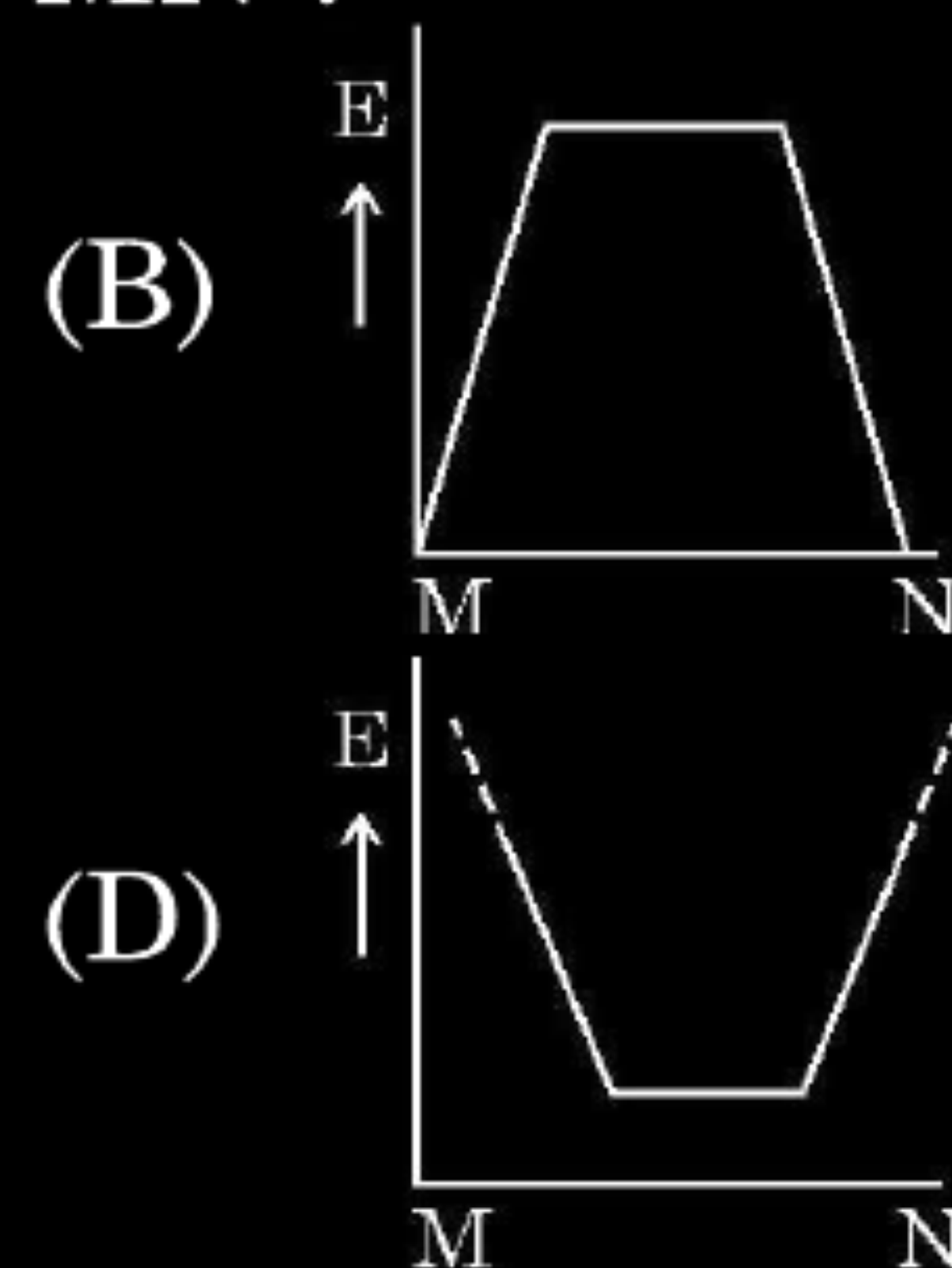
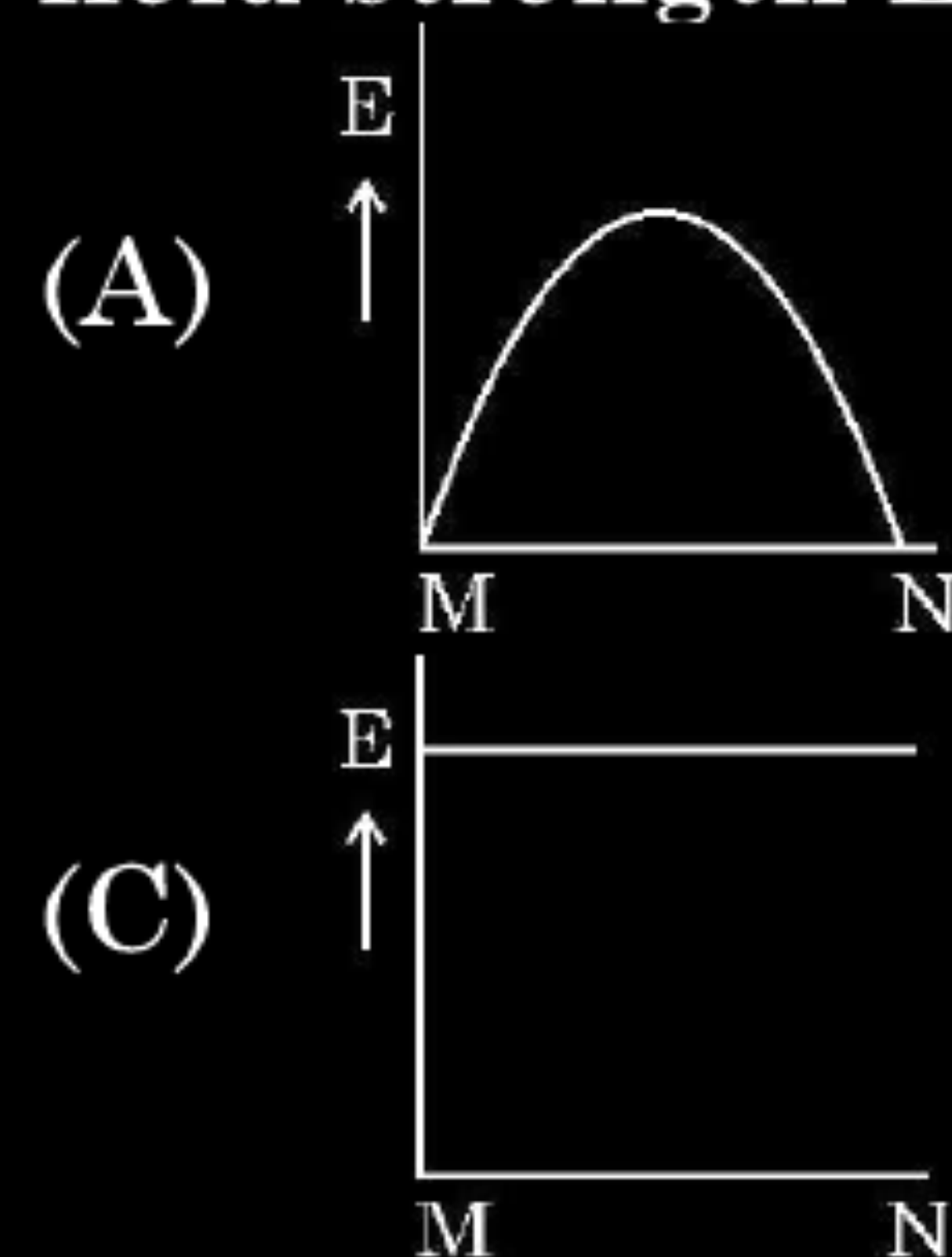
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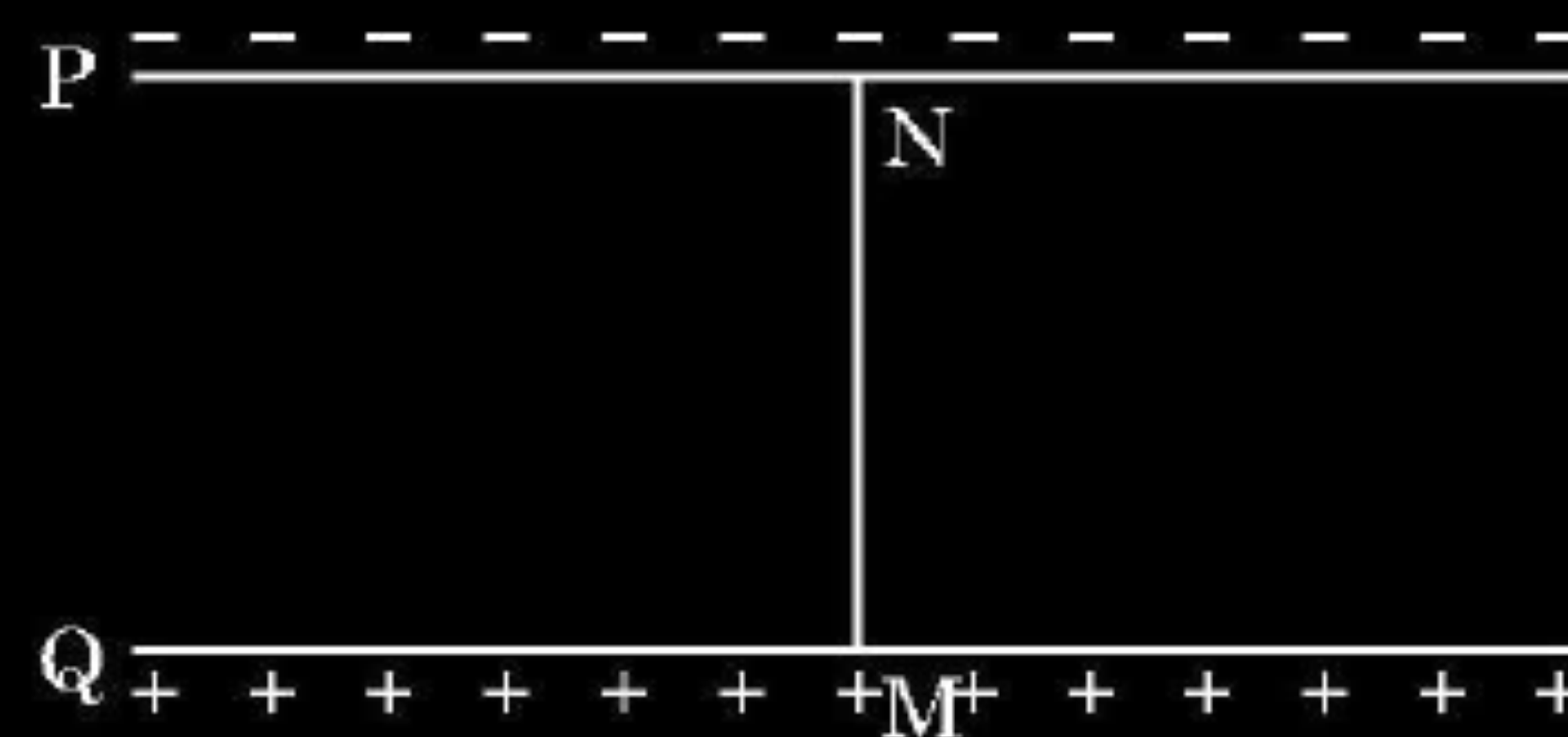
(iii)



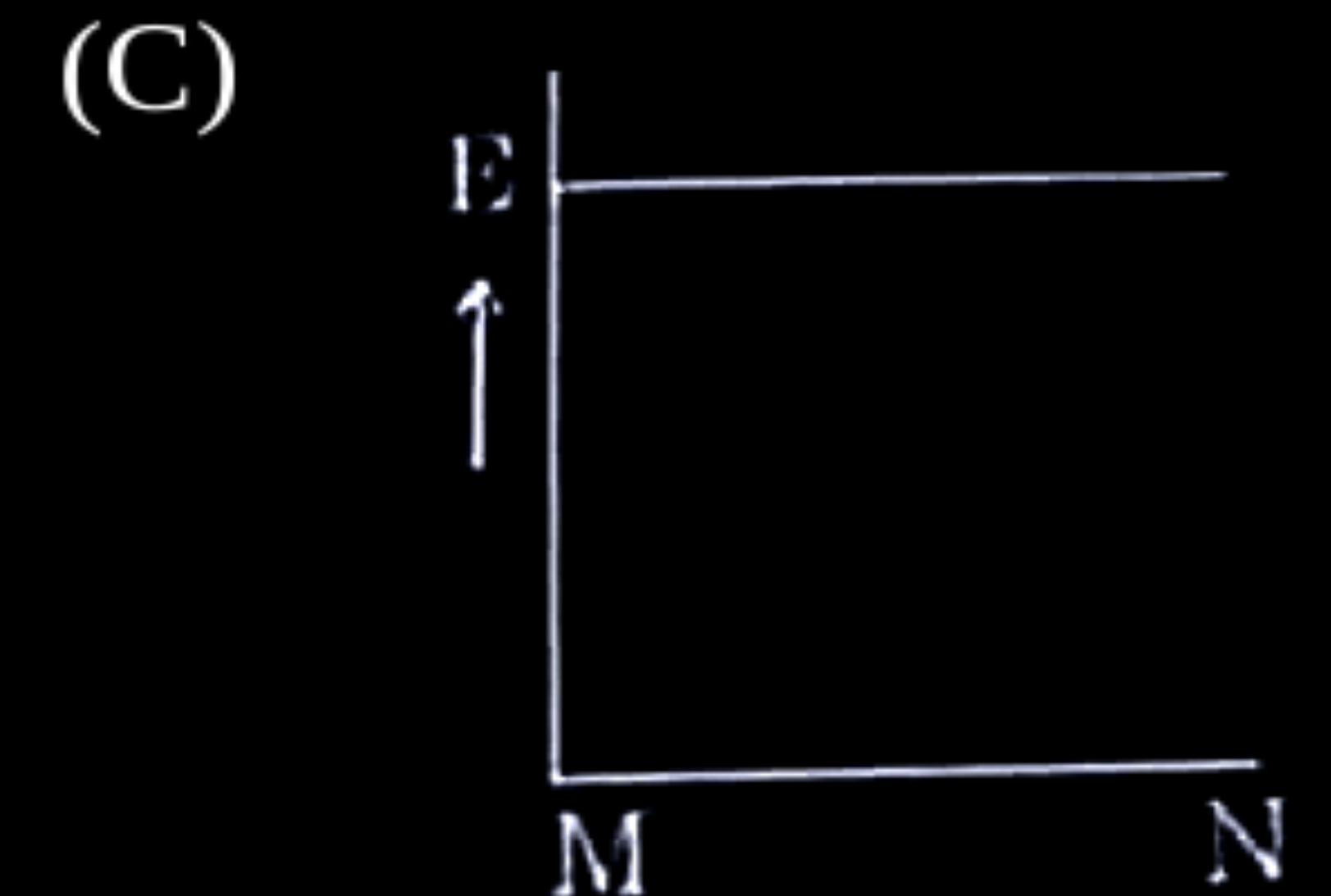
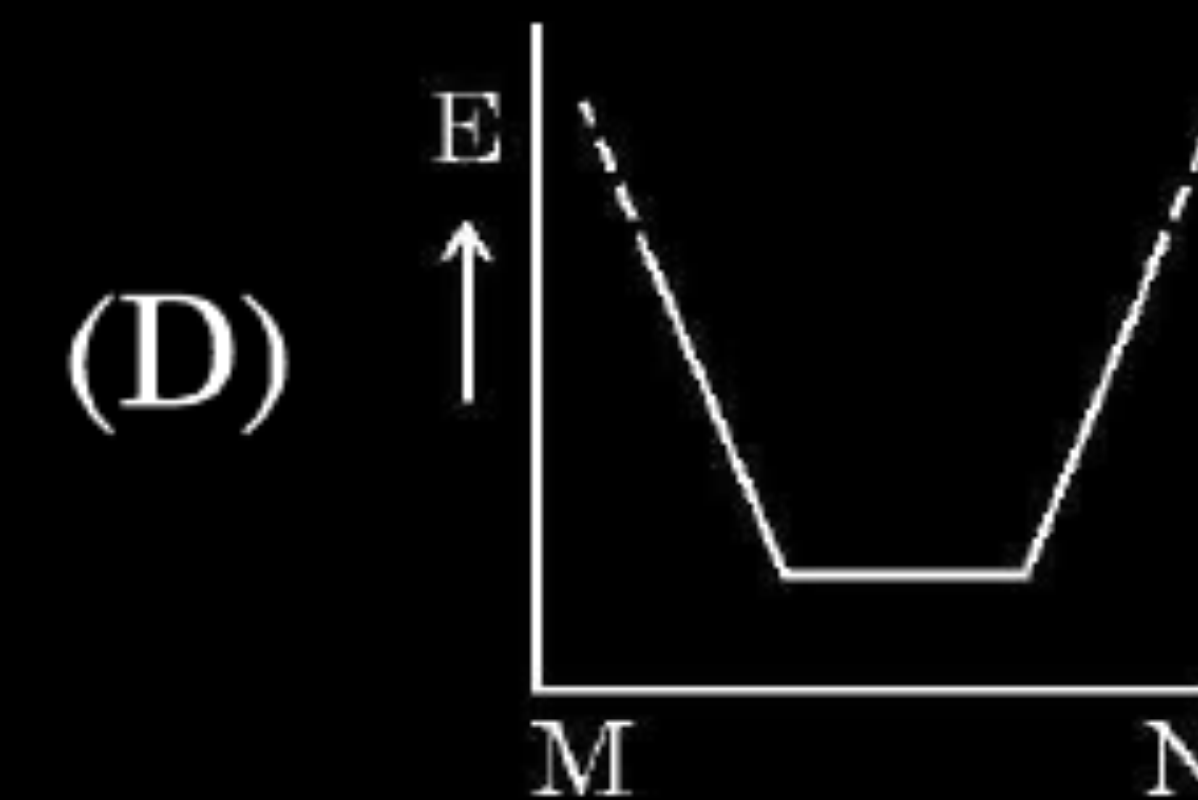
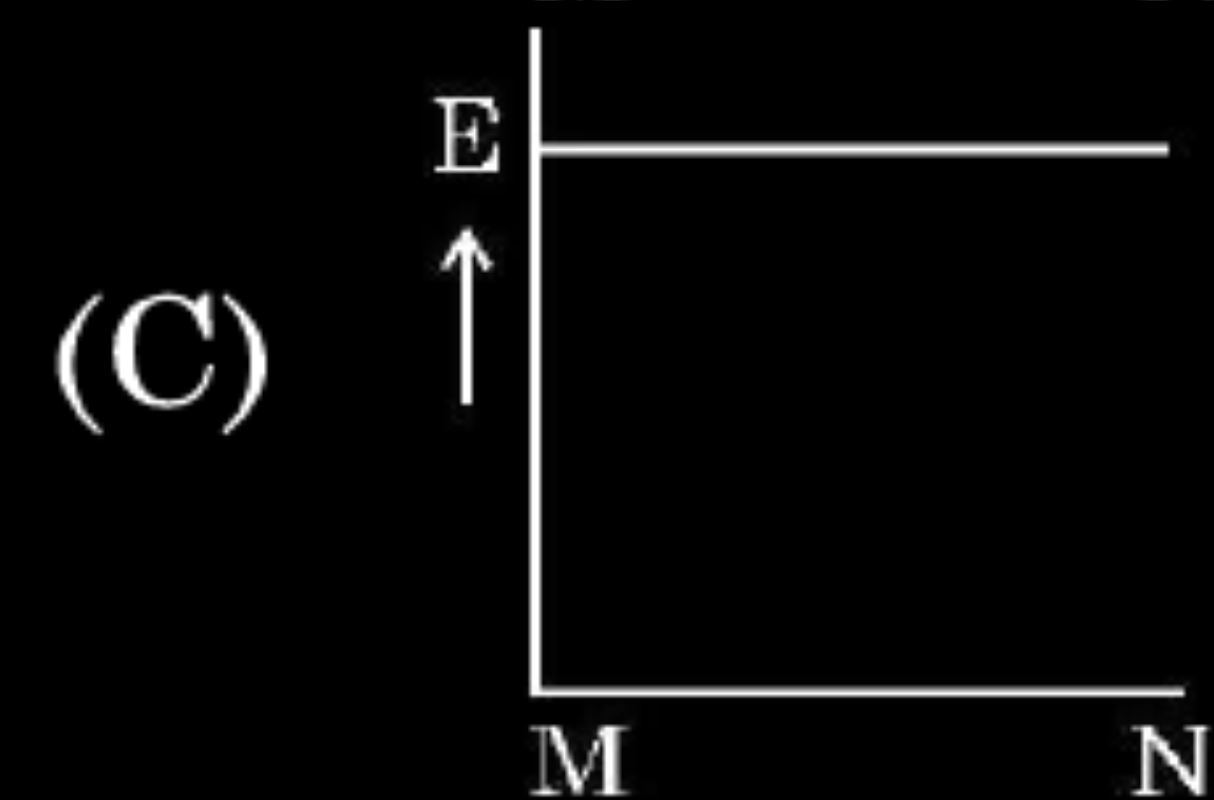
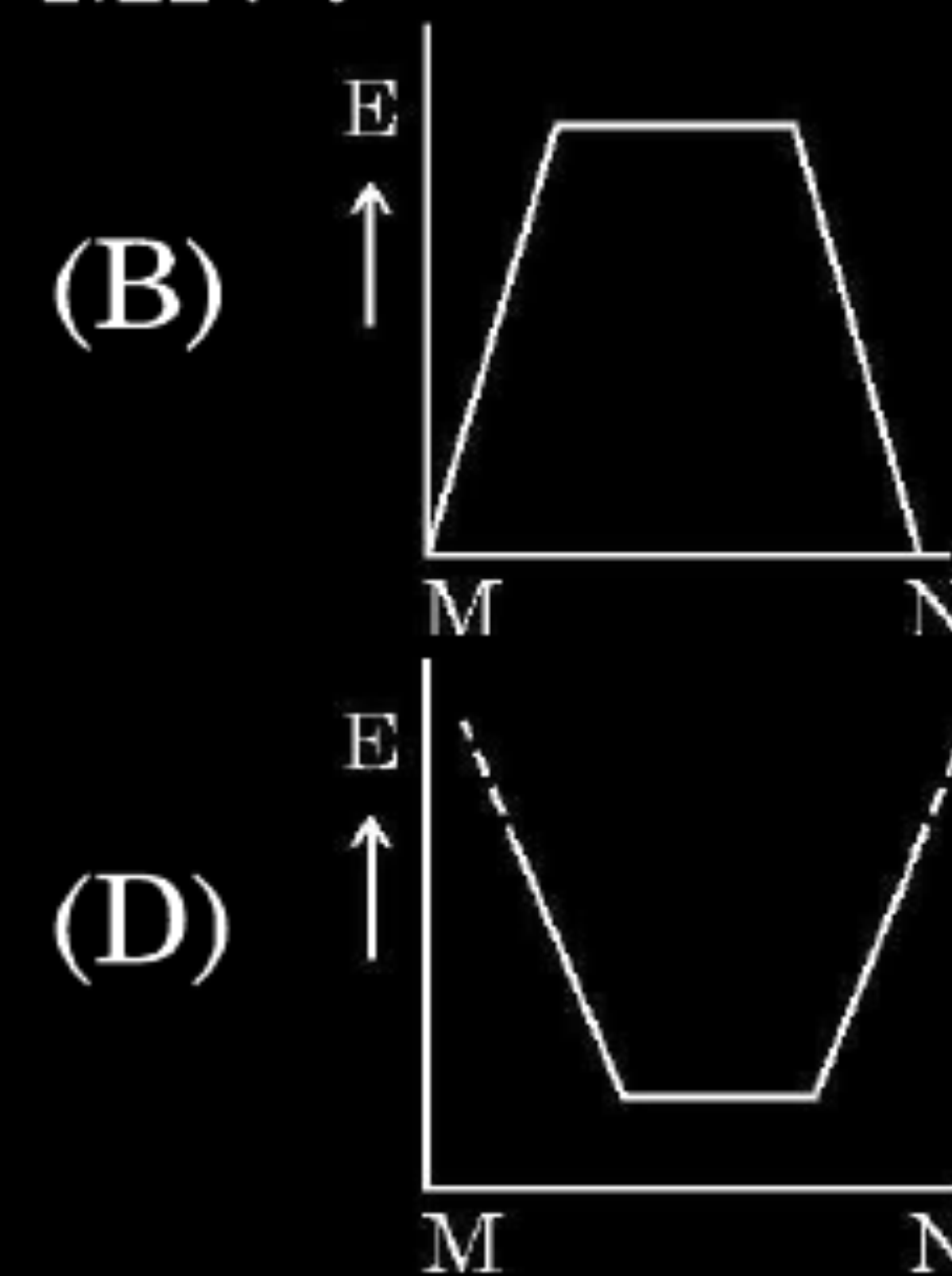
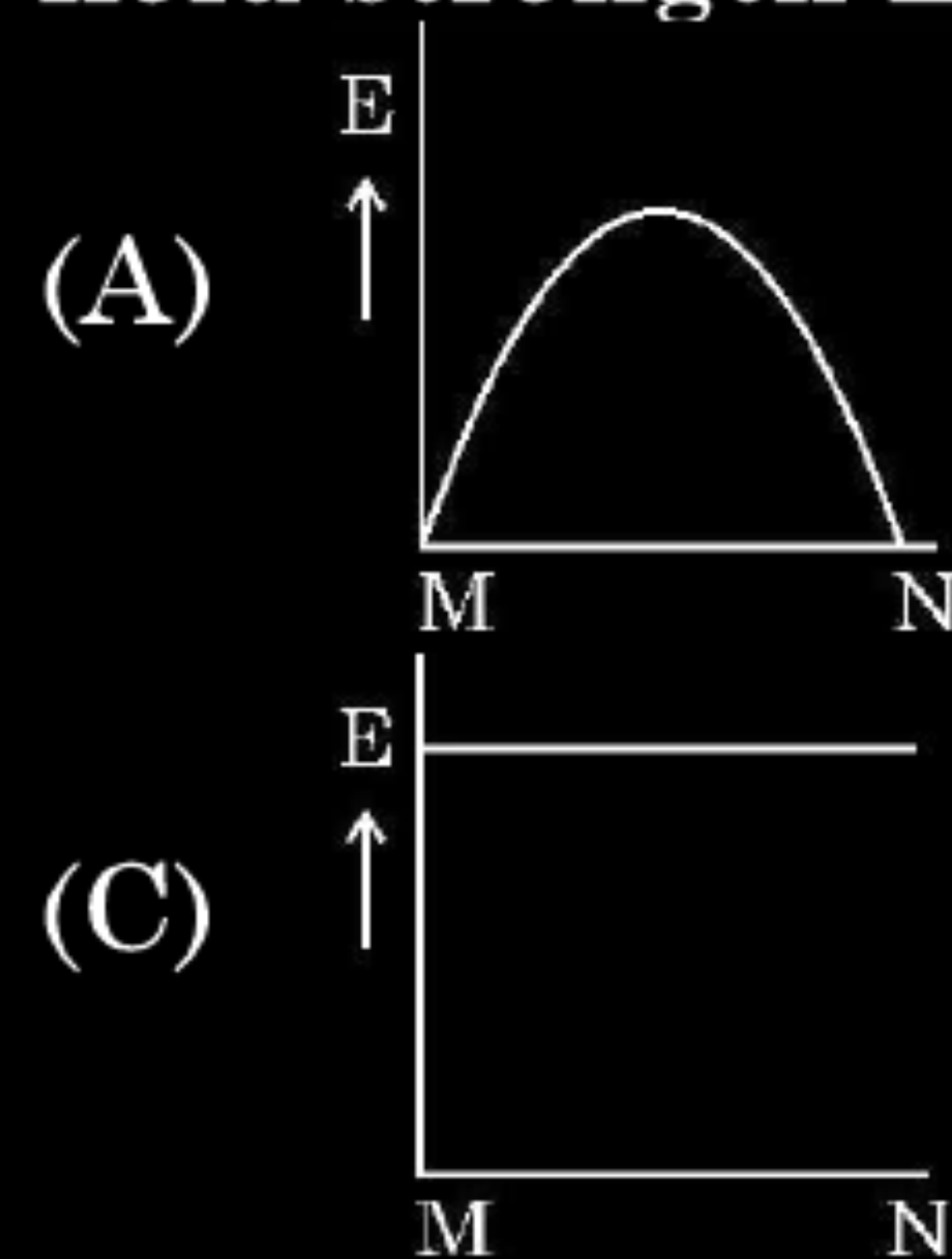
In the above figure P, Q are the two parallel plates of a capacitor. Plate Q is at positive potential with respect to plate P. MN is an imaginary line drawn perpendicular to the plates. Which of the graphs shows correctly the variations of the magnitude of electric field strength E along the line MN ?



(iii)



In the above figure P, Q are the two parallel plates of a capacitor. Plate Q is at positive potential with respect to plate P. MN is an imaginary line drawn perpendicular to the plates. Which of the graphs shows correctly the variations of the magnitude of electric field strength E along the line MN ?



(iv) Three parallel plates are placed above each other with equal displacement $\vec{d}$ between neighbouring plates. The electric field between the first pair of the plates is $\vec{E}_1$ and the electric field between the second pair of the plates is $\vec{E}_2$. The potential difference between the third and the first plate is –

(A) $(\vec{E}_1 + \vec{E}_2) \cdot \vec{d}$

(B) $(\vec{E}_1 - \vec{E}_2) \cdot \vec{d}$

(C) $(\vec{E}_2 - \vec{E}_1) \cdot \vec{d}$

(D) $\frac{d(E_1 + E_2)}{2}$

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(C) $(\vec{E}_2 - \vec{E}_1) \cdot \vec{d}$

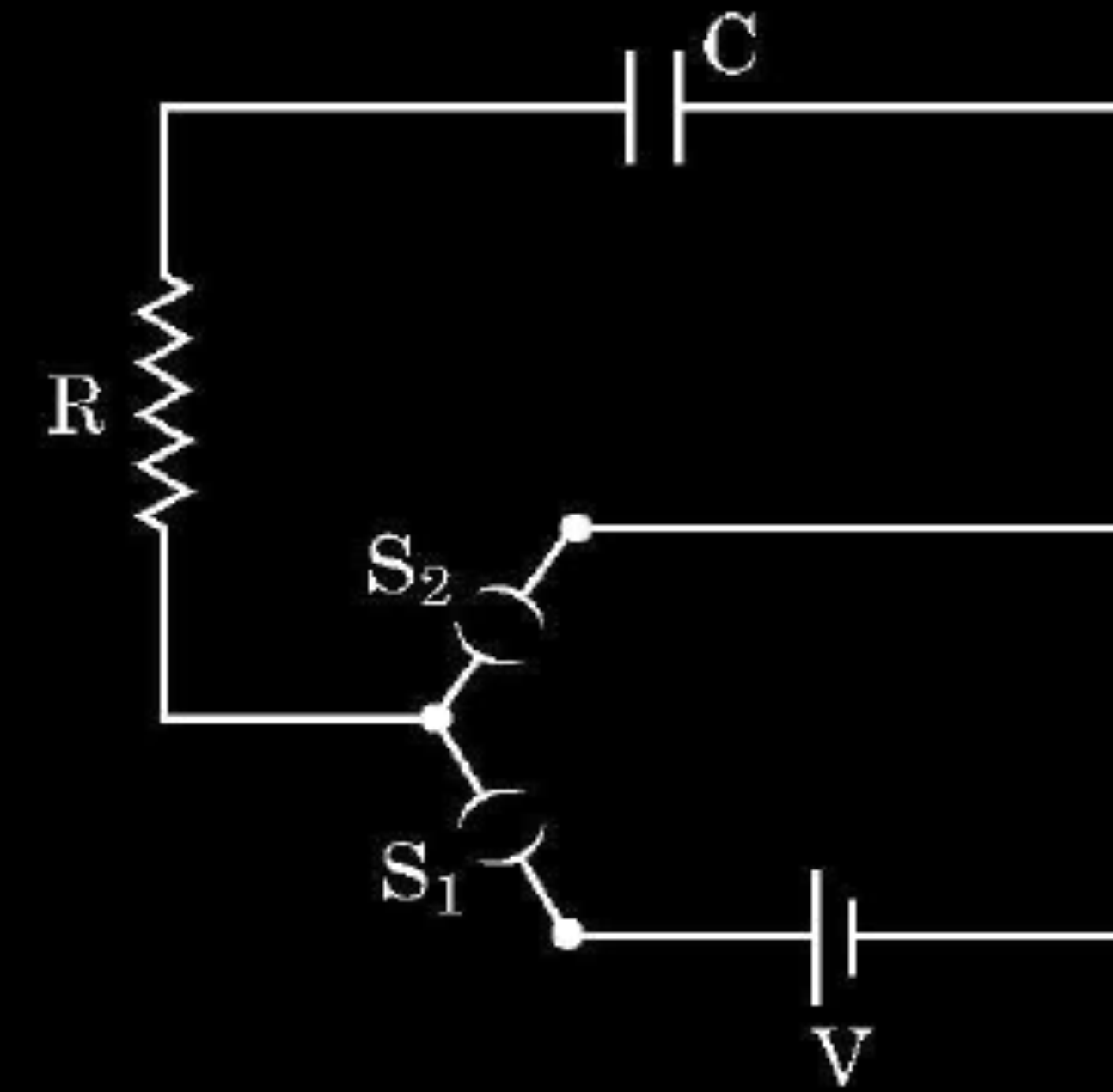
(D) $\frac{d(E_1 + E_2)}{2}$

(A) $(\vec{E}_1 + \vec{E}_2) \cdot \vec{d}$

(iv) A material of dielectric constant K is filled in a parallel plate capacitor of capacitance C . The new value of its capacitance becomes

- | | |
|----------|-------------------------------------|
| (A) C | (B) $\frac{C}{K}$ |
| (C) CK | (D) $C\left(1 + \frac{1}{K}\right)$ |

30. A circuit consisting of a capacitor C , a resistor of resistance R and an ideal battery of emf V , as shown in figure is known as RC series circuit. $4 \times 1 = 4$



CBSE 25

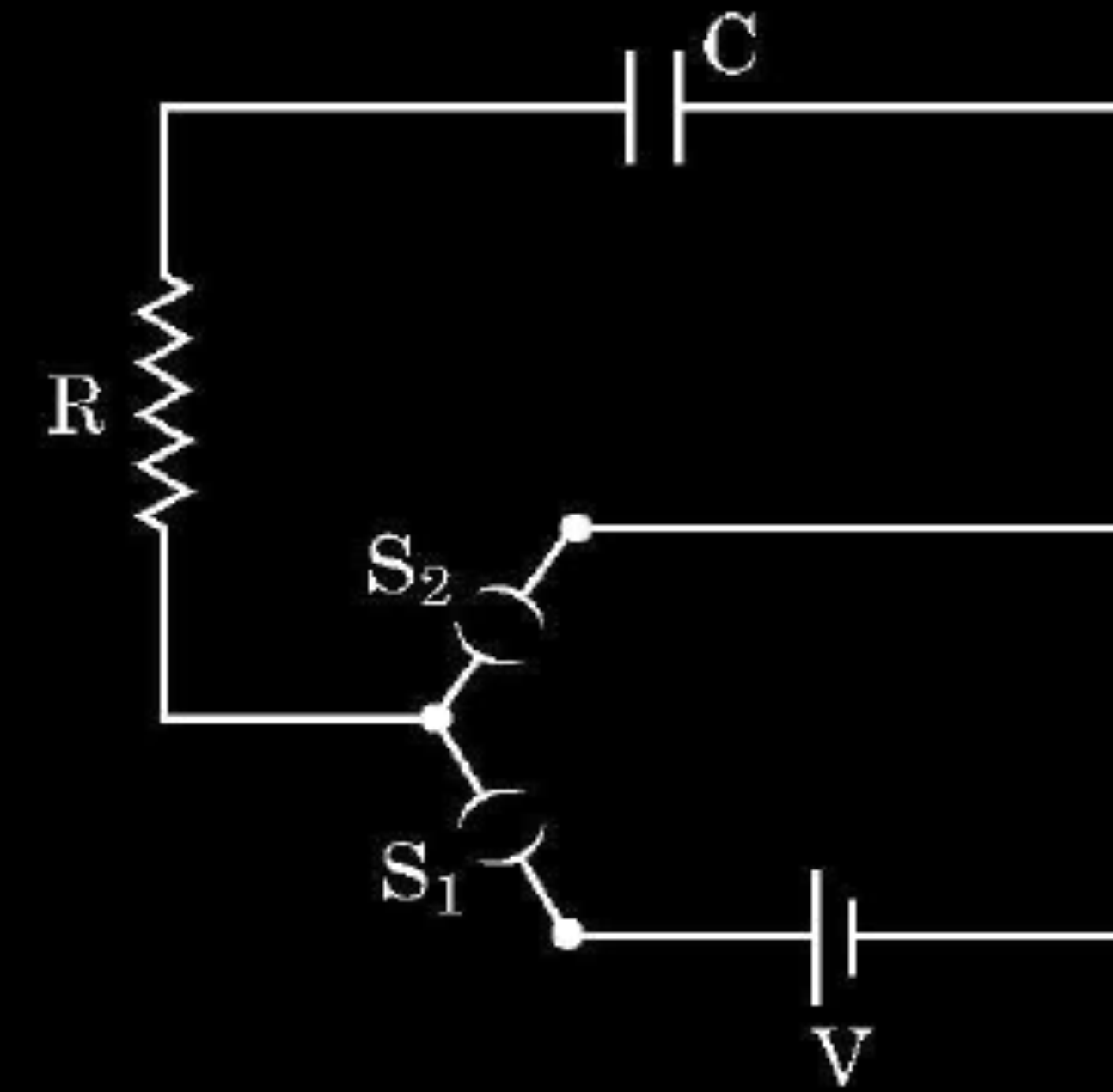
As soon as the circuit is completed by closing key S_1 (keeping S_2 open) charges begin to flow between the capacitor plates and the battery terminals. The charge on the capacitor increases and consequently the potential difference $V_c (= q/C)$ across the capacitor also increases with time. When this potential difference equals the potential difference across the battery, the capacitor is fully charged ($Q = VC$). During this process of charging, the charge q on the capacitor changes with time t as $q = Q[1 - e^{-t/RC}]$

The charging current can be obtained by differentiating it and using $\frac{d}{dx}(e^{mx}) = me^{mx}$.

Consider the case when $R = 20 \text{ k}\Omega$, $C = 500 \text{ }\mu\text{F}$ and $V = 10 \text{ V}$.

- (i) The final charge on the capacitor, when key S_1 is closed and S_2 is open, is
- | | |
|-----------------------------|---------------------|
| (A) $5 \text{ }\mu\text{C}$ | (B) 5 mC |
| (C) 25 mC | (D) 0.1 C |

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CBSE 25

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(A) $5 \text{ }\mu\text{C}$
(C) 25 mC

(B) 5 mC
(D) 0.1 C

(B) 5mC

- (ii) For sufficient time the key S_1 is closed and S_2 is open. Now key S_2 is closed and S_1 is open. What is the final charge on the capacitor ?
- (A) Zero (B) 5 mC
(C) 2.5 mC (D) 5 μ C

- (ii) For sufficient time the key S_1 is closed and S_2 is open. Now key S_2 is closed and S_1 is open. What is the final charge on the capacitor ?
- (A) Zero (B) 5 mC
(C) 2.5 mC (D) 5 μ C

(A) zero

(iii) The dimensional formula for RC is

(A) $[M L^2 T^{-3} A^{-2}]$ (B) $[M^0 L^0 T^{-1} A^0]$

(C) $[M^{-1} L^{-2} T^4 A^2]$ (D) $[M^0 L^0 T A^0]$

(iii) The dimensional formula for RC is

(A) $[M L^2 T^{-3} A^{-2}]$ (B) $[M^0 L^0 T^{-1} A^0]$

(C) $[M^{-1} L^{-2} T^4 A^2]$ (D) $[M^0 L^0 T A^0]$

(D) $[M^0 L^0 T A^0]$

(iv) The key S_1 is closed and S_2 is open. The value of current in the resistor after 5 seconds, is

- (A) $\frac{1}{2\sqrt{e}}$ mA (B) $\sqrt{e}$ mA (C) $\frac{1}{\sqrt{e}}$ mA (D) $\frac{1}{2e}$ mA

(iv) The key S_1 is closed and S_2 is open. The value of current in the resistor after 5 seconds, is

- (A) $\frac{1}{2\sqrt{e}}$ mA (B) $\sqrt{e}$ mA (C) $\frac{1}{\sqrt{e}}$ mA (D) $\frac{1}{2e}$ mA

iv) (A) $\frac{1}{2\sqrt{e}}$ mA

Note: 1 mark for this part may be given to all the students who have attempted other parts of the question.

- (iv) The key S_1 is closed and S_2 is open. The initial value of charging current in the resistor, is
- (A) 5 mA (B) 0.5 mA (C) 2 mA (D) 1 mA

(B) 0.5 mA

32. (a) (i) Two point charges $5\ \mu\text{C}$ and $-1\ \mu\text{C}$ are placed at points $(-3\ \text{cm}, 0, 0)$ and $(3\ \text{cm}, 0, 0)$ respectively. An external electric field $\vec{E} = \frac{A}{r^2} \hat{r}$ where $A = 3 \times 10^5\ \text{Vm}$ is switched on in the region. Calculate the change in electrostatic energy of the system due to the electric field.
- (ii) A system of two conductors is placed in air and they have net charge of $+80\ \mu\text{C}$ and $-80\ \mu\text{C}$ which causes a potential difference of $16\ \text{V}$ between them.
- (1) Find the capacitance of the system.
 - (2) If the air between the capacitor is replaced by a dielectric medium of dielectric constant 3, what will be the potential difference between the two conductors ?
 - (3) If the charges on two conductors are changed to $+160\ \mu\text{C}$ and $-160\ \mu\text{C}$, will the capacitance of the system change ? Give reason for your answer.

$$(i) \quad \vec{E} = \frac{3 \times 10^5}{r^2} \hat{r} \quad (\text{Given}) \quad dV = - \vec{E} \cdot d\vec{r}$$

$$V = 3 \times 10^5 / r$$

Electrostatic energy of the system in the absence of the field

$$U_i = \frac{Kq_1q_2}{r_{12}}$$

1/2

Electrostatic energy in the presence of the field

$$U_f = \frac{Kq_1q_2}{r_{12}} + q_1V(\vec{r}_1) + q_2V(\vec{r}_2)$$

$$\Delta U = U_f - U_i = q_1V(\vec{r}_1) + q_2V(\vec{r}_2)$$

1/2

$$\Delta U = \frac{5 \times 10^{-6} \times 3 \times 10^5}{3 \times 10^{-2}} - \frac{1 \times 10^{-6} \times 3 \times 10^5}{3 \times 10^{-2}}$$

1/2

$$= 40 \text{ J}$$

1/2

$$ii) 1) C = \frac{Q}{V} = \frac{80}{16} = 5 \mu\text{F}$$

1

$$2) C' = KC$$

$$= 3 \times 5 \mu\text{F} = 15 \mu\text{F}$$

1/2

$$V' = \frac{Q}{C'} = \frac{80 \mu\text{C}}{15 \mu\text{F}} = 5.33 \text{ V}$$

1/2

3) No,

1/2

The capacitance of the system depends on its geometry.

1/2

4. The capacitance of a parallel plate capacitor is $10\ \mu\text{F}$ when the distance between its plates is 8 cm. If the distance between the plates is halved, the capacitance will become :

(A) $10\ \mu\text{F}$

(B) $15\ \mu\text{F}$

(C) $20\ \mu\text{F}$

(D) $40\ \mu\text{F}$

CBSE 24 C

4. The capacitance of a parallel plate capacitor is $10\ \mu\text{F}$ when the distance between its plates is 8 cm. If the distance between the plates is halved, the capacitance will become :

(A) $10\ \mu\text{F}$

(B) $15\ \mu\text{F}$

(C) $20\ \mu\text{F}$

(D) $40\ \mu\text{F}$

CBSE 24 C

(C) $20\ \mu\text{F}$

12. A parallel plate capacitor is charged by a battery. The battery is then disconnected and the plates of the charged capacitor are then moved farther apart. In the process :

- (A) the charge on the capacitor increases.
- (B) the potential difference across the plates decreases.
- (C) the capacitance of the capacitor increases.
- (D) the electrostatic energy stored in the capacitor increases.

12. A parallel plate capacitor is charged by a battery. The battery is then disconnected and the plates of the charged capacitor are then moved farther apart. In the process :

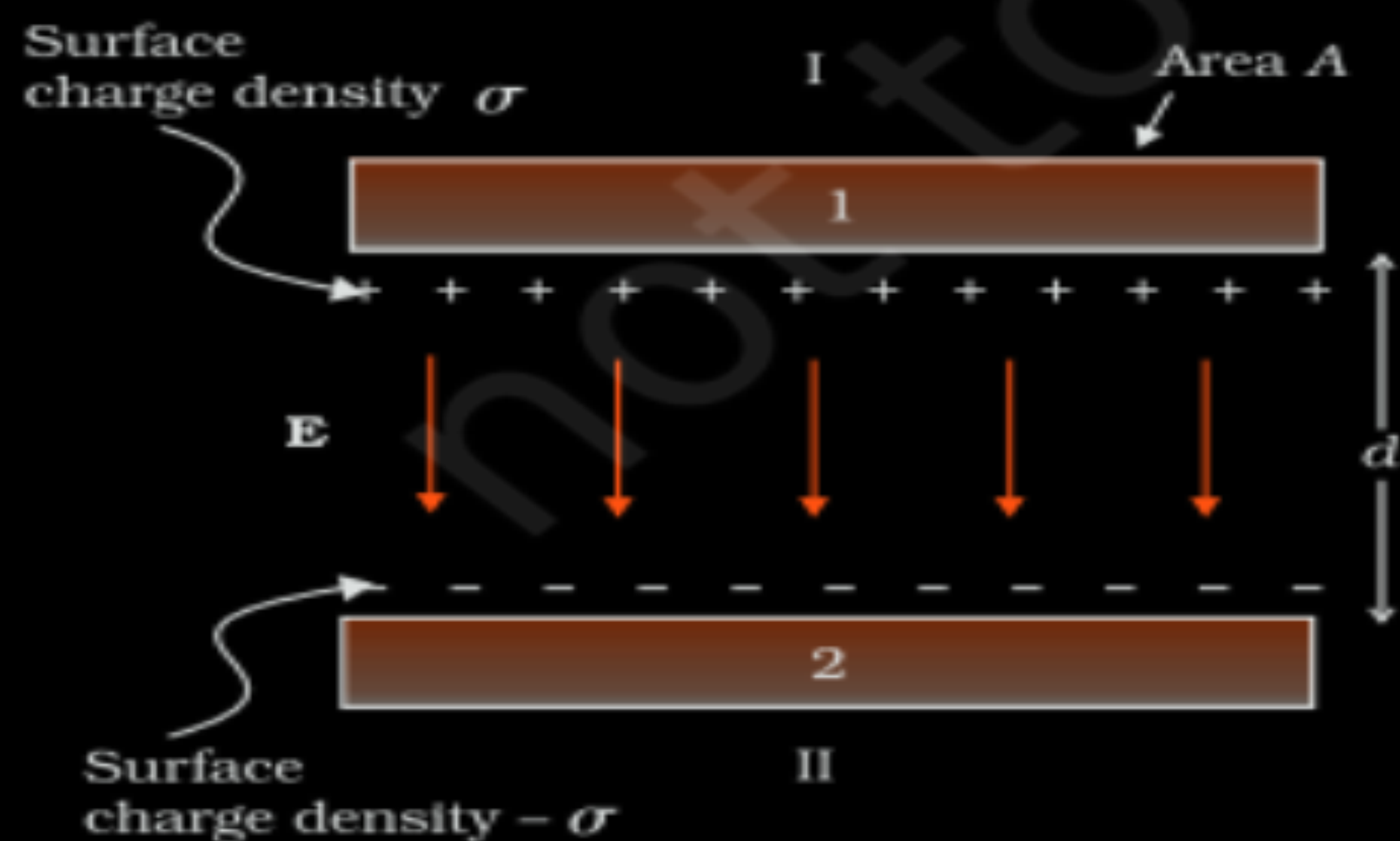
- (A) the charge on the capacitor increases.
- (B) the potential difference across the plates decreases.
- (C) the capacitance of the capacitor increases.
- (D) the electrostatic energy stored in the capacitor increases.

CBSE 24 C

(D) The electrostatic energy stored in the capacitor increases

- (b) (i) Obtain an expression for the capacitance of a parallel plate capacitor.
- (ii) Two identical plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$, respectively, are separated by a distance d in air and a dielectric of dielectric constant K is filled between them. Write the expressions for :
- (1) the potential difference between the plates
 - (2) capacitance of the capacitor so formed
 - (3) energy stored in this capacitor

5



(ii)

Electric field in between the plates of the capacitor

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A} \text{-----(1)}$$

The potential difference between the plates of the capacitor

$$V = Ed$$

The capacitance of the parallel plate capacitor

$$C = \frac{Q}{V}$$

$$C = \frac{\epsilon_0 A}{d}$$

(1) Potential difference when dielectric of constant K is filled = $\frac{\sigma d}{\epsilon_0 K}$

Alternatively:-

$$V' = \frac{V}{K}$$

(2) Capacitance of the capacitor formed = $\frac{KA\epsilon_0}{d}$

$$C' = KC$$

(3) Energy stored in the capacitor = $\frac{U}{K}$

30. Dielectrics play an important role in design of capacitors. The molecules of a dielectric may be polar or non-polar. When a dielectric slab is placed in an external electric field, opposite charges appear on the two surfaces of the slab perpendicular to electric field. Due to this an electric field is established inside the dielectric.

The capacitance of a capacitor is determined by the dielectric constant of the material that fills the space between the plates. Consequently, the energy storage capacity of a capacitor is also affected. Like resistors, capacitors can also be arranged in series and/or parallel.

- (i) Which of the following is a polar molecule ?



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The capacitance of a capacitor is determined by the dielectric constant of the material that fills the space between the plates. Consequently, the energy storage capacity of a capacitor is also affected. Like resistors, capacitors can also be arranged in series and/or parallel.

- (i) Which of the following is a polar molecule ?

(A) O_2

(B) H_2

(C) N_2

(D) HCl

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(D) HCl

- (ii) Which of the following statements about dielectrics is correct ?
- (A) A polar dielectric has a net dipole moment in absence of an external electric field which gets modified due to the induced dipoles.
 - (B) The net dipole moments of induced dipoles is along the direction of the applied electric field.
 - (C) Dielectrics contain free charges.
 - (D) The electric field produced due to induced surface charges inside a dielectric is along the external electric field.

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 - (D) The electric field produced due to induced surface charges inside a dielectric is along the external electric field.

(B) The net dipole moment of induced dipoles is along the direction of the applied electric field.

- (iii) When a dielectric slab is inserted between the plates of an isolated charged capacitor, the energy stored in it :
- (A) increases and the electric field inside it also increases.
 - (B) decreases and the electric field also decreases.
 - (C) decreases and the electric field increases.
 - (D) increases and the electric field decreases.

- (iii) When a dielectric slab is inserted between the plates of an isolated charged capacitor, the energy stored in it :
- (A) increases and the electric field inside it also increases.
 - (B) decreases and the electric field also decreases.
 - (C) decreases and the electric field increases.
 - (D) increases and the electric field decreases.

(B) decreases and the electric field also decreases.

- (iv) (a) An air-filled capacitor with plate area A and plate separation d has capacitance C_0 . A slab of dielectric constant K , area A and thickness $\left(\frac{d}{5}\right)$ is inserted between the plates. The capacitance of the capacitor will become

(A) $\left[\frac{4K}{5K+1}\right]C_0$

(B) $\left[\frac{K+5}{4}\right]C_0$

(C) $\left[\frac{5K}{4K+1}\right]C_0$

(D) $\left[\frac{K+4}{5K}\right]C_0$

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(C) $\left[\frac{5K}{4K+1}\right]C_0$

(D) $\left[\frac{K+4}{5K}\right]C_0$

(C) $\left[\frac{5K}{4K+1}\right]C_0$

(iv) (b) Two capacitors of capacitances $2 C_0$ and $6 C_0$ are first connected in series and then in parallel across the same battery. The ratio of energies stored in series combination to that in parallel is

(A) $\frac{1}{4}$

(B) $\frac{1}{6}$

(C) $\frac{2}{15}$

(D) $\frac{3}{16}$

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(C) $\frac{2}{15}$

(D) $\frac{3}{16}$

(D) $\frac{3}{16}$

31. (a) (i) A dielectric slab of dielectric constant 'K' and thickness 't' is inserted between plates of a parallel plate capacitor of plate separation d and plate area A. Obtain an expression for its capacitance.
- (ii) Two capacitors of different capacitances are connected first (1) in series and then (2) in parallel across a dc source of 100 V. If the total energy stored in the combination in the two cases are 40 mJ and 250 mJ respectively, find the capacitance of the capacitors.

5

31. (a) (i) A dielectric slab of dielectric constant 'K' and thickness 't' is inserted between plates of a parallel plate capacitor of plate separation d and plate area A. Obtain an expression for its capacitance.

a) (i)
Electric field in air between plates

$$E_0 = \frac{\sigma}{\epsilon_0}$$

Electric field inside the dielectric

$$E = \frac{\sigma}{\epsilon_0 K}$$

Potential difference between the plate

$$V = E_0(d-t) + Et$$

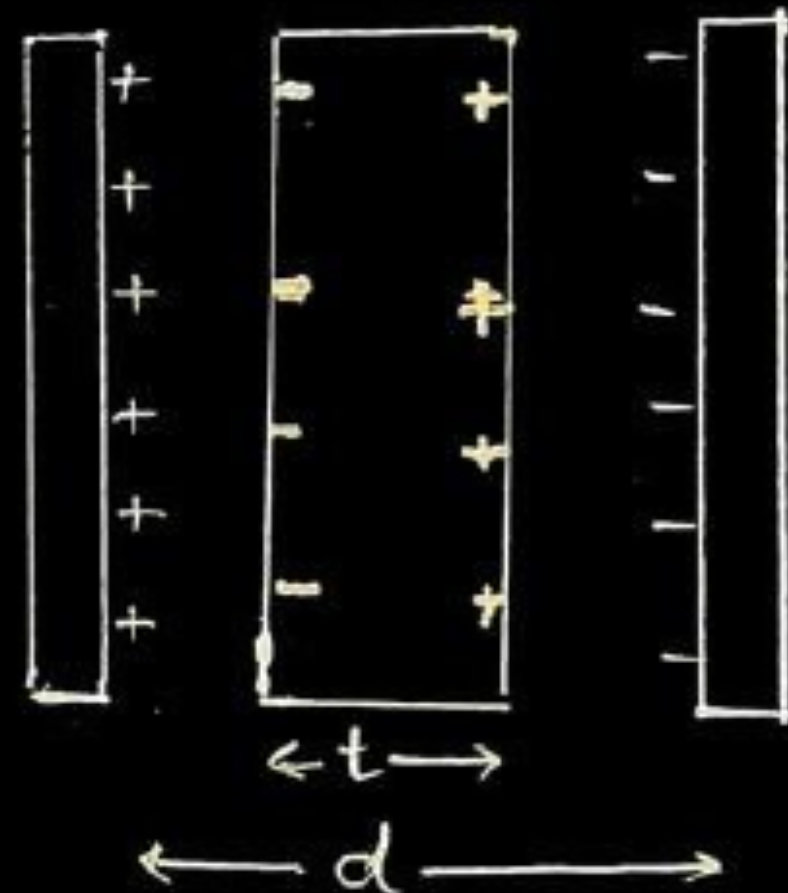
$$V = \frac{\sigma}{\epsilon_0} \left[d-t + \frac{t}{K} \right]$$

$$V = \frac{q}{A\epsilon_0} \left[d-t + \frac{t}{K} \right]$$

Capacitance

$$C = \frac{q}{V}$$

$$C = \frac{A\epsilon_0}{d-t + \frac{t}{K}}$$



(ii)

Two capacitors of different capacitances are connected first (1) in series and then (2) in parallel across a dc source of 100 V. If the total energy stored in the combination in the two cases are 40 mJ and 250 mJ respectively, find the capacitance of the capacitors.

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$$C = \frac{A\epsilon_0}{d-t \left(1 - \frac{1}{K} \right)}$$

ii) Total energy stored in series combination

$$\frac{1}{2} \left(\frac{C_1 C_2}{C_1 + C_2} \right) V^2 = 40 \times 10^{-3} \text{ J} \dots \dots \dots (1)$$

Energy stored in parallel combination

$$\frac{1}{2} (C_1 + C_2) V^2 = 250 \times 10^{-3} \text{ J} \dots \dots \dots (2)$$

Substituting value of V=100 V in eq (1) and (2) , on solving

$$C_1 = 4 \times 10^{-5} \text{ F or } 40 \mu\text{F}$$

$$C_2 = 1 \times 10^{-5} \text{ F or } 10 \mu\text{F}$$

1. The plates P_1 and P_2 of a $2\ \mu\text{F}$ capacitor are at potentials $25\ \text{V}$ and $-25\ \text{V}$ respectively. The charge on plate P_1 will be :

(A) $0.02\ \text{mC}$

(B) $0.1\ \text{mC}$

(C) $0.1\ \mu\text{C}$

(D) $1\ \mu\text{C}$

1. The plates P_1 and P_2 of a $2\ \mu\text{F}$ capacitor are at potentials $25\ \text{V}$ and $-25\ \text{V}$ respectively. The charge on plate P_1 will be :

(A) $0.02\ \text{mC}$

(B) $0.1\ \text{mC}$

(C) $0.1\ \mu\text{C}$

(D) $1\ \mu\text{C}$

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(B) 0.1mC

- 31.** (a) (i) Obtain the expression for the capacitance of a parallel plate capacitor with a dielectric medium between its plates.
- (ii) A charge of $6\ \mu\text{C}$ is given to a hollow metallic sphere of radius $0.2\ \text{m}$. Find the potential at (i) the surface and (ii) the centre of the sphere.

When a dielectric slab is inserted between the plates of capacitance there is induced charge density σ_p which opposes the original charge density (σ) on the plate of capacitance.

Electric field with dielectric medium is

$$E = \frac{(\sigma - \sigma_p)}{\epsilon_0}$$

$$V = E \times d = \frac{(\sigma - \sigma_p)}{\epsilon_0} d$$

$$(\sigma - \sigma_p) = \frac{\sigma}{K}$$

$$V = \frac{\sigma d}{\epsilon_0 K} = \frac{Qd}{A\epsilon_0 K}$$

$$C = \frac{Q}{V} = \frac{K\epsilon_0 A}{d}$$

(ii) Electric potential due to a point charge

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

(i) At the surface

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{9 \times 10^9 \times 6 \times 10^{-6}}{0.2}$$

$$V = 2.7 \times 10^5 \text{ V}$$

(ii) Since electric field inside the hollow sphere is zero, hence V remains constant throughout the volume.

$$V = 2.7 \times 10^5 \text{ V}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1. The capacitance of a parallel plate capacitor having a medium of dielectric constant $K = 4$ in between the plates is C . If this medium is removed, then the capacitance of the capacitor becomes :

(A) $4C$

(B) C

(C) $\frac{C}{4}$

(D) $2C$

1. The capacitance of a parallel plate capacitor having a medium of dielectric constant $K = 4$ in between the plates is C . If this medium is removed, then the capacitance of the capacitor becomes :

(A) $4C$

(B) C

(C) $\frac{C}{4}$

(D) $2C$

CBSE 24

(C) $\frac{C}{4}$

- 22.** An air-filled parallel plate capacitor with plate separation 1 mm has a capacitance of 20 pF. It is charged to 4.0 μC . Calculate the amount of work done to pull its plates to a separation of 5 mm. Assume the charge on the plates remains the same.

- 22.** An air-filled parallel plate capacitor with plate separation 1 mm has a capacitance of 20 pF. It is charged to 4.0 μC . Calculate the amount of work done to pull its plates to a separation of 5 mm. Assume the charge on the plates remains the same.

CBSE 24

3

$$C = \frac{A\epsilon_0}{d}; \quad C' = \frac{A\epsilon_0}{d'}$$

$$\Rightarrow \frac{C'}{C} = \frac{d}{d'} \Rightarrow C' = \frac{1}{5} \times 20$$

$$C' = 4\text{pF}$$

Work done = change in energy

$$= \frac{1}{2} \frac{Q^2}{C'} - \frac{1}{2} \frac{Q^2}{C}$$

$$= \frac{Q^2}{2} \left[\frac{1}{4} - \frac{1}{20} \right] \times 10^{12}$$

On solving ;

$$\text{Work done} = \frac{8}{5} J$$

$$= 1.6J$$

1

1

1

2. Ten capacitors, each of capacitance $1\ \mu\text{F}$, are connected in parallel to a source of $100\ \text{V}$. The total energy stored in the system is equal to :

(A) $10^{-2}\ \text{J}$

(B) $10^{-3}\ \text{J}$

(C) $0.5 \times 10^{-3}\ \text{J}$

(D) $5.0 \times 10^{-2}\ \text{J}$

2. Ten capacitors, each of capacitance $1\ \mu\text{F}$, are connected in parallel to a source of $100\ \text{V}$. The total energy stored in the system is equal to :

(A) $10^{-2}\ \text{J}$

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(D) $5.0 \times 10^{-2}\ \text{J}$

CBSE 24

(D) $5.0 \times 10^{-2}\ \text{J}$

31. (a) (i) A capacitor of capacitance C is charged to a potential difference V by a battery. The battery is disconnected and the separation between the plates is halved. How will the following quantities be affected ?

(1) Capacitance of the capacitor

(2) Electric field between the plates

(3) Energy stored in the capacitor

Justify your answer in each case.

(ii) A capacitor of $20\ \mu\text{F}$ is charged to $30\ \text{V}$. It is then connected to an uncharged $30\ \mu\text{F}$ capacitor. Calculate the charges on each capacitor in equilibrium.

(1)Capacitance becomes two times / increases.

Justification:

$$C = \epsilon_0 \frac{A}{d}$$

d is reduced to $\frac{d}{2}$.

Hence

$$C' = 2(\epsilon_0 \frac{A}{d}) = 2C$$

(2)Electric field remains constant.

Justification:

Electric field between the plates

$$E = \frac{\sigma}{\epsilon_0}$$

When battery is removed σ is constant, hence E is constant.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(3)Energy becomes half / decreases.

$$\text{Energy } U_E = \frac{1}{2} \frac{q^2}{C}$$

C becomes two times, hence U_E becomes half.

(ii) Common potential on the capacitors = V'

Total Charge

$$q = q_1 + q_2,$$

$$q = C_1 V_1 + C_2 V_2 = 20 \times 30 = 600 \mu C \dots\dots\dots (1)$$

$$\frac{q_1}{q_2} = \frac{C_1 V'}{C_2 V'} = \frac{2}{3} \dots\dots\dots (2)$$

Solving (1) and (2)

$$q_1 = 240 \mu C$$

$$q_2 = 360 \mu C$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

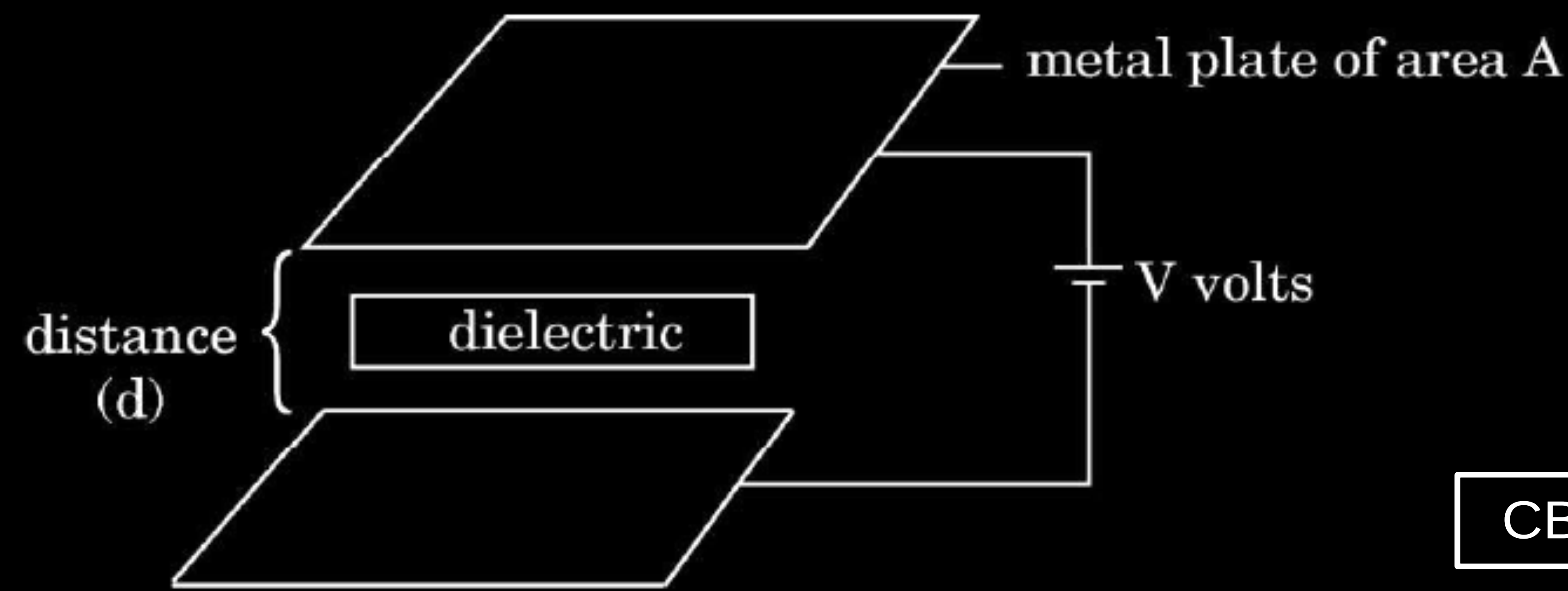
$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) How will the capacitance of a parallel plate capacitor change if :
- (1) the plates area is doubled ?
 - (2) the separation between the plates is doubled ?
- (ii) The effective capacitance of three capacitors of the same capacitance connected in series is $1\ \mu\text{F}$. Find the :
- (1) effective capacitance if they are connected in parallel.
 - (2) ratio of energy stored in the parallel combination of the capacitors to that in the series combination, if the combinations are connected to the same source one by one.

(i) (1) $C \propto A$ Capacitance gets doubled	1
(2) $C \propto \frac{1}{d}$ Capacitance reduces to half	1
(ii) $\frac{C}{3} = 1\mu F$ _____ (given) $C = 3\mu F$ (1) Effective capacitance in parallel $C_p = 3C = 3 \times 3\mu F$ $C_p = 9\mu F$ (2) $U = \frac{1}{2} CV^2$ $\frac{U_p}{P_s} = \frac{C_p}{C_s} = \frac{9}{3} = 3:1$	1½ 1½

34.



A parallel plate capacitor is an arrangement of two identical metal plates kept parallel, a small distance apart. The capacitance of a capacitor depends on the size and separation of the two plates and also on the dielectric constant of the medium between the plates. Like resistors, capacitors can also be arranged in series or parallel or a combination of both. By virtue of electric field between the plates, charged capacitors store energy.

(a) The capacitance of a parallel plate capacitor increases from $10\ \mu\text{F}$ to $80\ \mu\text{F}$ on introducing a dielectric medium between the plates. Find the dielectric constant of the medium. 1

(b) n capacitors, each of capacitance C , are connected in series. Find the equivalent capacitance of the combination. 1

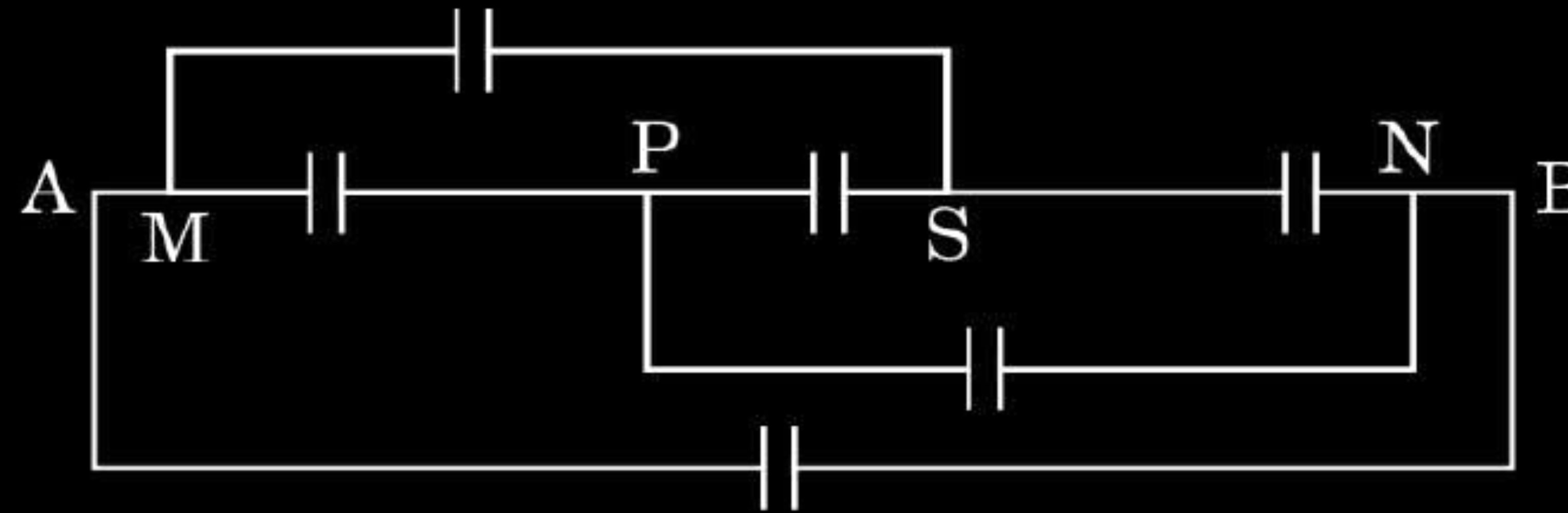
- (c) A capacitor is charged to a potential (V) by connecting it to a battery. After some time, the battery is disconnected and a dielectric is introduced between the plates. How will the potential difference between the plates, and the energy stored in it be affected ? Justify your answer.

2

OR

- (c) Find the equivalent capacitance between points A and B, if capacitance of each capacitor is C .

2



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(a) $K = \frac{C}{C_0}$	½		½
$K = \frac{80\mu F}{10\mu F} = 8$	½		½
(b) $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$ $\frac{1}{C_s} = \frac{n}{C}$ $C_s = \frac{C}{n}$	½	$U_2 = \frac{1}{2} \frac{Q^2}{C_2}$ $= \frac{1}{2} \frac{Q^2}{kC_1} = \frac{1}{k} \left(\frac{Q^2}{2C_2} \right)$ $U_2 = \frac{U_1}{K}$ Energy reduces by a factor of 1/K. OR	½
(c) Charge is constant $Q_1 = Q_2$	½	For calculating effective capacitance = 2 C.	2
$C_2 = KC_1$ $C_1 V_1 = K C_1 V_2$ $V_2 = \frac{V_1}{K}$ Potential diff decreases by a factor (1/K)	½		

- (b) (i) Define electric potential at a point and write its SI unit.
- (ii) Two capacitors are connected in series. Derive an expression of the equivalent capacitance of the combination.
- (iii) Two point charges $+q$ and $-q$ are located at points $(3a, 0)$ and $(0, 4a)$ respectively in x - y plane. A third charge Q is kept at the origin. Find the value of Q , in terms of q and a , so that the electrostatic potential energy of the system is zero.

5

(i) Electrical Potential – Electrostatic potential at any point in a region with electrostatic field is the work done in bringing a unit positive charge (without acceleration) from infinity to that point.

Alternatively:-

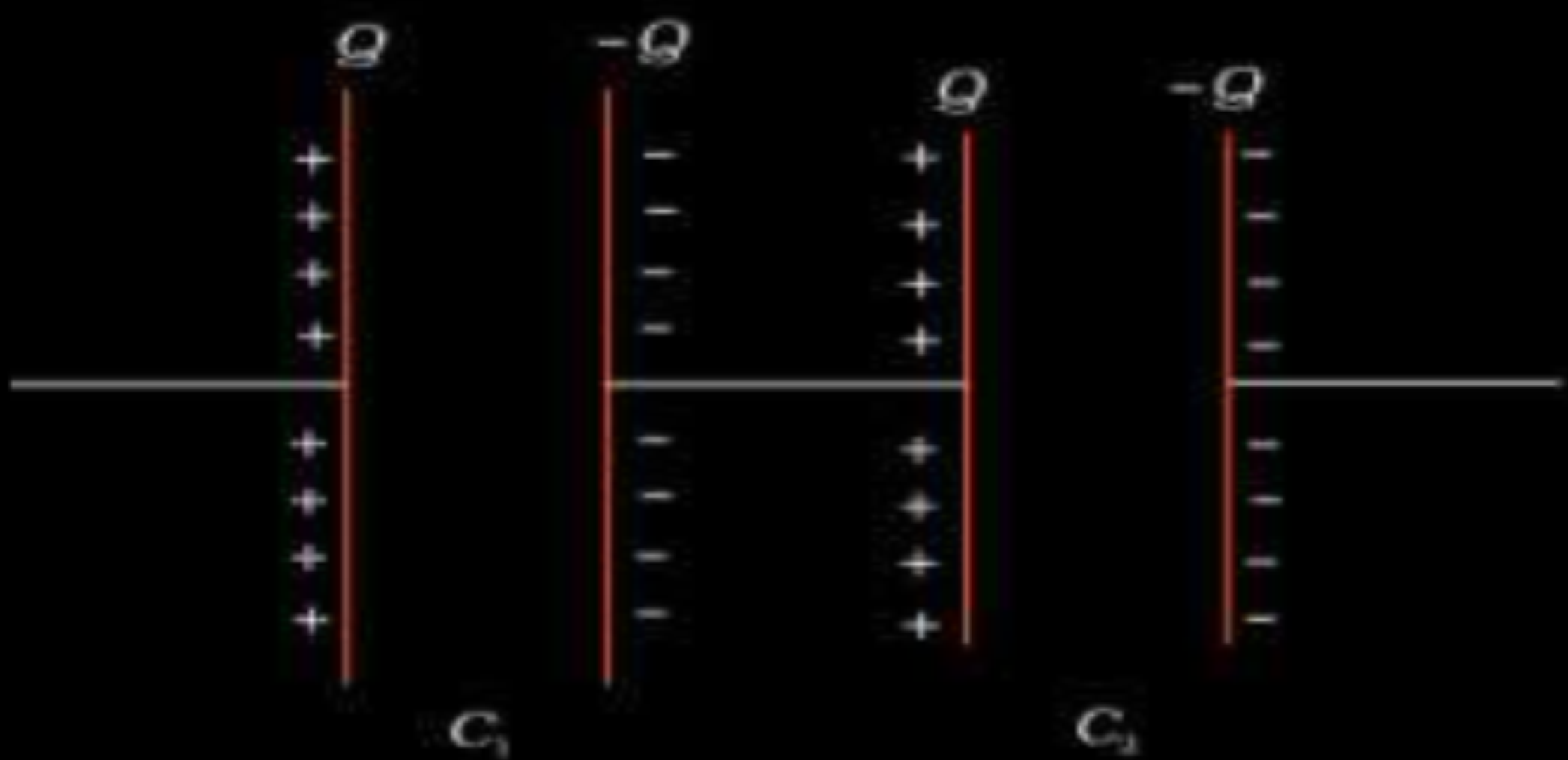
$$V = \frac{Work\ Done}{q}$$

$$V = -\int \vec{E} \cdot \vec{dl}$$

S.I. unit of electrostatic potential is volt.

Alternatively:-

S.I. unit is J/C.



$$V = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$\frac{Q}{C_{eq.}} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

$$\frac{1}{C_{eq.}} = \frac{1}{C_1} + \frac{1}{C_2}$$

½

½

½

½

½

½

(iii)

$$\text{Potential energy of the system} = K \left[\frac{Q(-q)}{4a} + \frac{Qq}{3a} - \frac{q^2}{5a} \right]$$

$$\text{Potential energy of the system} = 0$$

$$\Rightarrow K \left[\frac{-Qq}{4a} + \frac{Qq}{3a} - \frac{q^2}{5a} \right] = 0$$

$$\Rightarrow \frac{-Q}{4} + \frac{Q}{3} - \frac{q}{5} = 0$$

$$\Rightarrow +\frac{Q}{12} - \frac{q}{5} = 0$$

$$\Rightarrow Q = + \frac{12q}{5}$$

½

½

1

30. A $100\ \mu\text{F}$ capacitor is charged by a $12\ \text{V}$ battery.

- (a) How much electrostatic energy is stored by the capacitor ?
- (b) The capacitor is disconnected from the battery and connected in parallel to another uncharged $100\ \mu\text{F}$ capacitor. What is the electrostatic energy stored by the system ?

CBSE 23

$$C = 100 \mu F = 100 \times 10^{-6} F, V = 12V$$

$$\text{a) } U = \frac{1}{2} CV^2$$

$$= \frac{1}{2} \times 100 \times 10^{-6} \times (12)^2$$

$$= \frac{1}{2} \times 10^{-4} \times 144$$

$$= 72 \times 10^{-4} \text{ J} = 7.2 \text{ m J}$$

$$\text{b) } C_{eq} = C_1 + C_2$$

$$= 200 \mu F$$

$$Q = CV$$

$$= 100 \times 10^{-6} \times 12$$

$$= 12 \times 10^{-4} C$$

$$U = \frac{Q^2}{2C_{eq}} = \frac{(12 \times 10^{-4})^2}{2 \times 200 \times 10^{-6}}$$

$$= \frac{144}{4} \times 10^{-4}$$

$$= 36 \times 10^{-4} \text{ J} = 3.6 \text{ m J}$$

1/2

1/2

1/2

1/2

1/2

1/2

- (b) A parallel plate capacitor (A) of capacitance C is charged by a battery to voltage V . The battery is disconnected and an uncharged capacitor (B) of capacitance $2C$ is connected across A. Find the ratio of
- (i) final charges on A and B.
 - (ii) total electrostatic energy stored in A and B finally and that stored in A initially.

(b) A parallel plate capacitor (A) of capacitance C is charged by a battery to voltage V . The battery is disconnected and an uncharged capacitor (B) of capacitance $2C$ is connected across A. Find the ratio of

(i) final charges on A and B.

(ii) total electrostatic energy stored in A and B finally and that

stored in A initially.

CBSE 23

i)	Initially $Q = CV$	
	Finally $q_A = C_A V_1$ & $q_B = C_B V_1$	$\frac{1}{2}$
	$\frac{q_A}{q_B} = \frac{C_A}{C_B} = \frac{1}{2}$	$\frac{1}{2}$
ii)	$q_A + q_b = Q$	
	$\therefore q_A = \frac{Q}{3}$ & $q_B = \frac{2Q}{3}$	
	$\frac{U_f}{U_i} = \frac{U_A + U_B}{U_{Ai}}$	$\frac{1}{2}$
	$= \frac{\frac{q_A^2}{2C_A} + \frac{q_B^2}{2C_B}}{\frac{Q^2}{2C_A}}$	1
	$= \frac{1}{3}$	$\frac{1}{2}$

Alternatively ,
Common potential

$V_1 = \frac{Q_1 + Q_2}{C_1 + C_2}$	
$= \frac{Q}{3C} = \frac{V}{3}$	$\left[\because \frac{Q}{C} = V \right]$
$\frac{U_f}{U_i} = \frac{\frac{1}{2} C_{eq} V_1^2}{\frac{1}{2} C_A V^2}$	$\frac{1}{2}$
$= \frac{\frac{1}{2} 3C \times \left(\frac{V}{3} \right)^2}{\frac{1}{2} C V^2} = \frac{1}{3}$	$\frac{1}{2}$

12. The capacitors, each of $4\ \mu\text{F}$ are to be connected in such a way that the effective capacitance of the combination is $6\ \mu\text{F}$. This can be achieved by connecting

1

- (A) All three in parallel
- (B) All three in series
- (C) Two of them connected in series and the combination in parallel to the third.
- (D) Two of them connected in parallel and the combination in series to the third.

CBSE 23

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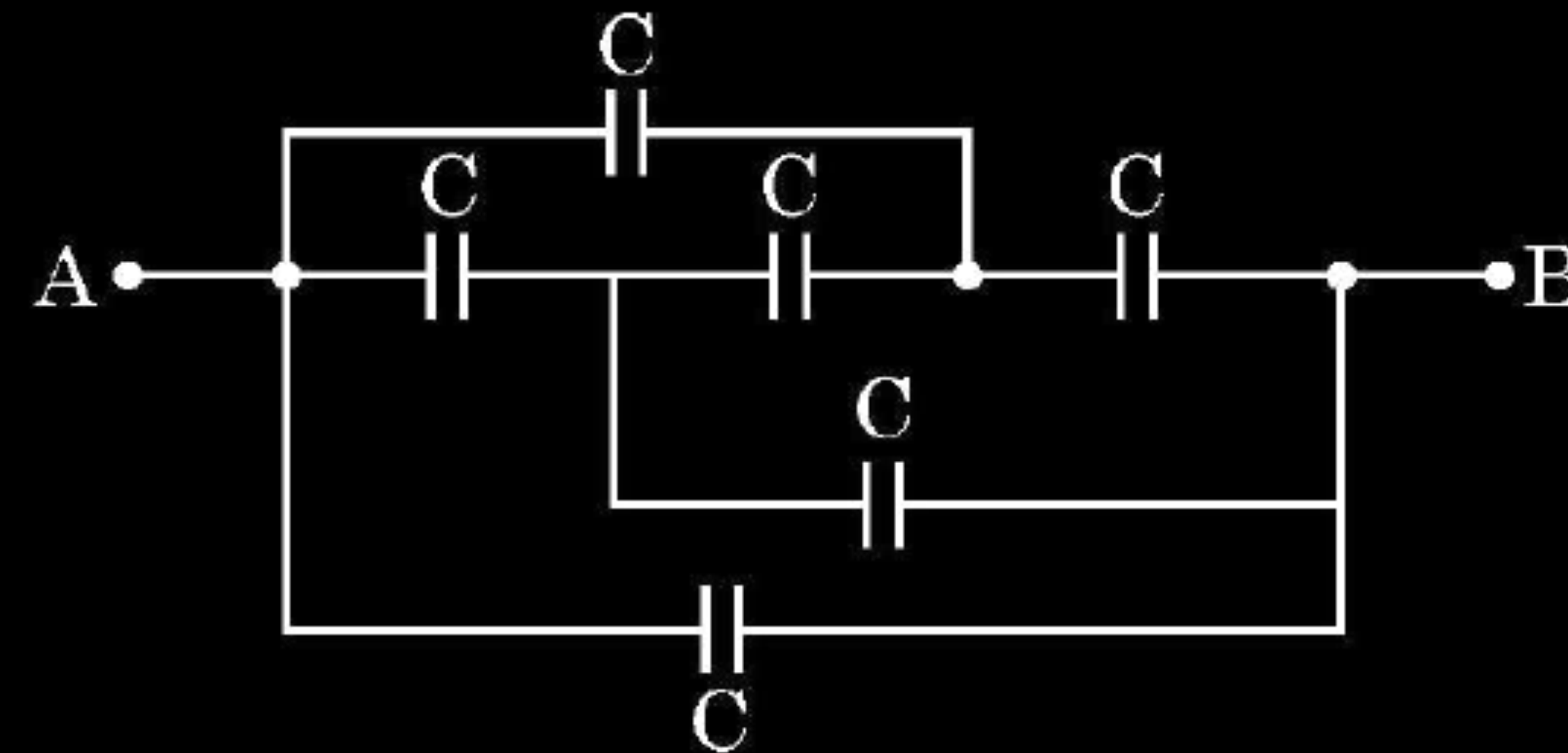
CBSE 23

(C) Two of them connected in series and the combination in parallel to the third

35. A capacitor is a system of two conductors separated by an insulator. The two conductors have equal and opposite charges with a potential difference between them. The capacitance of a capacitor depends on the geometrical configuration (shape, size and separation) of the system and also on the nature of the insulator separating the two conductors. They are used to store charges. Like resistors, capacitors can be arranged in series or parallel or a combination of both to obtain desired value of capacitance.

4

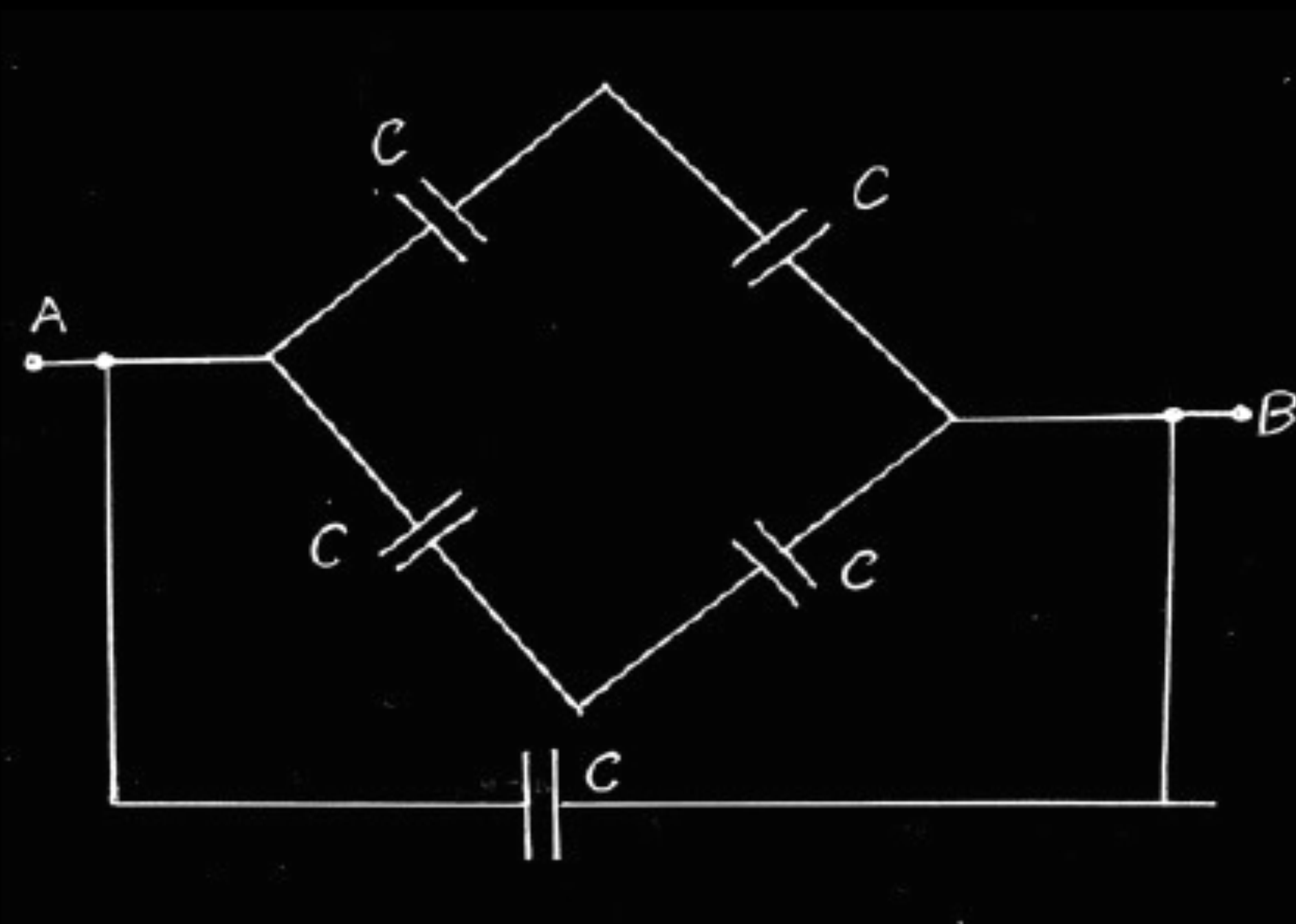
- (i) Find the equivalent capacitance between points A and B in the given diagram.



CBSE 23

- (ii) A dielectric slab is inserted between the plates of a parallel plate capacitor. The electric field between the plates decreases. Explain.
- (iii) A capacitor A of capacitance C , having charge Q is connected across another uncharged capacitor B of capacitance $2C$. Find an expression for (a) the potential difference across the combination and (b) the charge lost by capacitor A.

(i)

 $\frac{1}{2}$

$$C_{\text{net}} = C + C$$

$$= 2C$$

 $\frac{1}{2}$

(Note: Give full credit to a student if $2C$ written directly)

(ii) Within the dielectric slab, induced electric field due to polarization, decreases the electric field.

1

Alternatively: $E = E_0 - E_p$

Alternatively: $E = \frac{E_0}{K}$

(iii)

(a) $V' = \frac{Q_{\text{Total}}}{C_{\text{eqi}}} = \frac{Q}{3C}$

 $\frac{1}{2}$

$$V' = \frac{Q}{3C} = \frac{V}{3}$$

 $\frac{1}{2}$

(b)

$$Q_A' = C \times \frac{V}{3} = \frac{Q}{3}$$

 $\frac{1}{2}$

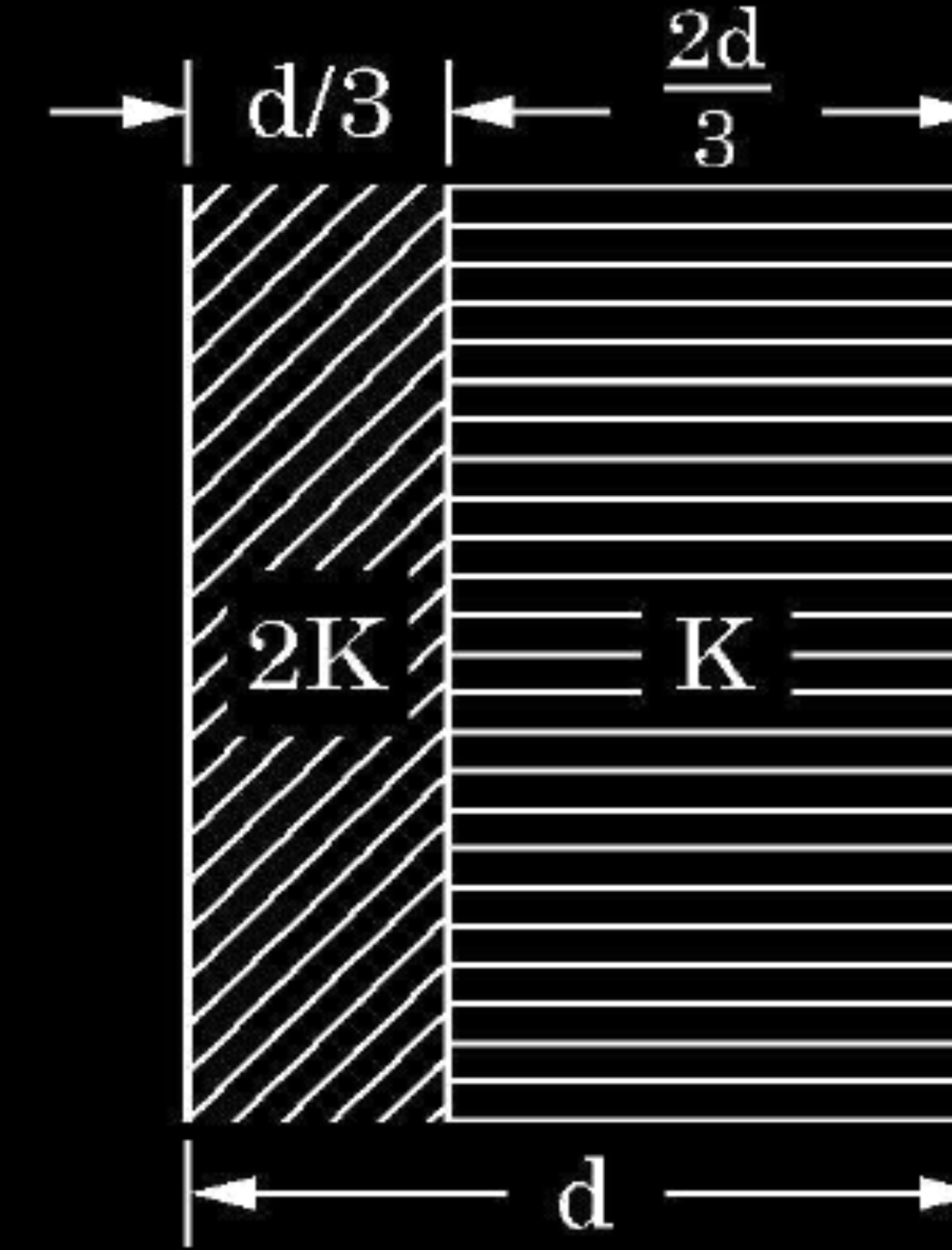
$$Q_A = CV = Q$$

Charge lost by capacitor A is

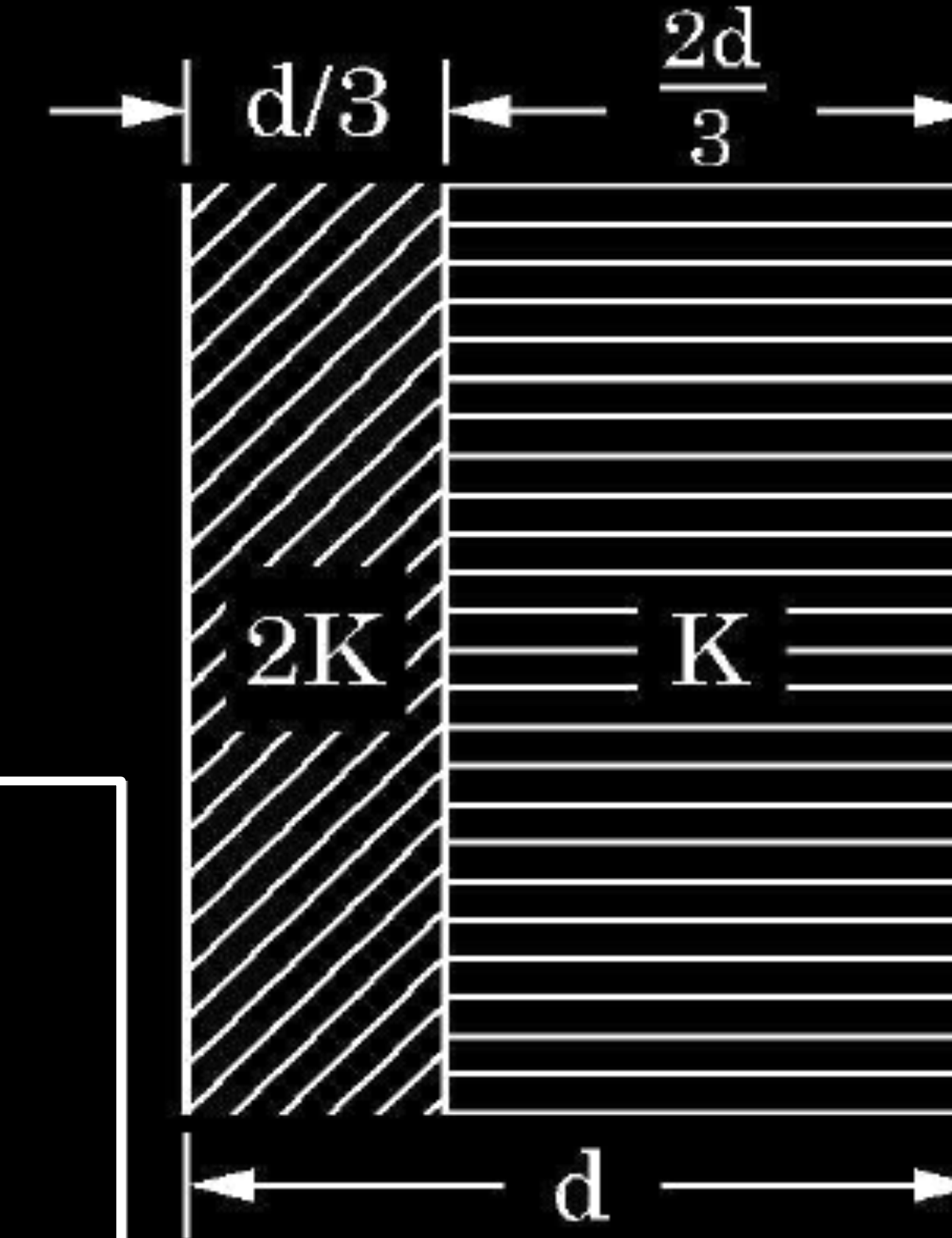
$$\Delta Q = Q - \frac{Q}{3} = \frac{2Q}{3}$$

 $\frac{1}{2}$

- (iii) Two slabs of dielectric constants $2K$ and K fill the space between the plates of a parallel plate capacitor of plate area A and plate separation d as shown in figure. Find an expression for capacitance of the system.



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CBSE 23

Capacitance of left portion, $C_1 = \frac{6K \epsilon_0 A}{d}$

Capacitance of right portion, $C_2 = \frac{3K \epsilon_0 A}{2d}$

As the capacitors are in series

$$\frac{1}{C_{eqi}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C_{eqi}} = \frac{d}{6K \epsilon_0 A} + \frac{2d}{3K \epsilon_0 A} = \frac{5d}{6KA \epsilon_0}$$

$$C_{eqi} = \frac{6KA \epsilon_0}{5d}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

Obtain the expression for the energy stored in a capacitor connected across a dc battery. Hence define energy density of the capacitor. CBSE 20 **2**

Obtain the expression for the energy stored in a capacitor connected across a dc battery. Hence define energy density of the capacitor.

CBSE 20

2

Let the charge on the capacitor plates at any instant, during charging process be q , amount of work done to supply further charge dq to the capacitor

$$dW = Vdq$$

where V is the potential difference and equals to $\frac{q}{C}$

Total work done to charge the capacitor upto charge Q

$$W = \int_0^Q Vdq$$

$$= \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C} \left(\frac{1}{2} CV^2 = \frac{1}{2} QV \right)$$

Since *Energy stored = work done*

$$\Rightarrow U = \frac{Q^2}{2C} \left(\frac{1}{2} CV^2 = \frac{1}{2} QV \right)$$

Energy density: Electrical energy stored per unit volume is known as energy density.

Alternatively:

$$\text{Energy density} = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \frac{\sigma^2}{\epsilon_0}$$

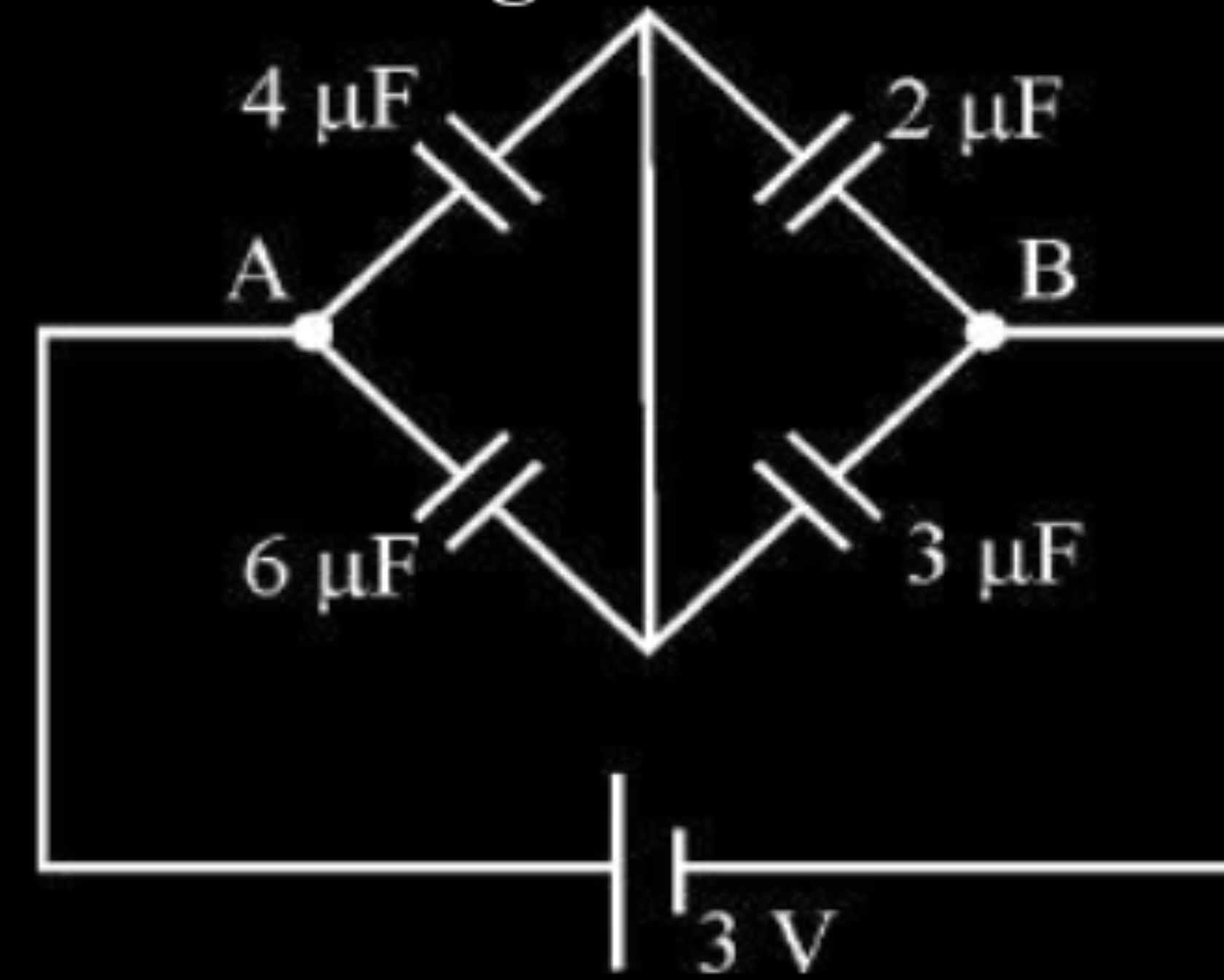
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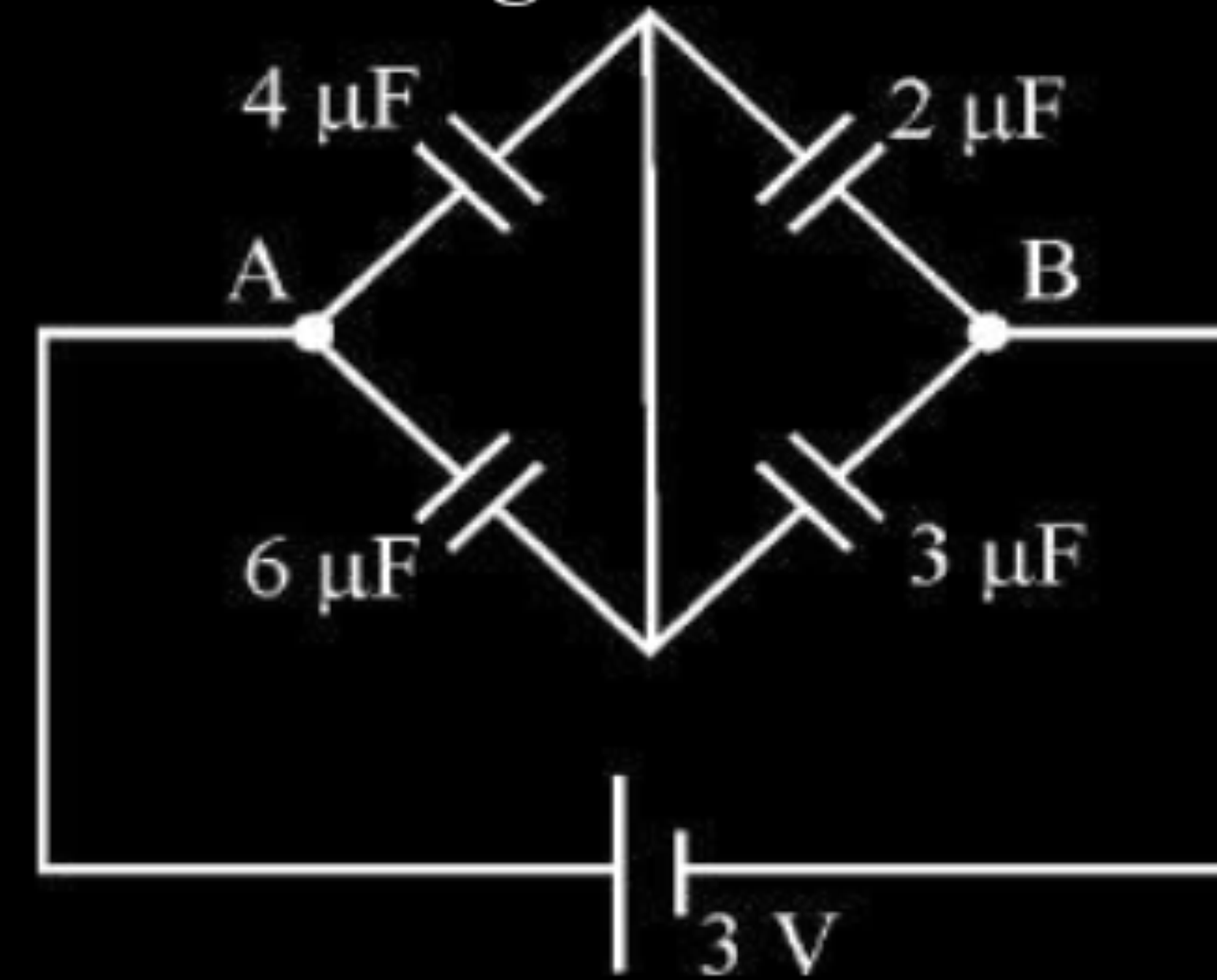
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21. Find the total charge stored in the network of capacitors connected between A and B as shown in figure :



CBSE 20

21. Find the total charge stored in the network of capacitors connected between A and B as shown in figure :



CBSE 20

2

(i)	Net Capacitance =>	$\frac{1}{C_{net}} = \frac{1}{10} + \frac{1}{5}$ $C_{net} = \frac{10}{3} \mu F$	$\frac{1}{2}$ $\frac{1}{2}$
(ii)	$q = C_{net} V = \frac{10}{3} \times 3 = 10 \mu C$		$\frac{1}{2} + \frac{1}{2}$

3. Two capacitors of capacitances C_1 and C_2 are connected in parallel. If a charge Q is given to the combination, the ratio of the charge on the capacitor C_1 to the charge on C_2 will be

CBSE 20

(A) $\frac{C_1}{C_2}$

(B) $\sqrt{\frac{C_1}{C_2}}$

(C) $\sqrt{\frac{C_2}{C_1}}$

(D) $\frac{C_2}{C_1}$

3. Two capacitors of capacitances C_1 and C_2 are connected in parallel. If a charge Q is given to the combination, the ratio of the charge on the capacitor C_1 to the charge on C_2 will be

CBSE 20

1

(A) $\frac{C_1}{C_2}$

(B) $\sqrt{\frac{C_1}{C_2}}$

(C) $\sqrt{\frac{C_2}{C_1}}$

(D) $\frac{C_2}{C_1}$

(A)

$\frac{C_1}{C_2}$

- (a) Derive an expression for the energy stored in a parallel plate capacitor of capacitance C when charged up to voltage V . How is this energy stored in the capacitor ?
- (b) A capacitor of capacitance $1\ \mu\text{F}$ is charged by connecting a battery of negligible internal resistance and emf $10\ \text{V}$ across it. Calculate the amount of charge supplied by the battery in charging the capacitor fully.

$$\begin{aligned}
 \text{a) Work done in adding a charge } dq &= dW \\
 &= Vdq \\
 &= \frac{q}{C} dq
 \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\therefore$ Total Amount of work(W)in charging a capacitor

$$W = \int dW = \frac{1}{C} \int_0^Q q dq$$

$\frac{1}{2}$

$$W = \frac{Q^2}{2C}$$

$\frac{1}{2}$

$$= \frac{(CV)^2}{2C} = \frac{1}{2} CV^2$$

$\frac{1}{2}$

The electrostatic Energy/ potential energy is stored in the electric field between the plates.

$\frac{1}{2}$

$$\text{b) } C = 1\mu F = 1 \times 10^{-6} F; \quad V = 10 \text{ volt}$$

$$Q = CV$$

1

$$= 1 \times 10^{-6} \times 10$$

$\frac{1}{2}$

$$= 10^{-5} \text{ coulomb}$$

$\frac{1}{2}$

In the figure given below, find the

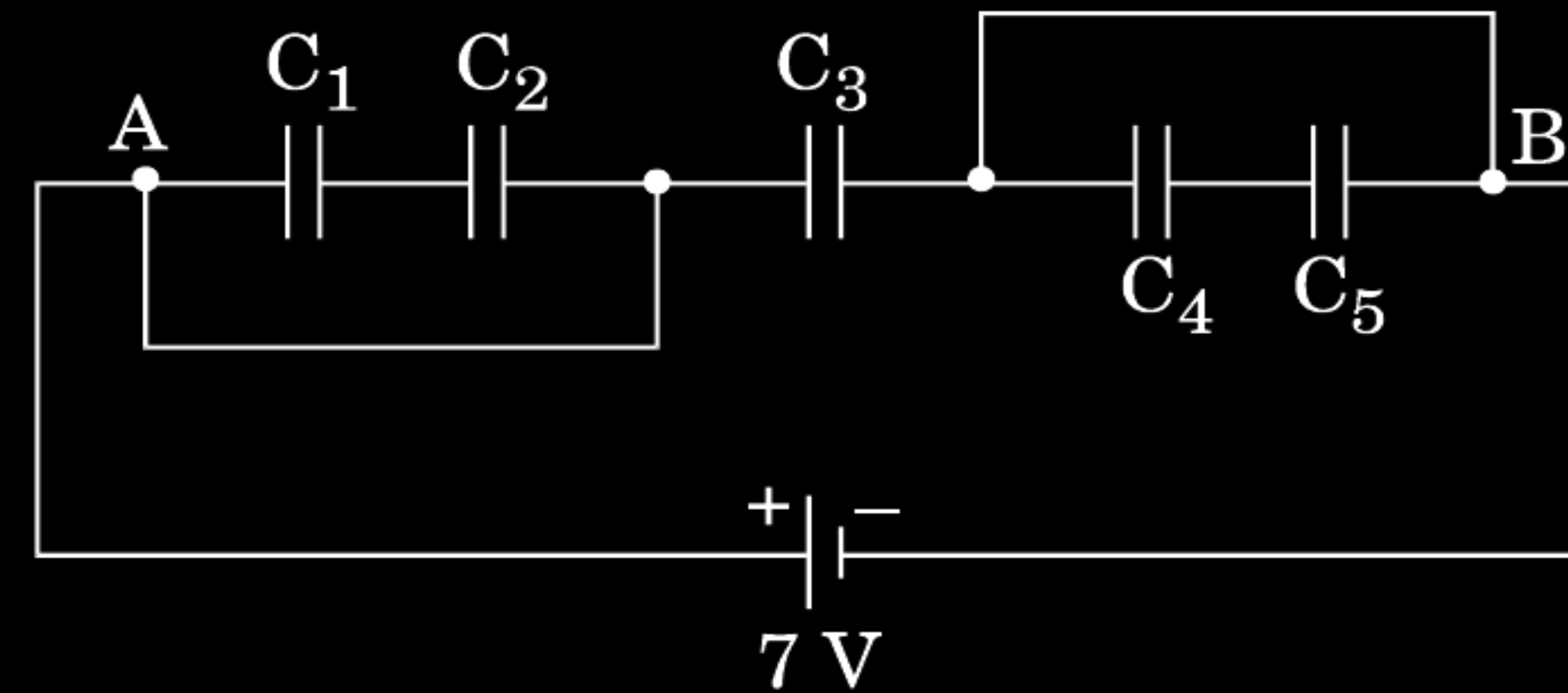
(a) equivalent capacitance of the network between points A and B.

Given : $C_1 = C_5 = 8 \mu\text{F}$, $C_2 = C_3 = C_4 = 4 \mu\text{F}$.

(b) maximum charge supplied by the battery, and

(c) total energy stored in the network.

3

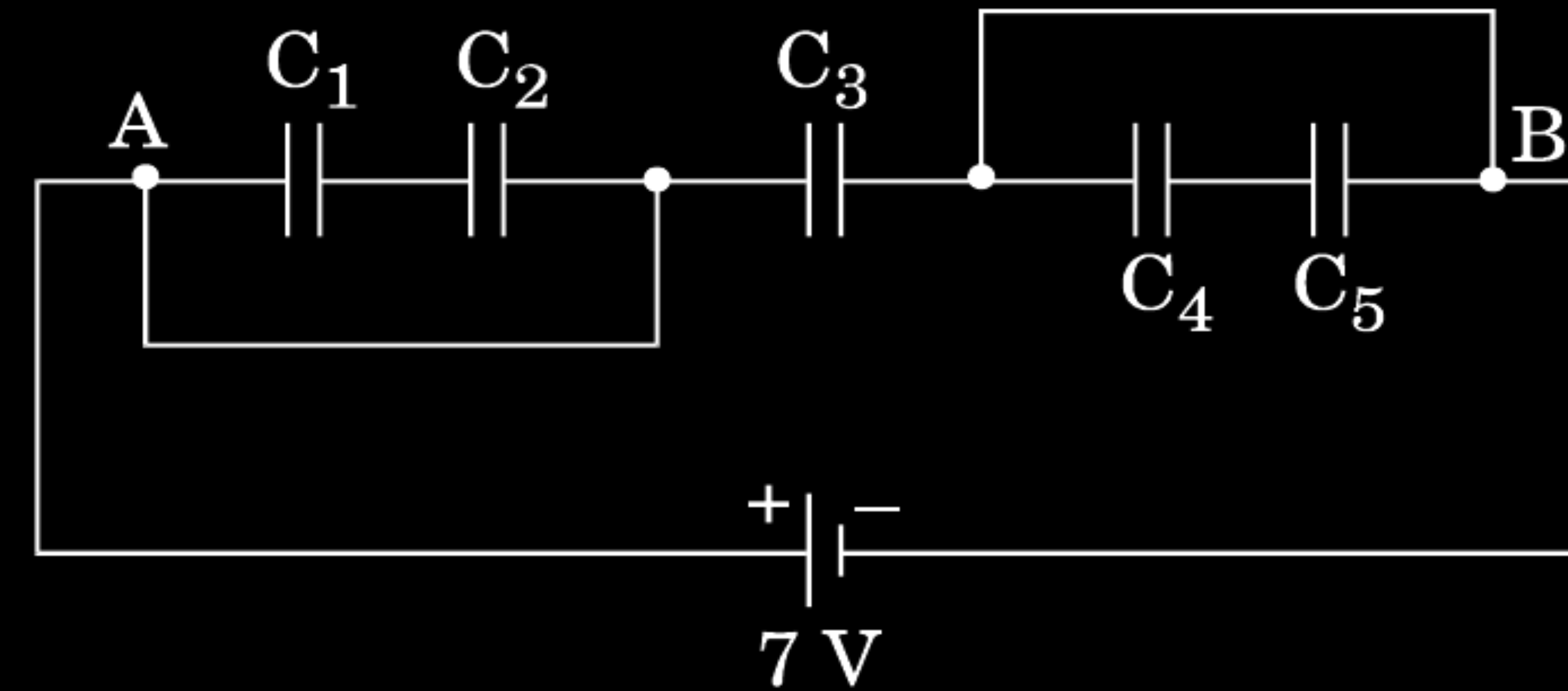


CBSE 20

In the figure given below, find the

- (a) equivalent capacitance of the network between points A and B.
Given : $C_1 = C_5 = 8 \mu\text{F}$, $C_2 = C_3 = C_4 = 4 \mu\text{F}$.
- (b) maximum charge supplied by the battery, and
- (c) total energy stored in the network.

3



CBSE 20

(a) $C = C_4 = 4\mu\text{F}$ (as C_1, C_2, C_4, C_5 are short circuited)

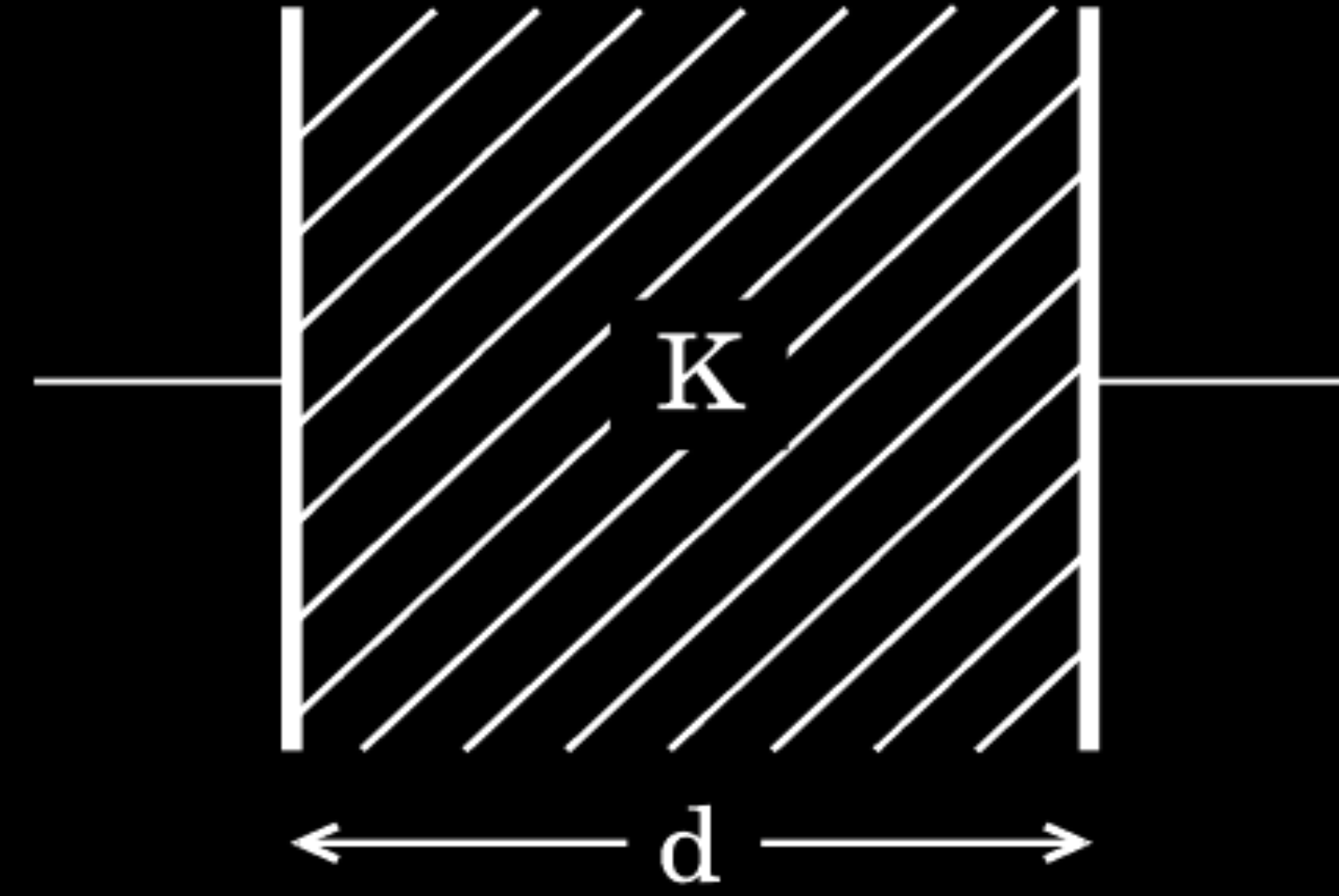
(b) $Q = CV = 4 \times 7 \mu\text{C}$
 $= 28 \mu\text{C}$

(c) $U = \frac{1}{2} CV^2$

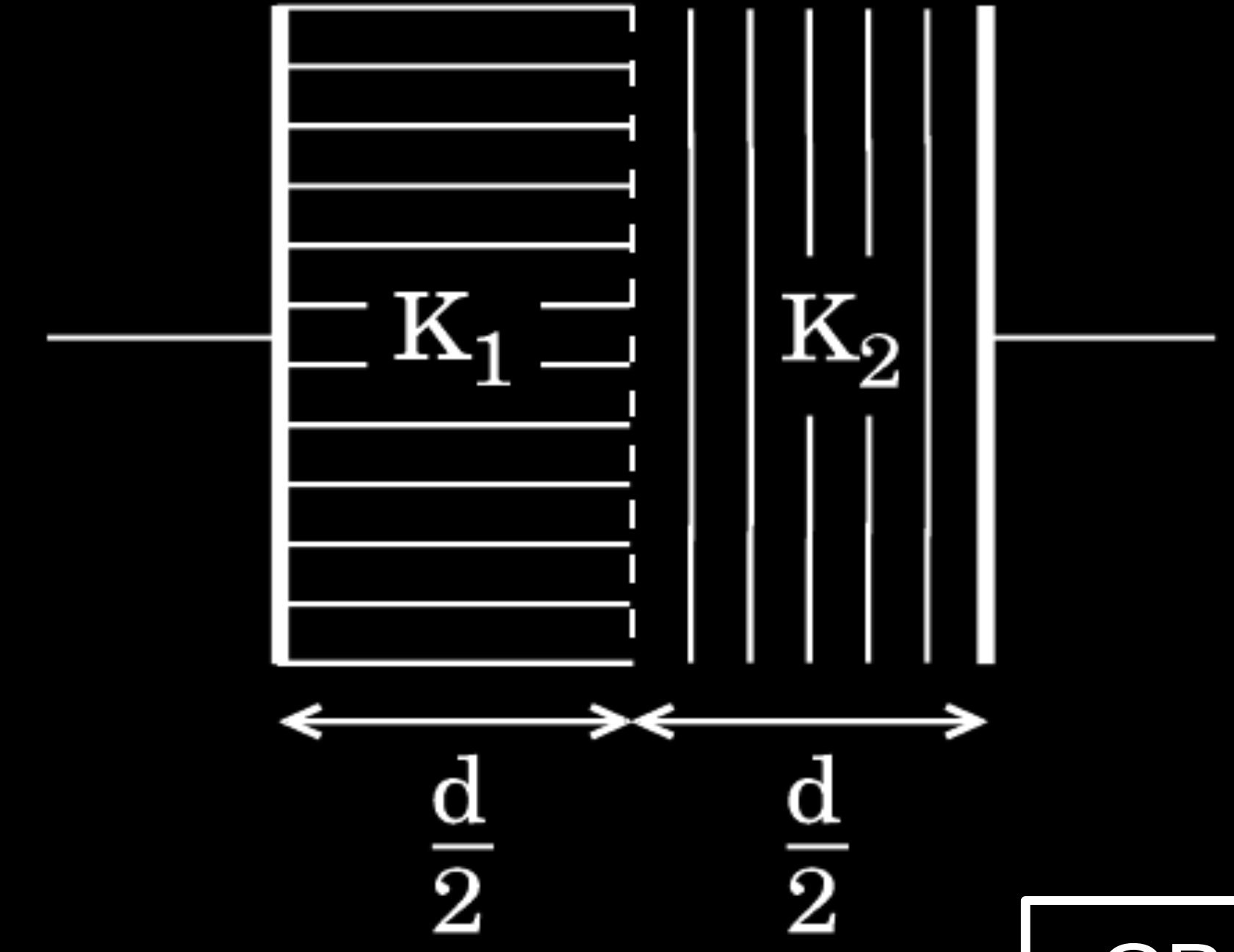
$$= \frac{1}{2} \times 4 \times 10^{-6} \times 7 \times 7 = 98 \times 10^{-6} \text{J}$$

22. The space between the plates of a parallel plate capacitor is completely filled in two ways. In the first case, it is filled with a slab of dielectric constant K . In the second case, it is filled with two slabs of equal thickness and dielectric constants K_1 and K_2 respectively as shown in the figure. The capacitance of the capacitor is same in the two cases. Obtain the relationship between K , K_1 and K_2 .

2



(Case 1)

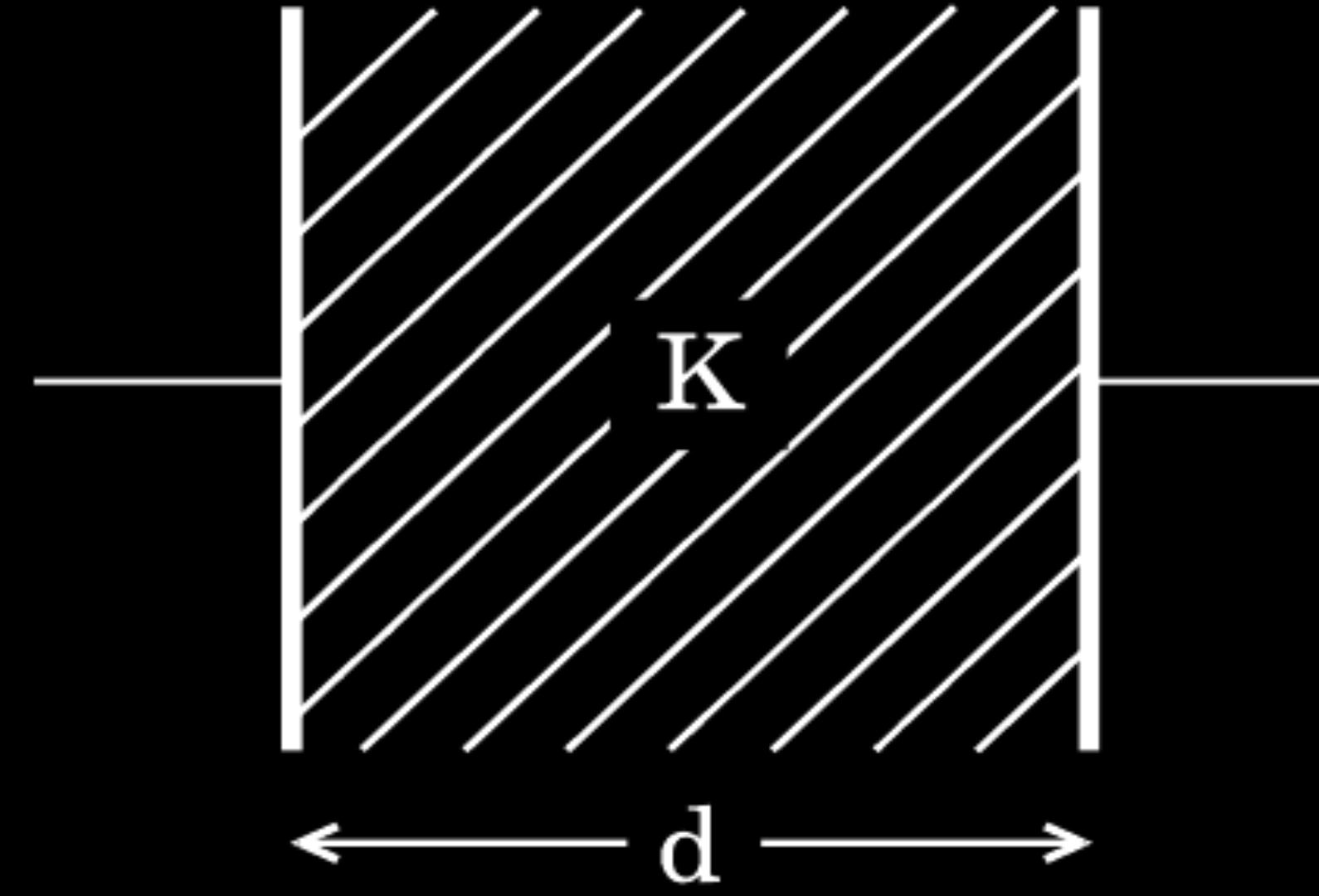


(Case 2)

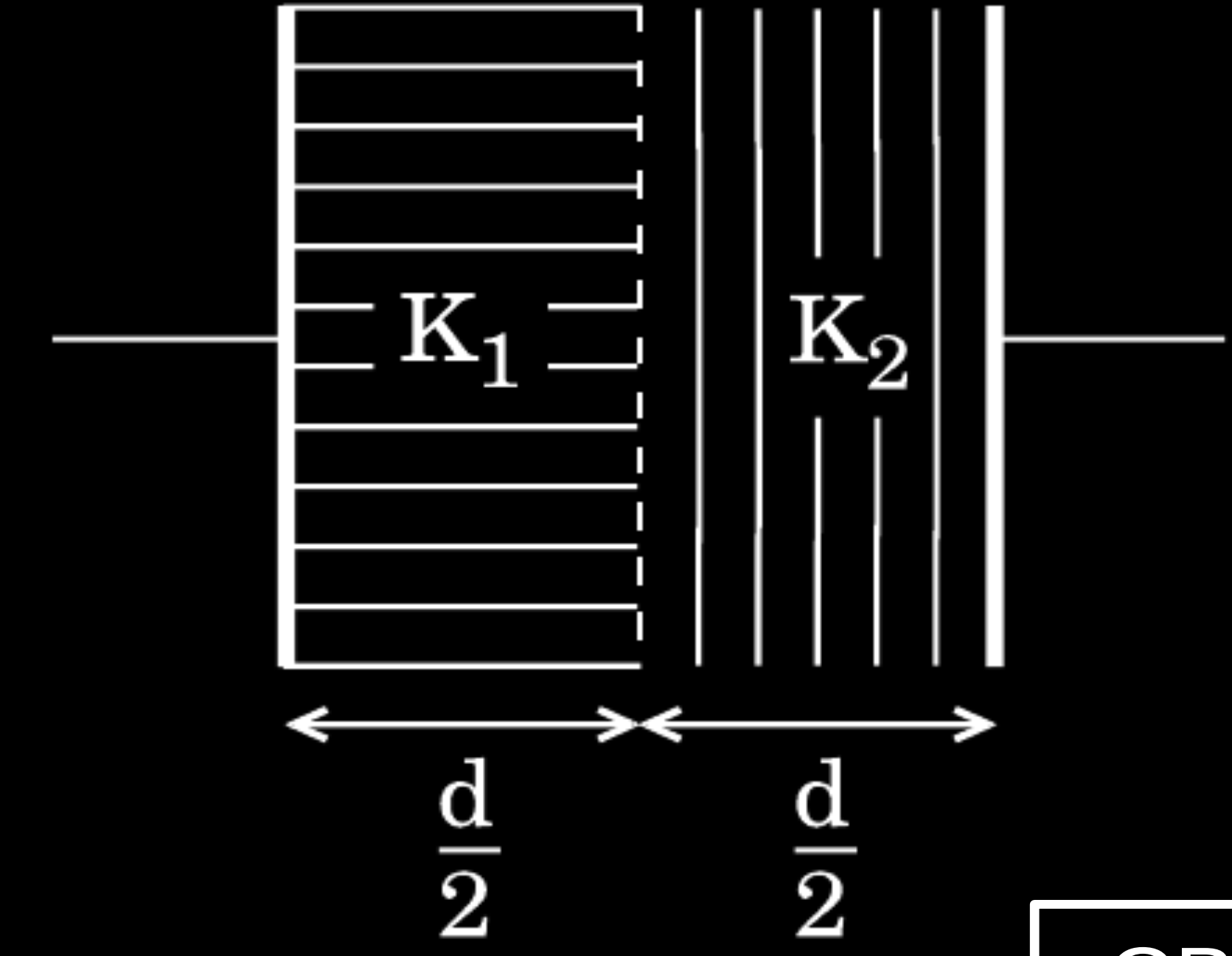
CBSE 20

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2



(Case 1)



(Case 2)

CBSE 20

$$C_1 = \frac{k\epsilon_0 A}{d}$$

Capacitor are connected in series

$$C_2 = \frac{C' C''}{C' + C''} = \left(\frac{2K_1 K_2}{K_1 + K_2} \right) \frac{\epsilon_0 A}{d}$$

$$C_1 = C_2$$

$$K = \frac{2K_1 K_2}{K_1 + K_2}$$

1/2

1

1/2

Two identical capacitors of 12 pF each are connected in series across a 50 V battery. Calculate the electrostatic energy stored in the combination. If these were connected in parallel across the same battery, find out the value of the energy stored in this combination.

Two identical capacitors of 12 pF each are connected in series across a 50 V battery. Calculate the electrostatic energy stored in the combination. If these were connected in parallel across the same battery, find out the value of the energy stored in this combination.

CBSE 19

$$C_s = 6 \text{ pF}$$

$$E_s = \frac{1}{2} C_s V^2 = \frac{1}{2} \times 6 \times 10^{-12} \times 50 \times 50 = 7500 \times 10^{-12} \text{ J}$$

$$= 7.5 \times 10^{-9} \text{ J}$$

$$C_p = 24 \text{ pF}$$

$$E_p = \frac{1}{2} C_p V^2 = \frac{1}{2} \times 24 \times 50 \times 50 \times 10^{-12}$$

$$= 3 \times 10^{-8} \text{ J}$$

1/2

1/2

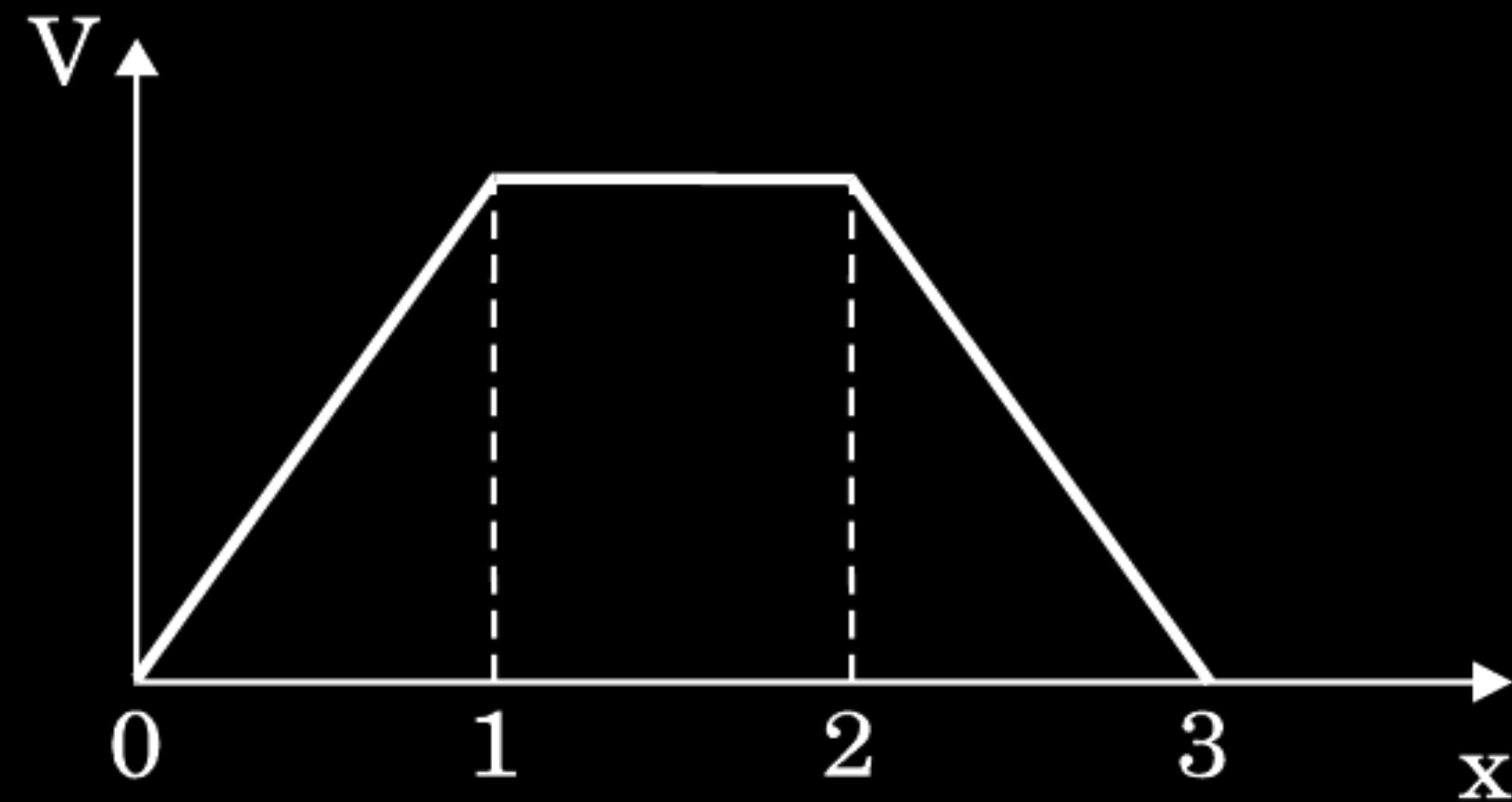
1/2

1/2

- (a) When a parallel plate capacitor is connected across a dc battery, explain briefly how the capacitor gets charged.
- (b) A parallel plate capacitor of capacitance ' C ' is charged to ' V ' volt by a battery. After some time the battery is disconnected and the distance between the plates is doubled. Now a slab of dielectric constant $1 < k < 2$ is introduced to fill the space between the plates. How will the following be affected ?
- (i) The electric field between the plates of the capacitor.
 - (ii) The energy stored in the capacitor.
- Justify your answer in each case.

CBSE 19

- (c) The electric potential as a function of distance ' x ' is shown in the figure. Draw a graph of the electric field E as a function of x .



- a) Charging of capacitor with dc battery whenever parallel plate capacitor is connected with dc source, plates start acquiring charge in accordance with the terminals of the battery till potential difference across the plate becomes equal to terminal potential of dc battery.

Note : Any other relevant explanation may also be accepted.

- b) i) The electric field between the plates of parallel plate capacitor

$$E_0 = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

If dielectric is inserted

$$E' = \frac{Q}{\epsilon_0 A \cdot K} = \frac{E_0}{K}$$

So, the electric field intensity decreases to $1/K$ times.

- ii) Since Energy stores in the capacitor

$$U = \frac{Q^2}{2C} = \frac{Q^2 d}{2\epsilon_0 A} \dots\dots\dots (1)$$

Similarly

$$U' = \frac{Q^2}{2C'} = \frac{Q^2 d_1}{2K\epsilon_0 A}$$

$$= \frac{2}{K} \left(\frac{Q^2 d_1}{2\epsilon_0 A} \right)$$

$$= \frac{2U}{K}$$

$i < k < 2$

Therefore energy stored between the plates increases

1

$\frac{1}{2}$

$\frac{1}{2}$

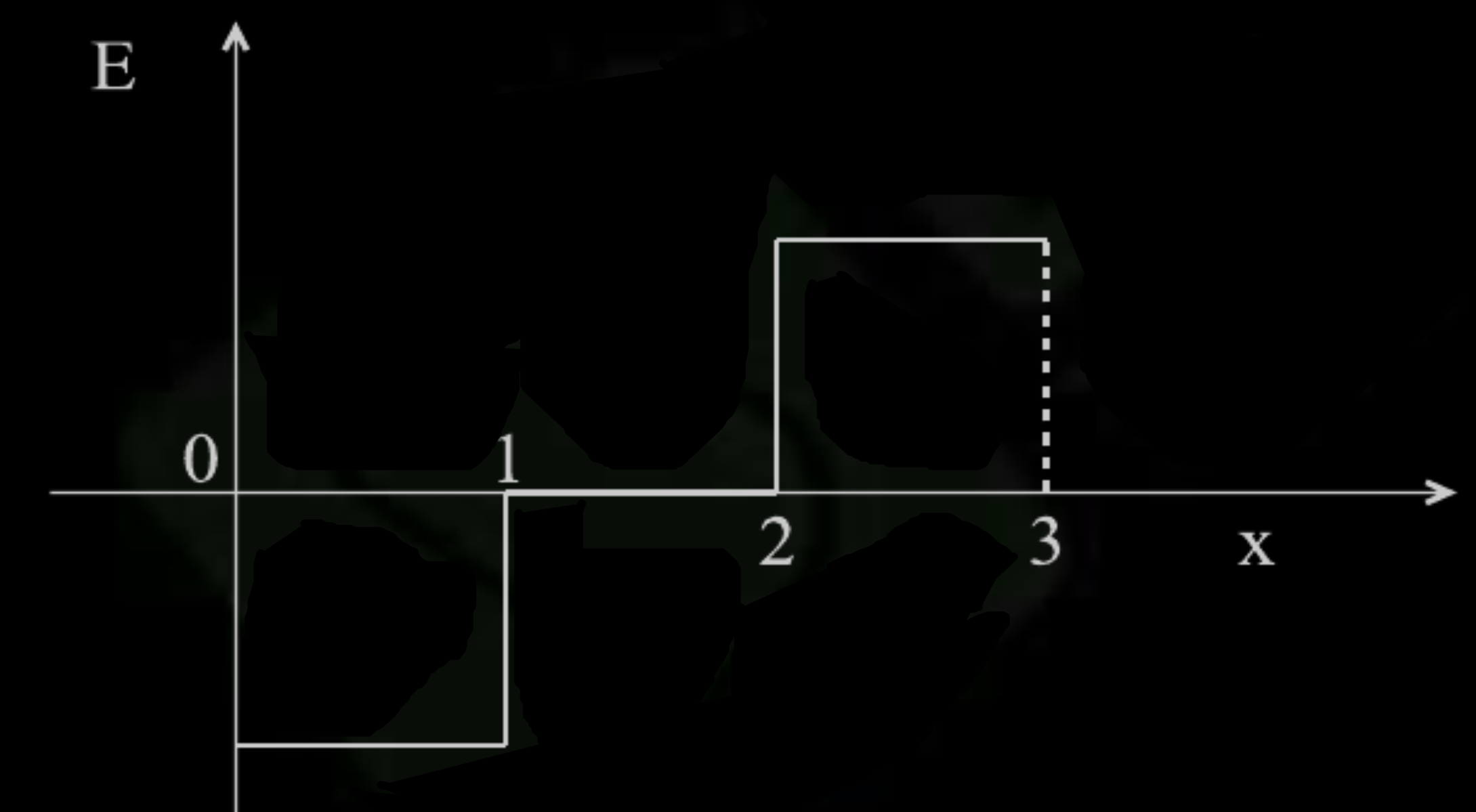
$\frac{1}{2}$

$\frac{1}{2}$

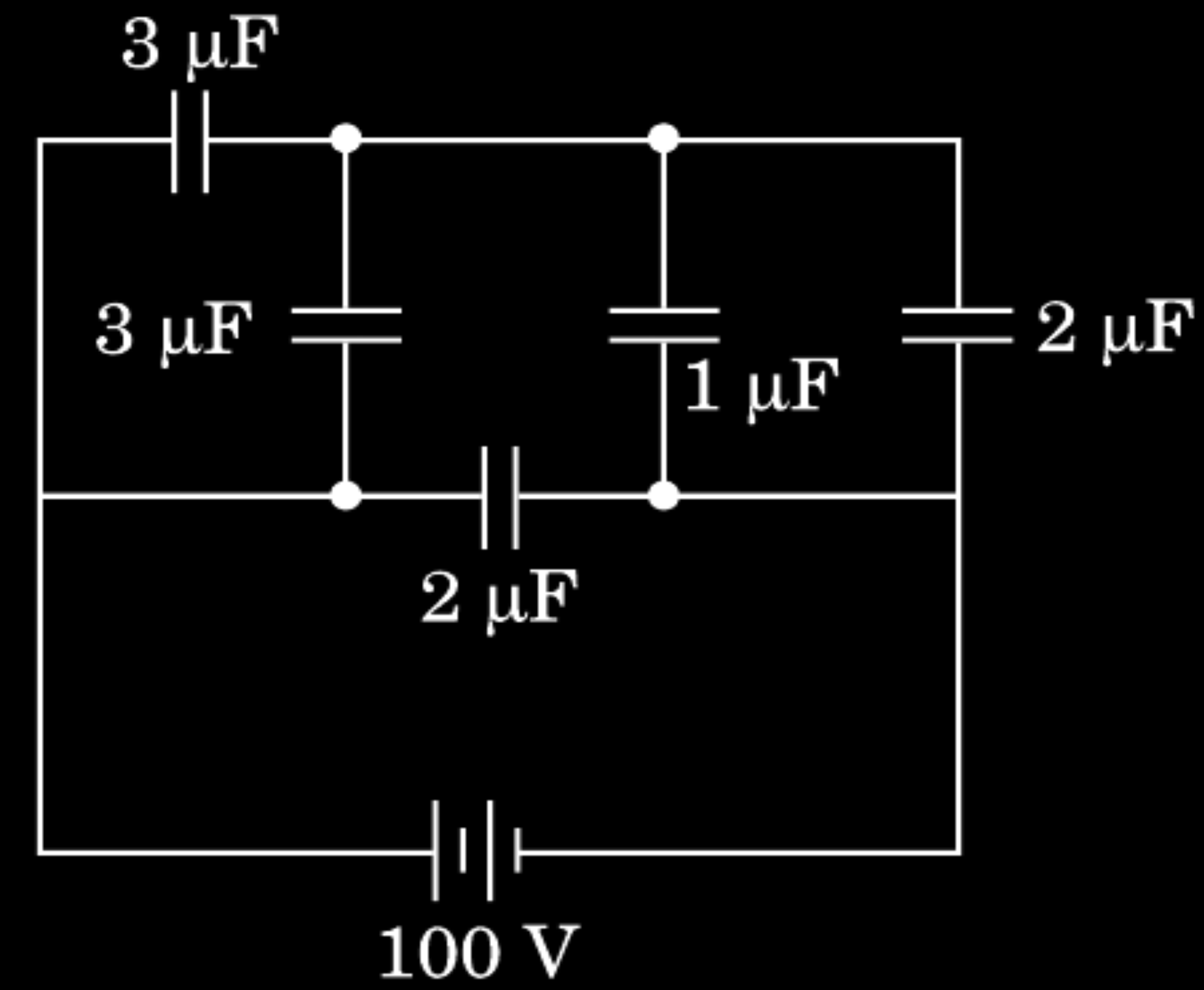
$\frac{1}{2}$

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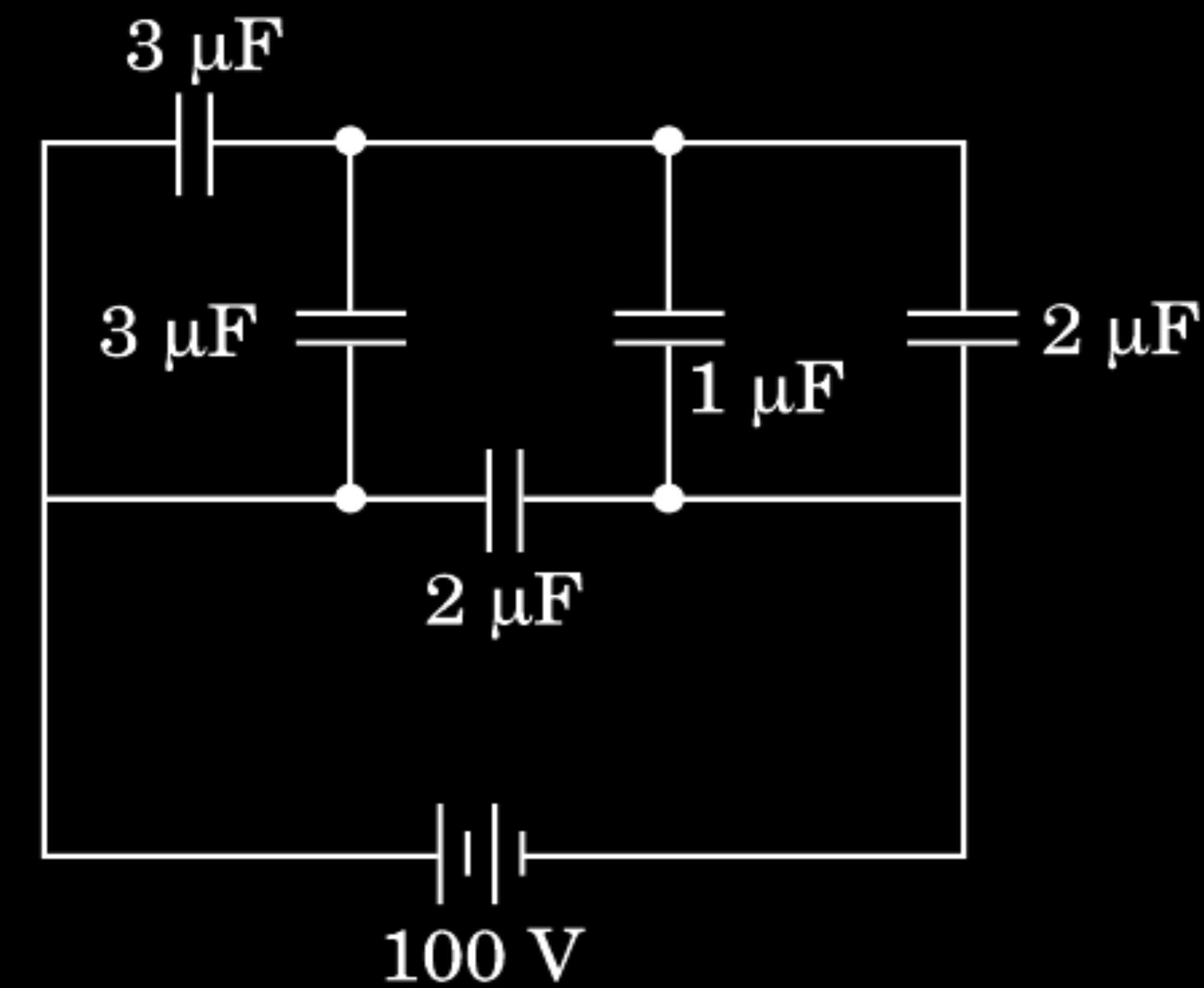
iii)



The figure shows a network of five capacitors connected to a 100 V supply. Calculate the total energy stored in the network.



The figure shows a network of five capacitors connected to a 100 V supply. Calculate the total energy stored in the network.



CBSE 19

- i) Equivalent capacitance of two $3\mu\text{F}$ capacitor in parallel $C_1 = 3 + 3 = 6\mu\text{F}$
 Similarly equivalent capacitance of $1\mu\text{F}$ and $2\mu\text{F}$ in parallel

$$C_2 = 1 + 2 = 3\mu\text{F}$$

Equivalent capacitance of

C_1 and C_2 in series

$$C_{12} = \frac{6 \times 3}{6 + 3} = 2\mu\text{F}$$

Net capacitance

$$C = 2 + 2 = 4\mu\text{F}$$

- ii) Energy stored $E = \frac{1}{2} CV^2$
 $= \frac{1}{2} \times 4 \times 10^{-6} \times (100)^2$
 $= 0.02 \text{ J}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

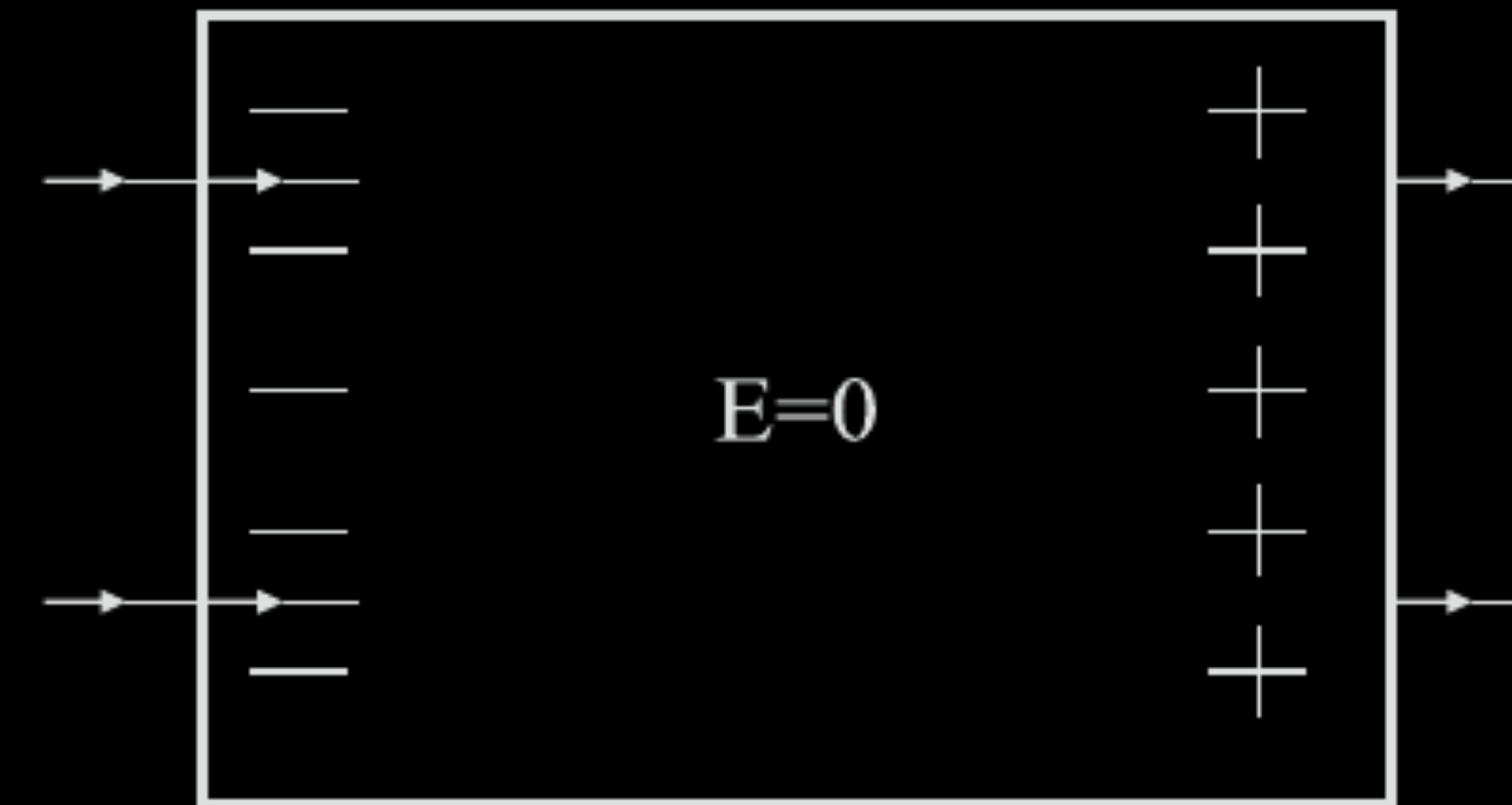
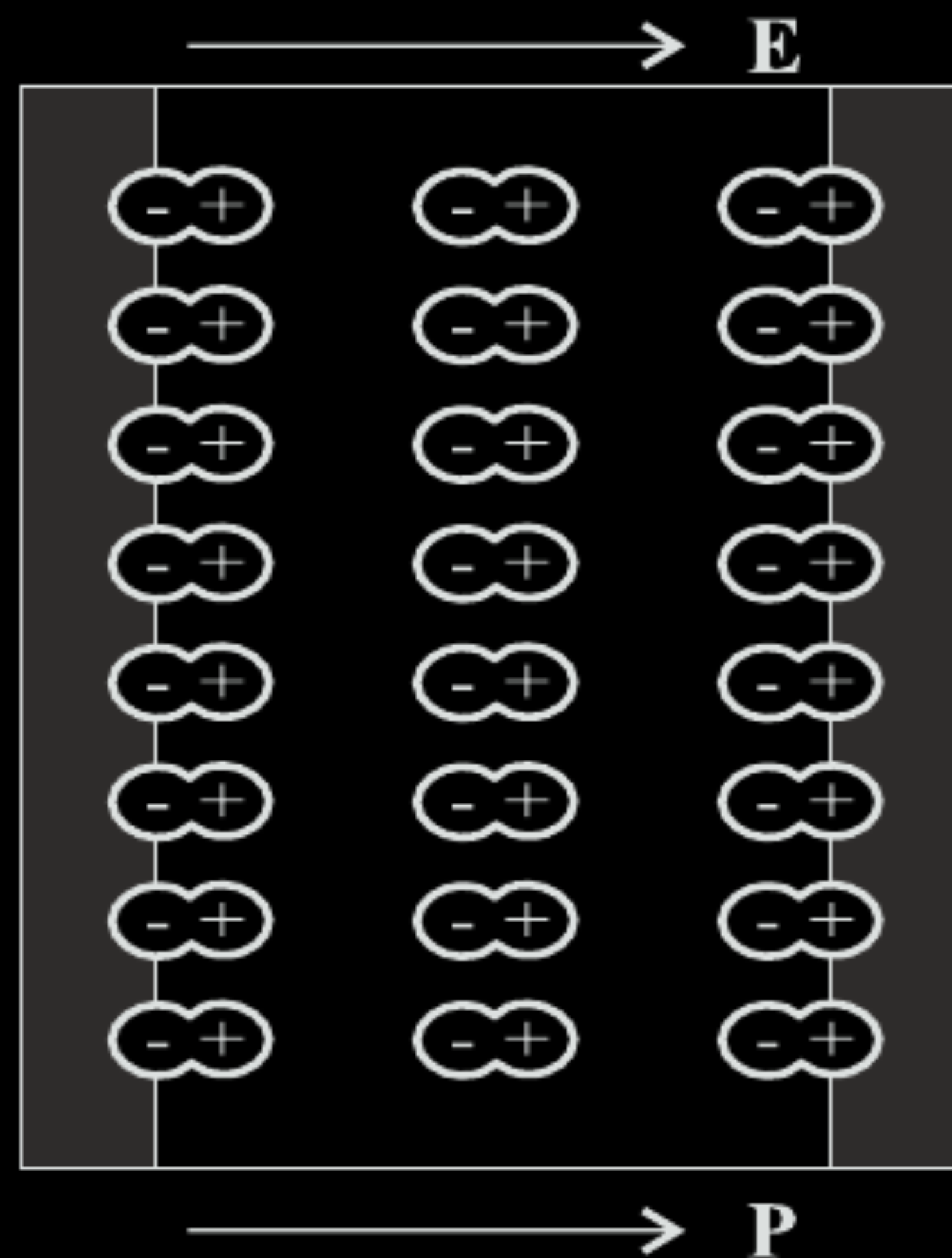
[Note: Award the 1 mark for correct calculation in part (ii) if the value of C_{eq} obtained in part (i) is correct]

- (a) Explain briefly, using a proper diagram, the difference in behaviour of a conductor and a dielectric in the presence of external electric field.
- (b) Define the term polarization of a dielectric and write the expression for a linear isotropic dielectric in terms of electric field.

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CBSE 19

(b) Define the term polarization of a dielectric and write the expression for a linear isotropic dielectric in terms of electric field.



For conductor

Due to induction the free electrons collect on the left face of slab creating equal positive charge on the right face. Internal electric field is equal and opposite to external field; hence net electric field (inside the conductor) is zero.

For dielectric

Due to alignment of atomic dipoles (permanent or induced) along $\vec{E}$, the net electric field within the dielectric decreases.

ii) The net dipole moment developed per unit volume in the presence of external electric field is called polarization vector $\vec{P}$.

Expression $\therefore$
 $\vec{P} = \chi_e \vec{E}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

A 200 μF parallel plate capacitor having plate separation of 5 mm is charged by a 100 V dc source. It remains connected to the source. Using an insulated handle, the distance between the plates is doubled and a dielectric slab of thickness 5 mm and dielectric constant 10 is introduced between the plates. Explain with reason, how the (i) capacitance, (ii) electric field between the plates, (iii) energy density of the capacitor will change ?

CBSE 19

Dielectric slab of thickness 5mm is equivalent to an air capacitor of thickness $= \frac{5}{10} \text{ mm}$.	$\frac{1}{2}$	(iii) $\frac{\text{New energy stored}}{\text{Original energy stored}} = \frac{\frac{1}{2} C' V^2}{\frac{1}{2} C V^2} = \frac{C'}{C} = \frac{10}{11}$	1
Effective separation between the plates with air in between is $= (5 + 0.50) \text{ mm} = 5.5 \text{ mm}$		New Energy density will be $\left(\frac{10}{11}\right)^2$ of the original energy density $= \frac{100}{121}$ of the original energy density.	
(i) Effective new capacitance $= 200 \mu\text{F} \times \frac{5 \text{ mm}}{5.5 \text{ mm}} = \frac{2000}{11} \mu\text{F}$ $\approx 182 \mu\text{F}$	1	Note: If the student writes $C = \frac{A \epsilon_0}{d}$ $C_m = \frac{K A \epsilon_0}{d}$ $E' = \frac{V}{d}$ $U = \frac{1}{2} \epsilon_0 E^2$	
(ii) Effective new electric field $= \frac{100 \text{ V}}{5.5 \times 10^{-3} \text{ m}} = \frac{20000}{11}$ $\approx 18182 \text{ V / m}$	$\frac{1}{2}$	Award full marks	

- (a) Describe briefly the process of transferring the charge between the two plates of a parallel plate capacitor when connected to a battery. Derive an expression for the energy stored in a capacitor.
- (b) A parallel plate capacitor is charged by a battery to a potential difference V . It is disconnected from battery and then connected to another uncharged capacitor of the same capacitance. Calculate the ratio of the energy stored in the combination to the initial energy on the single capacitor.

The electrons are transferred to the positive terminal of the battery from the metallic plate connected to the positive terminal, leaving behind positive charge on it. Similarly, the electrons move on to the second plate from negative terminal, hence it gets negatively charged. Process continuous till the potential difference between two plates equals the potential of the battery.

[Note: award this $\frac{1}{2}$ mark, If the student writes, there will be no transfer of charge between the plates]

Let 'dw' be the work done by the battery in increasing the charge on the capacitor from q to (q+ dq).

$$dW = V dq$$

Where $V = \frac{q}{C}$

$$\therefore dW = \frac{q}{C} dq$$

Total work done in charging up the capacitor

$$W = \int dw = \int_0^Q \frac{q}{C} dq$$

$$\therefore W = \frac{Q^2}{2C}$$

Hence energy stored = $W = \frac{Q^2}{2C} (= \frac{1}{2} CV^2 = \frac{1}{2} QV)$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b) Charge stored on the capacitor $q=CV$

When it is connected to the uncharged capacitor of same capacitance, sharing of charge takes place between the two capacitor till the potential of both the capacitor becomes $\frac{V}{2}$

$\frac{1}{2}$

$$\text{Energy stored on the combination } (u_2) = \frac{1}{2} C \left(\frac{V}{2} \right)^2 + \frac{1}{2} C \left(\frac{V}{2} \right)^2 = \frac{CV^2}{4}$$

$\frac{1}{2}$

Energy stored on single capacitor before connecting

$\frac{1}{2}$

$$U_1 = \frac{1}{2} CV^2$$

Ratio of energy stored in the combination to that in the single capacitor

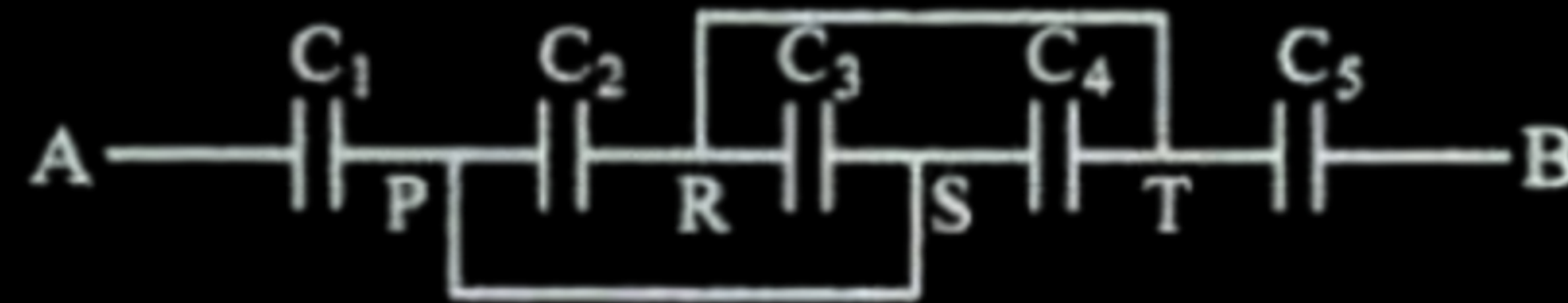
$$\frac{U_2}{U_1} = \frac{CV^2/4}{CV^2/2} = 1:2$$

$\frac{1}{2}$

A capacitor of capacitance C_1 is charged to a potential V_1 while another capacitor of capacitance C_2 is charged to a potential difference V_2 . The capacitors are now disconnected from their respective charging batteries and connected in parallel to each other.

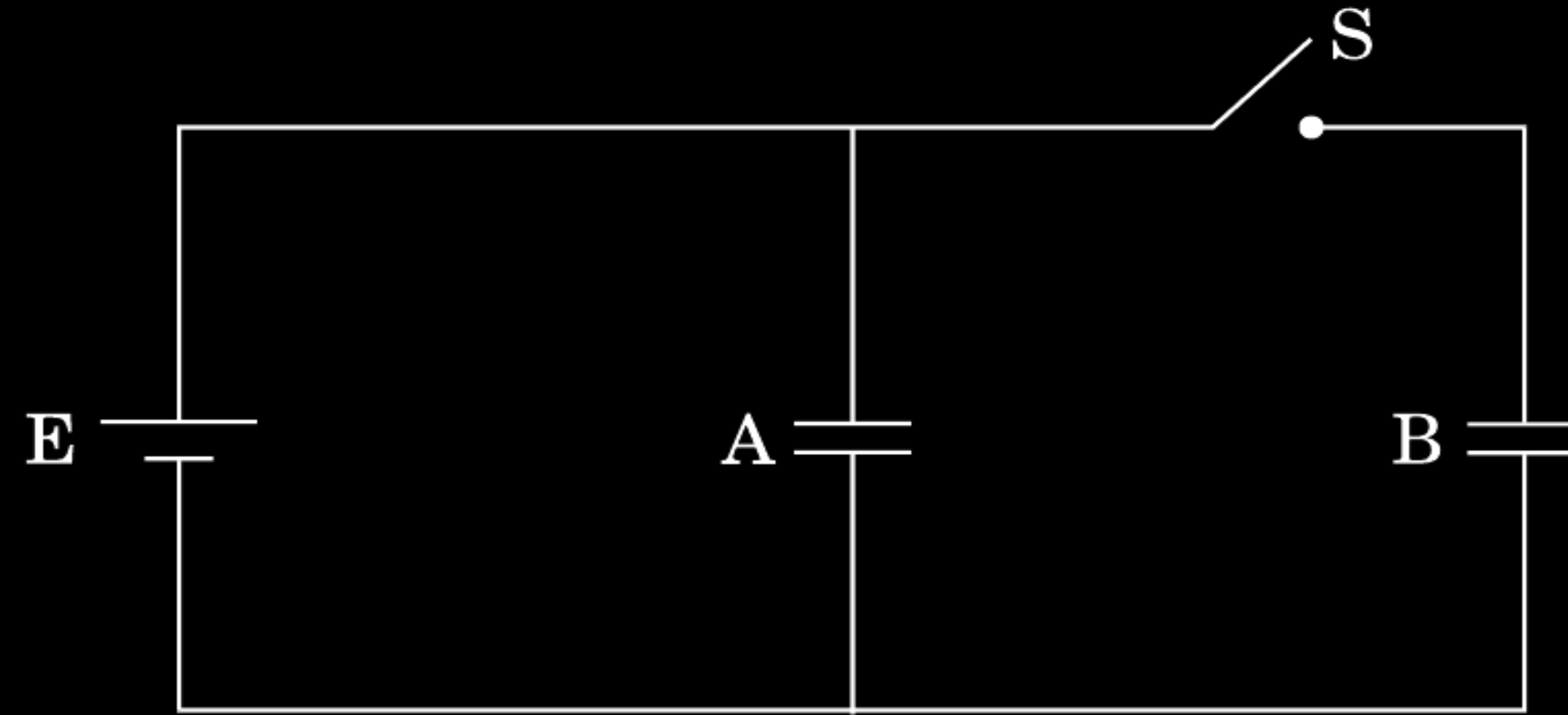
- (a) Find the total energy stored in the two capacitors before they are connected.
- (b) Find the total energy stored in the parallel combination of the two capacitors.
- (c) Explain the reason for the difference of energy in parallel combination in comparison to the total energy before they are connected.

- (i) Find equivalent capacitance between A and B in the combination given below.
Each capacitor is of $2\ \mu\text{F}$ capacitance.



- (ii) If a dc source of $7\ \text{V}$ is connected across AB, how much charge is drawn from the source and what is the energy stored in the network ?

Two identical parallel plate capacitors A and B are connected to a battery of V volts with the switch S closed. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant K . Find the ratio of the total electrostatic energy stored in both capacitors before and after the introduction of the dielectric.



$$\text{Energy stored} = \frac{1}{2} CV^2 \left(= \frac{1}{2} \frac{Q^2}{C} \right)$$

$\frac{1}{2}$

$$\text{Net capacitance with switch S closed} = C + C = 2C$$

$\frac{1}{2}$

$$\therefore \text{Energy stored} = \frac{1}{2} \times 2C \times V^2 = CV^2$$

$\frac{1}{2}$

$$\text{After the switch S is opened, capacitance of each capacitor} = KC$$

$$\therefore \text{Energy stored in capacitor A} = \frac{1}{2} KCV^2$$

For capacitor B,

$\frac{1}{2}$

$$\text{Energy stored} = \frac{1}{2} \frac{Q^2}{KC} = \frac{1}{2} \frac{C^2 V^2}{KC} = \frac{1}{2} \frac{CV^2}{K}$$

$$\therefore \text{Total Energy stored} = \frac{1}{2} KCV^2 + \frac{1}{2} \frac{CV^2}{K} = \frac{1}{2} CV^2 \left(K + \frac{1}{K} \right)$$

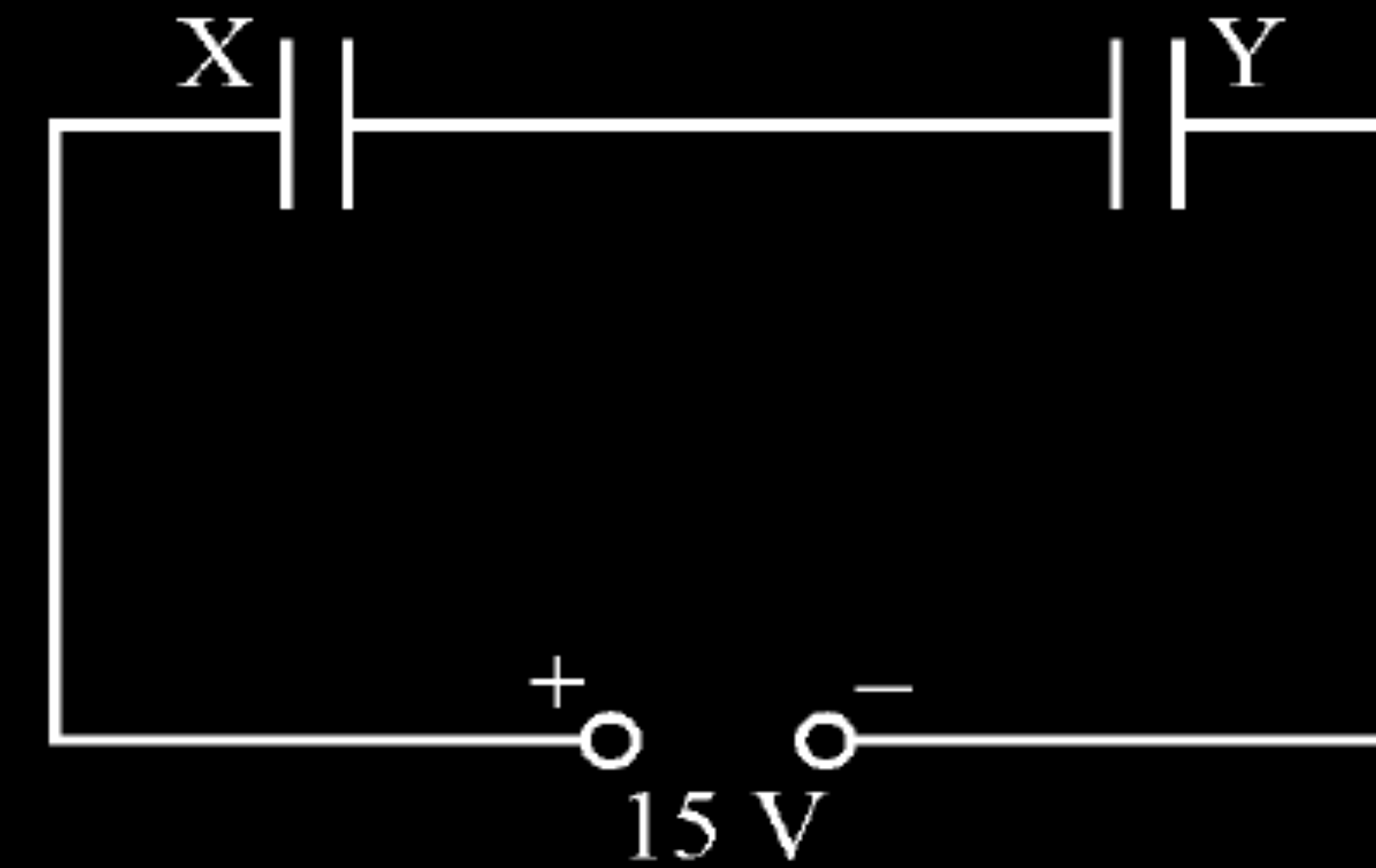
$$= \frac{1}{2} CV^2 \left(\frac{K^2 + 1}{K} \right)$$

$\frac{1}{2}$

$$\therefore \text{Required ratio} = \frac{2CV^2 \cdot K}{CV^2(K^2 + 1)} = \frac{2K}{(K^2 + 1)}$$

$\frac{1}{2}$

Two parallel plate capacitors X and Y have the same area of plates and same separation between them. X has air between the plates while Y contains a dielectric medium of $\epsilon_r = 4$.



- (i) Calculate capacitance of each capacitor if equivalent capacitance of the combination is $4\ \mu\text{F}$.
- (ii) Calculate the potential difference between the plates of X and Y.
- (iii) Estimate the ratio of electrostatic energy stored in X and Y.

i) Let $C_X = C$ $C_Y = 4C$ (as it has a dielectric medium of $\epsilon_r = 4$) For series combination of two capacitors $\frac{1}{C} = \frac{1}{C_X} + \frac{1}{C_Y}$ $\Rightarrow \frac{1}{4\mu F} = \frac{1}{C} + \frac{1}{4C}$ $\frac{1}{4\mu F} = \frac{5}{4C}$ $\Rightarrow C = 5\mu F$ Hence $C_X = 5\mu F$ $C_Y = 20\mu F$	$\frac{1}{2}$ $\frac{1}{2}$
ii) Total charge $Q = CV$ $= 4\mu F \times 15 \text{ V} = 60\mu C$ $V_X = \frac{Q}{C_X} = \frac{60\mu C}{5\mu F} = 12 \text{ V}$ $V_Y = \frac{Q}{C_Y} = \frac{60\mu C}{20\mu F} = 3 \text{ V}$ iii) $\frac{E_x}{E_y} = \frac{\frac{Q^2}{2C_X}}{\frac{Q^2}{2C_Y}} = \frac{C_Y}{C_X} = \frac{20}{5} = 4 : 1$ (Also accept any other correct alternative method)	$\frac{1}{2}$ $\frac{1}{2}$ 1

What is electrostatic shielding ? How is this property used in actual practice ? Is the potential in the cavity of a charged conductor zero ?

What is electrostatic shielding ? How is this property used in actual practice ? Is the potential in the cavity of a charged conductor zero ?

CBSE 16

The field inside a conductor is zero.

$\frac{1}{2}$

Sensitive instruments are shielded from outside electrical influences by enclosing them in a hollow conductor .

1

(any other relevant answer.)

Potential inside the cavity is not zero/ potential is constant.

$\frac{1}{2}$

- (a) A parallel plate capacitor (C_1) having charge Q is connected, to an identical uncharged capacitor C_2 in series. What would be the charge accumulated on the capacitor C_2 ?
- (b) Three identical capacitors each of capacitance $3\mu\text{F}$ are connected, in tern, in series and in parallel combination to the common source of V volt. Find out the ratio of the energies stored in two configurations.

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CBSE 16

a) Zero

$\frac{1}{2}$

b) We have $C_{\text{series}} = \frac{3\mu\text{F}}{3} = 1\mu\text{F}$

Also, $C_{\text{parallel}} = (3+3+3) = 9\mu\text{F}$

Energy stored $= \frac{1}{2} CV^2$

$\therefore$ Energy in series combination $= \frac{1}{2} 1 \times 10^{-6} \times V^2$

$\frac{1}{2}$

$\frac{1}{2}$

Energy in parallel combination $= \frac{1}{2} 9 \times 10^{-6} \times V^2$

$\frac{1}{2}$

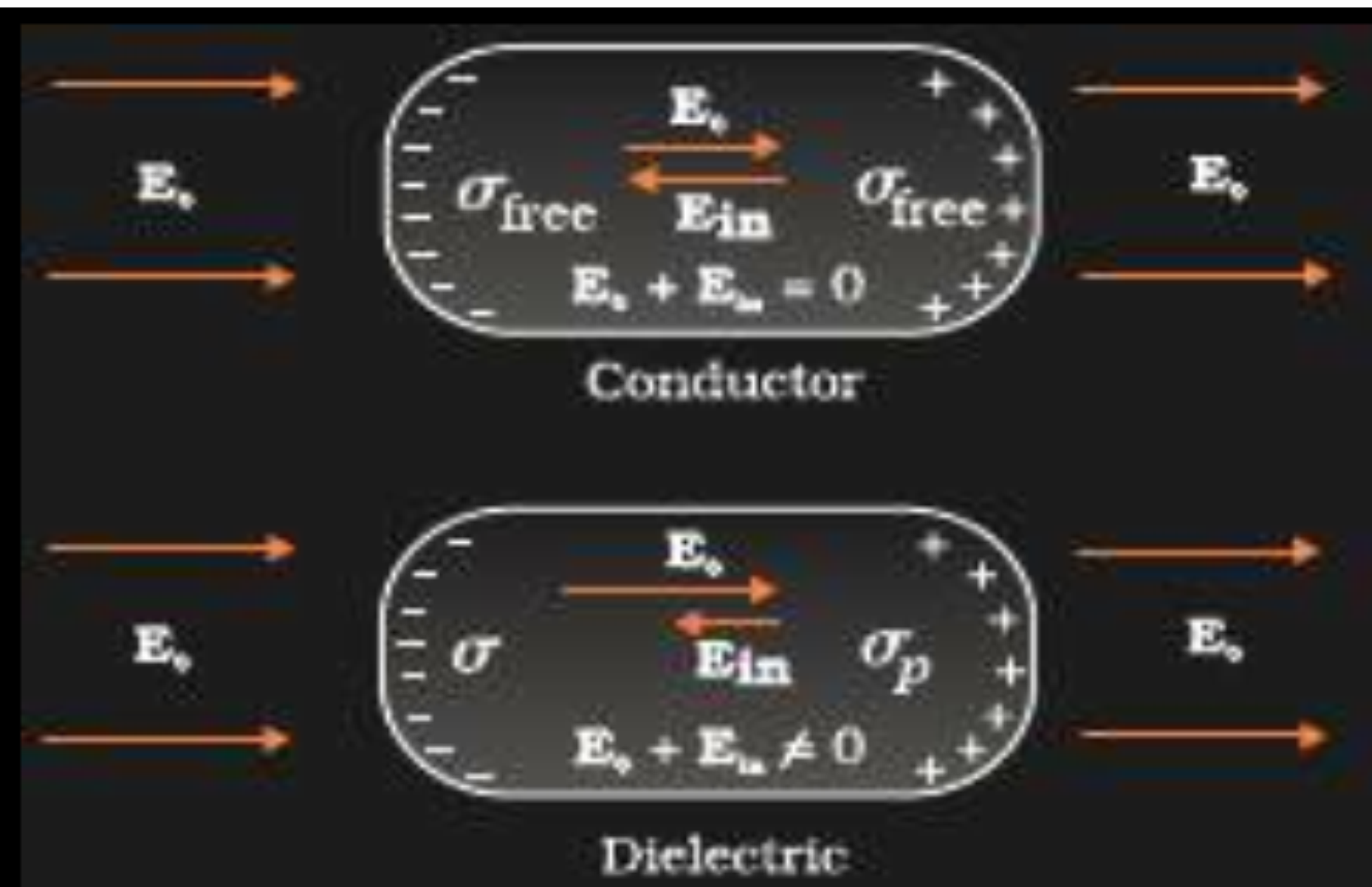
$\frac{1}{2}$

$\therefore$ Ratio = 1:9

$\frac{1}{2}$

- (a) Distinguish, with the help of a suitable diagram, the difference in the behaviour of a conductor and a dielectric placed in an external electric field. How does polarised dielectric modify the original external field ?
- (b) A capacitor of capacitance C is charged fully by connecting it to a battery of emf E . It is then disconnected from the battery. If the separation between the plates of the capacitor is now doubled, how will the following change ?
- (i) charge stored by the capacitor.
 - (ii) field strength between the plates.
 - (iii) energy stored by the capacitor.
- Justify your answer in each case.

a)

 $\frac{1}{2} + \frac{1}{2}$

No electric field inside a conductor .

(Give full credit to diagram. Give $\frac{1}{2}$ mark if explanation only is given without diagram)

Induced electric field ,due to polarisation of dielectric, is in opposite direction to the applied field.

1

$$E_{net} = E_0 - E_p$$

(b)

(i) Charge remains same, as after disconnecting capacitor no transfer of charge take place.

 $\frac{1}{2} + \frac{1}{2}$

(ii) Electric field, $E = \frac{\sigma}{\epsilon_0} = \frac{q}{\epsilon_0 A}$ remain same, as there is no change in charge.

 $\frac{1}{2} + \frac{1}{2}$

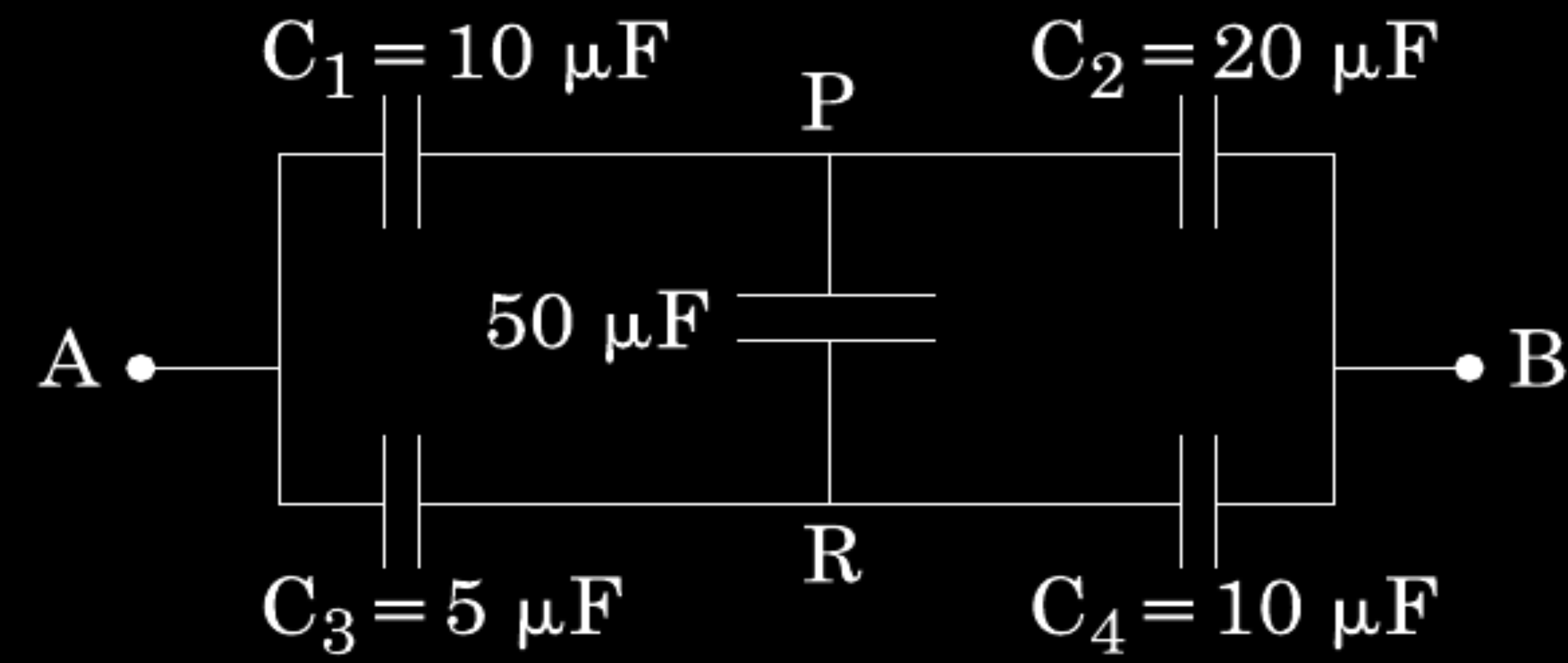
(iii) Energy stored $= \frac{q^2}{2C} = \frac{q^2}{2\left(\frac{\epsilon_0 A}{d}\right)} = \frac{q^2 d}{2\epsilon_0 A}$

 $\frac{1}{2}$

a. Energy will be doubled as separation between the plates(d) is doubled.

 $\frac{1}{2}$

Calculate the equivalent capacitance between points A and B in the circuit below. If a battery of 10 V is connected across A and B, calculate the charge drawn from the battery by the circuit.



$$\therefore \frac{C_1}{C_2} = \frac{C_3}{C_4}$$

$\frac{1}{2}$

This is the condition of balance so there will be no current across PR (50 μ F capacitor)

Now C_1 and C_2 are in series

$$C_{12} = \frac{C_1 C_2}{C_1 + C_2} = \frac{10 \times 20}{10 + 20} = \frac{200}{30} = \frac{20}{3} \mu\text{F}$$

$\frac{1}{2}$

$\therefore C_3$ and C_4 are in series

$$C_{34} = \frac{C_3 C_4}{C_3 + C_4} = \frac{5 \times 10}{5 + 10} = \frac{50}{15} = \frac{10}{3} \mu\text{F}$$

$\frac{1}{2}$

Equivalent capacitance between A and B is

$$C_{AB} = C_{12} + C_{34} = \frac{20}{3} + \frac{10}{3} = 10 \mu\text{F}$$

$\frac{1}{2}$

Charge drawn from battery (q) = CV

$\frac{1}{2}$

$$= 10 \times 10 \mu\text{C}$$

$$= 100 \mu\text{C or } 10^{-4} \text{C}$$

$\frac{1}{2}$

- (ii) Find the ratio of the potential differences that must be applied across the parallel and series combination of two capacitors C_1 and C_2 with their capacitances in the ratio 1 : 2 so that the energy stored in the two cases becomes the same.

- (ii) Find the ratio of the potential differences that must be applied across the parallel and series combination of two capacitors C_1 and C_2 with their capacitances in the ratio 1 : 2 so that the energy stored in the two cases becomes the same.

CBSE 16

$$U_s = \frac{1}{2} C_s V_s^2$$

$$\frac{1}{2}$$

$$U_p = \frac{1}{2} C_p V_p^2$$

$$\frac{1}{2}$$

$$\Rightarrow \frac{V_{series}}{V_{parallel}} = \sqrt{\frac{C_{equivalent parallel}}{C_{equivalent series}}}$$

$$= \sqrt{\frac{\frac{C_1 + C_2}{C_1 C_2}}{C_1 + C_2}}$$

$$\frac{1}{2}$$

$$= \frac{C_1 + C_2}{\sqrt{C_1 C_2}} = \frac{3}{\sqrt{2}}$$

$$\frac{1}{2}$$

- (i) If two similar large plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$ are separated by a distance d in air, find the expressions for
- (a) field at points between the two plates and on outer side of the plates. Specify the direction of the field in each case.
 - (b) the potential difference between the plates.
 - (c) the capacitance of the capacitor so formed.
- (ii) Two metallic spheres of radii R and $2R$ are charged so that both of these have same surface charge density σ . If they are connected to each other with a conducting wire, in which direction will the charge flow and why ?

Inside

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

$$= \frac{\sigma + \sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

Outside

$$\vec{E} = \vec{E}_2 - \vec{E}_1$$

$$= \frac{\sigma - \sigma}{2\epsilon_0} = 0$$

b. Potential difference between plates

$$V = Ed = \frac{1}{\epsilon_0} \frac{Qd}{A}$$

c. Capacitance

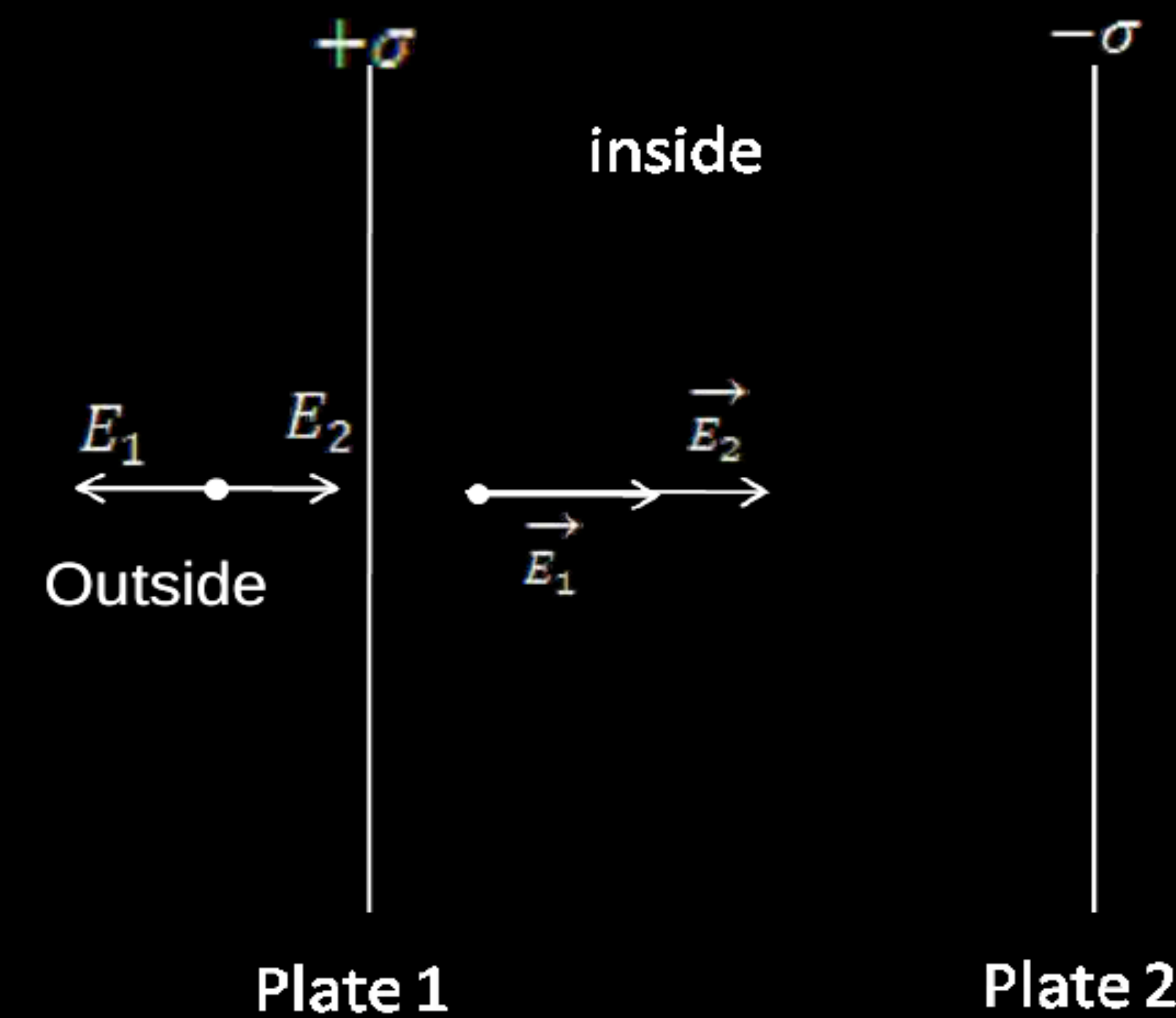
$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}$$

ii. As potential on and inside a charged sphere is given

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{4\pi r^2 \sigma}{r}$$

$$\therefore V \propto r$$

Hence, the bigger sphere will be at higher potential, so charge will flow from bigger sphere to smaller sphere.



$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$